

HOW COMMON ARE COMMON ENVELOPES? QUANTIFYING THEIR ROLE IN FORMING GRAVITATIONAL-WAVE SOURCES

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Abstract

A central goal of gravitational-wave astronomy is to use merging binary black hole (BBH), black hole–neutron star (BHNS), and binary neutron star (BNS) systems as fossils to reconstruct the formation and evolution of massive stars across cosmic time. In practice, this inference relies on population-synthesis models that map populations of massive stellar binaries to merging compact objects. However, these models currently disagree on the dominant orbital-hardening mechanisms within the isolated binary-evolution paradigm, particularly on whether common-envelope (CE) evolution is required. To address this, we compile and systematically compare formation-channels from more than 200 isolated binary evolution population synthesis simulations. We organize these models within a unified hierarchical taxonomy that separates systems evolving with and without CE phases. We find that for BBH and BHNS formation pathways, contributions with and without CE phases span the full allowed range (0–100%) and often yield similar merger rates. This demonstrates a fundamental degeneracy: merger-rate measurements alone do not uniquely constrain the underlying evolutionary pathways. In contrast, BNS formation in current population-synthesis studies universally proceeds through at least one CE phase, with ≥ 90 –100% of systems undergoing CE. This suggests that, within present isolated binary-evolution frameworks, CE evolution plays a central role in forming merging BNS and may strongly impact the relative merger rates across compact-object populations. We find that the relative contributions of with-CE and without-CE formation pathways are largely dependent on key assumptions governing mass-transfer stability and angular-momentum loss, in particular the critical mass ratio and the population-synthesis parameters α (common-envelope efficiency), β (mass-transfer efficiency), and γ (angular-momentum loss), with additional contributions from supernova remnant mass prescriptions and σ (natal kick magnitude). Our results demonstrate that formation-channel contributions within isolated binary evolution remain highly uncertain. Progress will require systematic cross-code comparisons, explicit quantification of formation-channel fractions, and targeted exploration of key physical uncertainties (e.g., mass-transfer stability, angular-momentum loss, and CE physics) to break degeneracies and robustly connect gravitational-wave observations to the astrophysical processes shaping massive binary evolution.

Subject headings: Massive Star Evolution, Gravitational Wave Astrophysics, Common Envelope

1. INTRODUCTION

The number of detected gravitational-wave sources from merging stellar-mass BBH, BHNS, and BNS systems is rapidly increasing, revealing new insights into their masses, spins, and merger rates (e.g., [The LIGO Scientific Collaboration et al. 2025a,c, 2026a,b](#)). These observations provide a unique opportunity to study massive stars across cosmic time where merging compact-object binaries detected through gravitational waves can be used as *fossils* of massive binary evolution,

encoding information about the formation, evolution, and explosive deaths of their progenitor stars and the environments in which they formed (e.g. [Mandel & Farmer 2022](#); [Costa et al. 2023](#); [Chruślińska 2024](#)).

One of the leading formation scenarios for these systems is the isolated binary evolution channel, in which two massive stars are born in a binary system that evolves isolated from dynamical interactions with other stars (e.g. [Mandel & Farmer 2018](#); [Mapelli 2021](#); [Mandel & Broekgaarden 2022](#)). This channel is typically characterized by initially wide binaries that can evolve through episodes of mass transfer and/or

other orbital-altering processes, such as supernova kicks, that tighten the binary and eventually produce compact-object systems that merge through gravitational-wave emission and can be observed today (e.g. Mandel & Farmer 2022; Marchant & Bodensteiner 2024). The isolated binary evolution channel has been shown to reproduce several observed properties of the BBH, BHNS, and BNS populations (e.g., Belczynski et al. 2016; Stevenson et al. 2017; Santoliquido et al. 2020; Bavera et al. 2021; Briel et al. 2021; Broekgaarden & Berger 2021; van Son et al. 2022; Olejak et al. 2022a; Dorozzmai & Toonen 2024; Hendriks et al. 2023; Xing et al. 2024a; The LIGO Scientific Collaboration et al. 2026b).

However, the physical details of the processes that tighten massive binaries within this scenario remain highly uncertain (e.g. Marchant et al. 2021b; Gallegos-Garcia et al. 2021; Iorio et al. 2023b; Willcox et al. 2023; Romagnolo et al. 2025; Xu et al. 2025; Klencki et al. 2026). A key ingredient in this problem is the role of binary interactions, particularly mass transfer, in transforming wide binaries into compact systems that can merge within a Hubble time (e.g. Belczynski et al. 2002; Marchant et al. 2021b; Li & Chen 2024). Mass transfer governs orbital evolution and strongly affects the final properties of the compact-object binary, including masses, spins, and orbital separations (e.g. Dominik et al. 2012; Neijssel et al. 2019; Bavera et al. 2021; van Son et al. 2022; Briel et al. 2023; Olejak et al. 2024; Klencki et al. 2026; Xu et al. 2025). It also plays a central role in shaping a broad range of observable astrophysical phenomena (e.g. Soker et al. 2019; Evans et al. 2020; Pavlovskii et al. 2017; Metzger 2022; Korb et al. 2024; Wang et al. 2024; Klencki & Metzger 2025). Yet the stability and outcome of mass transfer in massive binaries remain poorly understood, and many key phases are difficult to observe directly (e.g. Sana et al. 2012; Schneider et al. 2015; Agrawal et al. 2023; Marchant & Bodensteiner 2024). Gravitational-wave observations may therefore provide a powerful probe of these otherwise elusive evolutionary stages, provided we can robustly connect observed populations to their formation pathways (e.g. Mandel & Farmer 2017; Marchant & Bodensteiner 2024).

Historically, it was widely assumed that dynamically unstable mass transfer leading to a CE phase is required to produce merging compact-object binaries (e.g. Smarr & D. Blandford 1976; Paczyński 1976; Tutukov & Yungelson 1993; van den Heuvel 1994; Taam & Sandquist 2000; Tauris & van den Heuvel 2006; Dominik et al. 2012; Ivanova et al. 2013a, 2020). During a CE phase, unstable mass transfer leads to the engulfment of the companion in a shared envelope, driving rapid orbital shrinkage. If the envelope is successfully ejected, this can produce short-period binaries that merge within a Hubble time. As a result, CE evolution has long been considered a critical ingredient in forming gravitational-wave sources (Iaconi & De Marco 2019a; Ivanova et al. 2020; Mandel & Broekgaarden 2022; Tauris & van den Heuvel 2023, and references therein).

More recently, however, an alternative picture has emerged. A growing body of work suggests that CE onset and the successful (complete) ejection of the shared envelope, particularly in massive binaries, may occur under more restrictive conditions than traditionally assumed (Mennekens & Vanbeveren 2014; Kruckow et al. 2016; Eldridge & Stanway 2016; Fragos et al. 2019; Klencki et al. 2020, 2021a; Marchant et al. 2021b; Gallegos-Garcia et al. 2021). At the same time, there is increasing evidence that mass transfer in massive binaries may be more stable than previously assumed (van den Heuvel et al.

2017a; Inayoshi et al. 2017; Ge et al. 2015, 2020a; Pavlovskii et al. 2017; Olejak et al. 2021; Bhattacharjee et al. 2026). As a result, sequences of stable mass-transfer episodes may in some cases shrink binaries sufficiently without requiring a CE phase (e.g. van den Heuvel et al. 2017a; Inayoshi et al. 2017; Neijssel et al. 2019; Bavera et al. 2021; Marchant et al. 2021b; Gallegos-Garcia et al. 2021; Olejak et al. 2024; Picco et al. 2024; Xu et al. 2025). In these models, stable mass transfer can become a viable, and in some cases dominant, formation channel for BBH and BHNS mergers (e.g. Neijssel et al. 2019; Gallegos-Garcia et al. 2021; van Son et al. 2022; Briel et al. 2023; Iorio et al. 2023b), though many uncertainties remain (e.g. Klencki et al. 2026; Briel et al. 2026). In contrast, BNS systems appear more consistent across models, with most studies finding that their formation involves at least one CE phase, with only a small parameter space allowing for alternative pathways (e.g. Chruślińska et al. 2018; Vigna-Gómez et al. 2018; Wagg et al. 2022; Gallegos-Garcia et al. 2023; Iorio et al. 2023b; Tanaka et al. 2023; van Son et al. 2024; Picco et al. 2024; Chattaraj et al. 2026). These results highlight a major question in massive binary evolution: whether CE evolution is (truly) required to form the majority of gravitational-wave sources (e.g., Klencki et al. 2026).

More fundamentally, the diversity highlighted above reflects qualitatively different physical assumptions regarding how interacting binaries redistribute angular momentum and dissipate orbital energy. In CE evolution, rapid inspiral, hydrodynamic drag, and envelope ejection dominate the orbital hardening (e.g., Paczyński 1976; Taam & Sandquist 2000; Ivanova et al. 2013a, 2020), whereas stable mass transfer (SMT) pathways rely on stable angular-momentum redistribution through Roche-lobe overflow without the onset of a dynamically unstable engulfment phase (e.g., Soberman et al. 1997; Pavlovskii et al. 2017; van den Heuvel et al. 2017a; Marchant et al. 2021b; Dorozzmai & Toonen 2024). As a result, the relative contributions of CE and without-CE formation channels are not merely a quantitative question of efficiency, but reflect fundamentally different physical pictures of how massive binaries respond to mass transfer and evolve toward compact-object mergers (e.g., Gallegos-Garcia et al. 2021; Marchant et al. 2021a; Klencki et al. 2026). Disentangling these physical pictures is therefore a prerequisite for robustly connecting gravitational-wave observations to the astrophysical processes shaping massive binary evolution.

Despite extensive work investigating the formation channels of BBH, BHNS, and BNS mergers within the isolated binary evolution scenario, most studies so far focus on a single compact-object class, a single population synthesis code, and/or a limited exploration of parameter variations. As a result, it remains difficult to disentangle differences arising from physical assumptions within a model from those driven by the underlying population-synthesis framework itself. A systematic comparison across studies is therefore needed to map the true diversity of the isolated binary evolution formation pathways.

In this work, we compile and compare formation-channel predictions from more than two hundred population-synthesis simulations in the isolated binary evolution paradigm. By bringing together results across different codes and mapping out their assumptions, we aim to disentangle parameter-driven variations from framework-dependent differences and to provide a global view of the current theoretical landscape. Specifically, we address three key questions in the context of isolated binary evolution population synthesis models:

1. *Is common-envelope evolution required to form BBH, BHNS, and BNS mergers?*
2. *What fraction of BBH, BHNS, and BNS mergers typically forms through pathways involving a CE phase?*
3. *What drives the disagreement? Which physical assumptions and model choices are responsible for the large differences between studies?*

By systematically organizing isolated binary evolution formation-channel predictions within a unified hierarchical taxonomy, analyzing their assumptions, and comparing them across the literature, this work aims to provide a foundation for interpreting gravitational-wave observations and identifying the key uncertainties that currently limit our understanding of massive binary evolution. This is further accomplished by making all data and code to reproduce all results and figures in this work publicly available.

2. METHOD: DATA COLLECTION

We compile intrinsic compact-object merger rates by formation pathway from the isolated binary-evolution literature. We restrict the sample to studies published within the past decade that (i) report **formation-channel-specific** merger rates and (ii) provide **intrinsic (astrophysical)** rates rather than detection-weighted (“detectable”) rates. We exclude studies that only report detectable rates because such rates depend on assumed detector sensitivities and selection functions, which vary across analyses and make comparison challenging. For transparency and reproducibility, the rate-extraction procedure for each study is documented in Appendix A. We organize formation pathways into two hierarchical levels, illustrated in Figure 1.

2.1. Level 1 classification

At Level 1, systems are classified into three broad categories according to whether they experience zero CEs (“without CE”), at least one CE episode (“with CE”), or belong to a residual “other” category where the presence or absence of a CE phase is not specified. This distinction between with- and without-CE evolution is chosen as a broad category to focus on the question whether the formation pathway involves a CE. The “other” category is included for studies that aggregate a small residual fraction of systems into unspecified or unlabeled pathways, for example when many detailed channels are presented individually but a remainder is left without an explicit statement on CE involvement (e.g. Olejak et al. 2021; Neijssel et al. 2019; Mestichelli et al. 2025). In practice, this contribution is typically subdominant and only appears in our compilation for the formation-channel classification of Olejak et al. (2021). When a study reports only finer subdivisions (Level 2), we map each sub-channel to Level 1 and sum sub-channels to obtain the total “without CE” and “with CE” contributions. If a study reports rates for only a subset of channels, we treat unreported channels as negligible and set their rates to zero for that study. This convention is applied consistently across the compilation and is recorded alongside the extracted values.

2.2. Without-CE versus stable mass-transfer channel terminology

Importantly, throughout this work the term “without CE” refers strictly to the absence of a CE episode and should not be

interpreted as synonymous with a SMT channel. Distinguishing binaries that avoid CE evolution from those that specifically evolve through stable mass transfer has become increasingly relevant given growing interest in whether gravitational-wave sources can form entirely without a CE phase (e.g. Bavera et al. 2021; Briel et al. 2023; Gallegos-Garcia et al. 2021, 2023). Many studies therefore report results at the level of “CE” versus “stable”, “SMT”, or “only stable mass transfer” channels (e.g. Bavera et al. 2021; van Son et al. 2023; Hendriks et al. 2023).

However, these terms are not always used consistently across the literature. In several population-synthesis studies, the population labeled as SMT is identified operationally as all systems that avoid a CE phase during their evolution (e.g. Bavera et al. 2021; van Son et al. 2023). As a result, these selections can include not only binaries undergoing stable Roche-lobe overflow, but also systems that never experience mass transfer at all. Moreover, the term SMT itself is also used inconsistently. In some studies it refers narrowly to only systems involving stable mass transfer both before *and* after the first supernova (Neijssel et al. 2019; Broekgaarden et al. 2022c; Iorio et al. 2023b) (as we will see, this is sometimes referred to as the ‘classic SMT’ or ‘SMT+SMT’ pathway), whereas in others it refers more broadly to any pathway involving stable mass transfer while avoiding CE evolution (van Son et al. 2022; Briel et al. 2021; Bavera et al. 2021). Throughout this work we therefore reserve “without CE” for the broader Level 1 classification of any formation pathway that does not involve a CE, while using SMT specifically for Level 2 channels involving stable mass transfer episodes.

An important example of such a non-interacting pathway is the ‘NON+NON’ or “lucky kick” channel, in which binaries avoid both Roche-lobe overflow and CE evolution entirely. These systems can nevertheless form merging compact-object binaries through favorable natal kicks that tighten the orbit sufficiently for the binary to remain bound and merge despite initially wide separations (e.g. Vigna-Gómez et al. 2018; Broekgaarden et al. 2021; Chattaraj et al. 2026; Mestichelli et al. 2025). Similar non-interacting pathways have also been proposed for the formation of ultra-wide pulsar systems (e.g. Stevenson et al. 2022; van der Wateren et al. 2024).

Another pathway that typically avoids a CE phase and is sometimes modeled within the same simulation is chemically homogeneous evolution (CHE) (e.g. de Mink & Mandel 2016; Mandel & de Mink 2016). Although often assumed to be subdominant, several population synthesis studies find that CHE may contribute a non-negligible fraction of the observable BBH population (Riley et al. 2021; Stevenson et al. 2022; van Son et al. 2024; Li et al. 2025a). We further note that recent work suggests that CHE binaries may evolve through prolonged contact phases (Marchant et al. 2016), blurring the distinction between interaction-free and interacting channels. Nevertheless, because the two studies included in our compilation that contain CHE Li et al. (2025a); van Son et al. (2024) explicitly classify CHE systems as avoiding Roche-lobe overflow and CE evolution, we group them within the broader level 1 “without-CE” category as a no mass transfer subchannel throughout this work.

This diversity in terminology and classification criteria further motivates the need for a systematic framework to compare and categorize isolated binary evolution channels, a point we revisit in Section 4.3.

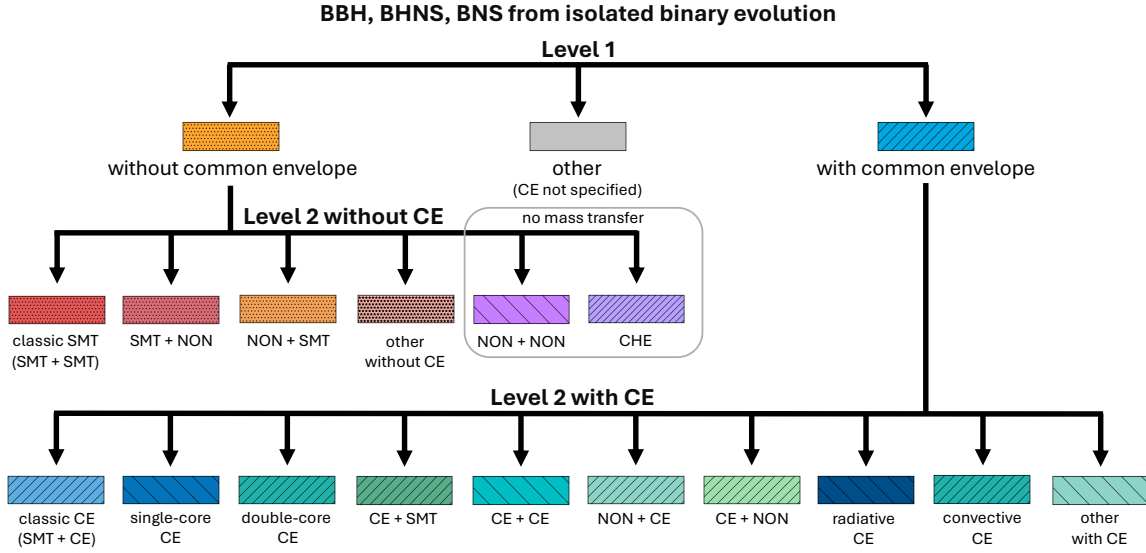


FIG. 1.— Hierarchical formation-channel taxonomy used in this work to compile intrinsic merger rates from isolated binary evolution simulations of BBH, BHNS, and BNS systems. **Level 1** partitions mergers into systems evolving *without CE* (no CE episodes), *with CE* (one or more CE episodes), and an *other* category for residual contributions where the original study does not further specify the CE involvement. Throughout this work we primarily focus on the more consistently defined Level 1 classification. **Level 2** illustrates the broader diversity of formation-channel definitions used across the literature. The Level 2 channels follow the original definitions adopted in each study and are therefore not always mutually exclusive. For the *without-CE* branch, we distinguish the formation pathway involving stable mass transfer before and after formation of the first compact object (classic SMT; SMT+SMT), channels with stable mass transfer only before or after compact-object formation (SMT+NON and NON+SMT), systems without any mass-transfer episodes (NON+NON), CHE, and an “other without CE” residual category. In the studies considered here, the NON+NON and CHE pathways correspond to channels without any mass-transfer episodes. For the *with-CE* branch, we include the classic CE pathway (SMT+CE), channels distinguished by the CE type prior to compact-object formation (single-core CE and double-core CE), channels categorized by whether the CE occurs before and/or after formation of the first compact object (CE+SMT, CE+CE, NON+CE, and CE+NON), envelope-structure subclasses (radiative and convective CE), and an “other with CE” residual category. Because different studies adopt different classification schemes, Level 2 channels can overlap conceptually. For example, systems classified as radiative or convective CE in one study may correspond to binaries grouped into single-core CE, double-core CE, classic CE, or other with-CE categories in another study. See text and Appendix A for further discussion.

2.3. Level 2 classification

We further subdivide both the “without CE” and “with CE” Level 1 pathways using the Level 2 subchannel taxonomy illustrated in Figure 1. Here, SN1 denotes the formation of the first compact object (via supernova or direct collapse). We use the shorthand SMT for stable mass transfer, CE for common-envelope evolution, and NON for the absence of mass transfer in the relevant evolutionary phase. The Level 2 taxonomy is intended as a flexible comparative framework to organize the heterogeneous channel definitions used throughout the literature; additional discussion of these study-dependent classifications is provided in Section 2.4.

- **Level 2 without CE (no CE episodes):** For studies that provide interaction-sequence information around SN1, we distinguish the following subchannels: (i) SMT+SMT, i.e. stable mass transfer both before and after SN1 (often termed the “only stable mass transfer” or “classic” SMT pathway; e.g., Olejak et al. 2021; Briel et al. 2023; Broekgaarden et al. 2021; Sgalletta et al. 2024; Iorio et al. 2023b; Boesky et al. 2024a); (ii) SMT+NON and NON+SMT, i.e. systems with stable mass transfer in only one of the two phases relative to SN1; and (iii) NON+NON, i.e. systems that do not undergo mass transfer in either phase around SN1. When a study reports CHE as a distinct formation pathway, we record it separately as CHE (in the studies considered here by definition involving no Roche-lobe overflow/SMT and no CE; Li et al. 2025a). Any residual or study-specific categories that do not map cleanly onto the above bins are recorded as “other without CE”. This channel is only included in simulations such as

Broekgaarden et al. (2022c) where the authors split up the without-CE formation channel contribution into the classic stable mass transfer channel (SMT + SMT) and everything else without a CE phase.

- **Level 2 with CE (one or more CE episodes):** When studies provide CE-subtype information, we retain it at Level 2. The most common refinement separates a “classic CE” channel from CE channels where at CE onset both stars are evolved or from different sequences of mass transfer: (i) *classic CE*, commonly corresponding to an interaction sequence in which SMT occurs prior to SN1 and a CE episode occurs after SN1 (SMT + CE in the pre-/post-SN1 shorthand; e.g., Broekgaarden et al. 2021; Olejak et al. 2021; Boesky et al. 2024a; Sgalletta et al. 2024; Iorio et al. 2023b); (ii) *single-core CE* and *double-core CE* (as the first mass transfer), where one or both stars, respectively, are evolved at the onset of the CE episode (e.g., Vigna-Gómez et al. 2018; Broekgaarden et al. 2021; Romagnolo et al. 2023; Sgalletta et al. 2025; Iorio et al. 2023b). The double-core CE channel typically requires both stars to evolve on a similar timescale, and as a result is mostly effective for BNS progenitors with very equal mass ratio $q \gtrsim 0.9$ and has historically been discussed in various contexts involving NSs (e.g. Brown 1995; Bethe & Brown 1998; Dewi et al. 2006; Belczynski et al. 2010a; Hwang et al. 2015; Ivanova et al. 2020; Vigna-Gómez et al. 2020). (iii) Recent studies instead split CE episodes by envelope structure at onset; we follow the authors and record these as *radiative CE* and *convective CE* when reported (Dorozzmai & Toonen 2024). (iv) For studies that provide interaction-sequence splits relative to SN1, we ad-

ditionally map CE systems into the combinations SMT + CE, CE + SMT, CE + CE, NON + CE, and CE + NON (e.g., [Briel et al. 2023](#)). (v) Finally, we retain “other with CE” for residual CE-involving pathways, including accretion-induced collapse channels (AIC with CE) when explicitly reported (e.g., [Pellouin et al. 2025](#)). The contributions of this channel are typically very small.

2.4. Level 2 bookkeeping framework

The Level 2 categories shown in Figure 1 should not be interpreted as a single universally adopted or mutually exclusive classification scheme used throughout the literature. Instead, they represent a bookkeeping framework constructed to map heterogeneous author-defined formation channels onto a common taxonomy. Different studies typically classify systems according to different evolutionary criteria and often report only a limited subset of these channels (commonly 2–4 out of the broader set shown here). As a result, apparent overlaps between Level 2 categories usually do not occur within a single study, but instead arise because different works partition the evolutionary space using different definitions or levels of detail. For example, one study may distinguish systems by interaction sequence relative to SN1 (e.g., SMT+CE versus CE+SMT), while another separates CE systems according to envelope structure (radiative versus convective CE) or according to whether the CE is single-core or double-core. The taxonomy adopted here is therefore intended as a flexible comparative framework rather than a claim that all studies track the same mutually exclusive channel decomposition.

2.5. Local merger rates

All reported rates are converted to local intrinsic merger rates (redshift $z \approx 0$ or $z \approx 0.2$ in some studies) in units of $\text{Gpc}^{-3} \text{ yr}^{-1}$, which facilitates comparison with gravitational-wave measurements and removes dependence on detector sensitivity. We denote these rates as $\mathcal{R}_m$. While these local rates are most commonly published, we note that formation-channel contributions can evolve significantly with redshift ([Olejak et al. 2022b](#); [van Son et al. 2022](#); [Boesky et al. 2024b](#)). Variations in assumed star-formation histories or in the precise definition of “local” redshift likely introduce minor differences in the inferred rates, as demonstrated in Section 3.2, and should be investigated in more detail in future work.

We compare the merger rates predicted by the population-synthesis simulations to the observationally inferred merger rates for BBH, BHNS, and BNS systems from [The LIGO Scientific Collaboration et al. \(2026b\)](#), derived from the gravitational-wave detections reported in the GWTC-5 catalog ([The LIGO Scientific Collaboration et al. 2026a](#)). Specifically, we adopt the merger-rate constraints inferred using the FullPop and PixelPop population models reported in Table 2 of [The LIGO Scientific Collaboration et al. \(2026b\)](#). These constraints already incorporate both statistical and population-model uncertainties.

For BBHs, [The LIGO Scientific Collaboration et al. \(2026b\)](#) report merger rates at $z \approx 0.2$, corresponding to the redshift range where the BBH merger rate is currently best constrained observationally. We therefore adopt a BBH merger-rate range of $27.5\text{--}49.4 \text{ Gpc}^{-3} \text{ yr}^{-1}$. In contrast, the BNS and BHNS merger rates are reported at $z = 0$, for which we adopt ranges of $5.1\text{--}154.7 \text{ Gpc}^{-3} \text{ yr}^{-1}$ and $6.7\text{--}32.8 \text{ Gpc}^{-3} \text{ yr}^{-1}$, respectively.

3. RESULTS

3.1. Overview of Formation Channel Diversity

Figure 2 presents the compiled formation-channel contributions across more than one hundred population-synthesis simulations for BBH, BHNS, and BNS mergers. We first summarize the Level 1 classification, distinguishing formation pathways into those with and without CE evolution; shown in Figures 3–5. We then follow with the more detailed Level 2 decomposition, for which the figures are presented in Appendix ??.

3.1.1. Level 1: With CE versus without-CE pathways

We begin by comparing the relative importance of systems that experience at least one CE phase to those that avoid CE entirely across the compiled population-synthesis simulations. To provide a global overview of the diversity in formation pathways across all simulations, Figure 2 shows the fraction of systems forming without a CE phase for BBH, BHNS, and BNS mergers. Each point corresponds to a single population-synthesis model in our compilation.

This immediately illustrates the central result of this work: BBH and BHNS formation pathways span nearly the full allowed range from CE-dominated to without-CE-dominated formation pathways, whereas BNS systems cluster strongly toward CE-dominated pathways. The BHNS simulations also show a small clustering around small without-CE fractions.

For BBH mergers, the Level 1 classification exhibits the greatest diversity among the compact object merger classes as shown in detail in Figure 3. Both CE-dominated and without-CE-dominated regimes are present, with many models occupying intermediate fractions, highlighting the strong sensitivity of BBH formation to assumptions governing binary interaction physics. The associated local intrinsic merger rates span $\sim [1, 500] \text{ Gpc}^{-3} \text{ yr}^{-1}$ and show no clear correlation with the CE channel contribution. This reveals a fundamental degeneracy: models with similar channel fractions can yield substantially different merger rates, while comparable total merger rates can arise from either CE- or without-CE-dominated pathways. This implies that merger rates alone do not uniquely constrain the underlying evolutionary history, as qualitatively different formation pathways can give rise to similar observable populations.

The BHNS population displays somewhat less diversity than BBHs, as shown in Figure 4. In many simulations, CE pathways dominate, typically contributing $\geq 70\%$ of BHNS mergers. However, a significant subset of models predicts substantial contributions from systems evolving without CE, and in some cases (particularly for COMPAS models with low CE efficiencies of $\alpha \sim 0.1\text{--}0.5$ and low mass transfer efficiencies of $\beta \sim 0.25\text{--}0.5$) BHNS formation is instead dominated ($\geq 60\%$) by without-CE pathways. This demonstrates that no single evolutionary channel consistently dominates across studies. The predicted local intrinsic merger rates span $\sim [1, 400] \text{ Gpc}^{-3} \text{ yr}^{-1}$, comparable to the BBH population.

In contrast, BNS mergers exhibit markedly more uniform behavior, as shown in detail in Figure 5. Across the compiled simulations (and additional literature), all models consistently predict that the dominant formation pathway involves at least one CE phase, accounting for $\geq 90\%$ – 100% of systems, with only a small minority ($\leq 10\%$) forming through channels without CE. Compared to the BBH and BHNS populations, the spread in channel fractions is significantly smaller, indicating strong agreement on the central role of CE evolution in BNS formation. The corresponding local intrinsic merger rates

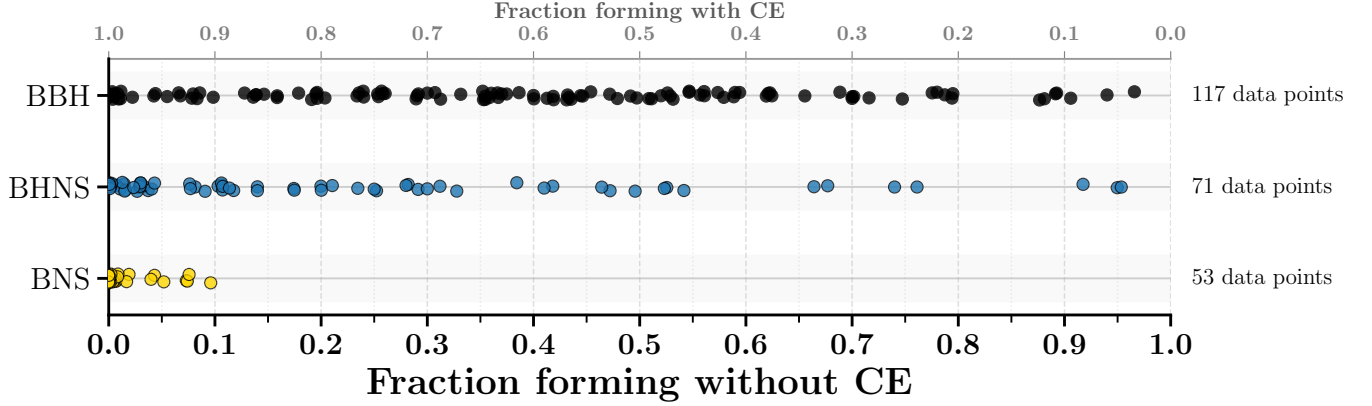


FIG. 2.— Global overview of Level 1 formation-channel diversity across the compiled population-synthesis simulations. Each point shows the fraction of systems forming without a CE phase for a given model, for BBH, BHNS, and BNS mergers. Data points are given a small random vertical offset for visualization clarity. BBH and BHNS populations span nearly the full range from CE-dominated to without-CE-dominated formation, whereas BNS mergers show a strong preference ($\geq 90\%$ of all formed BNS) for channels involving CE evolution. The BHNS population additionally shows tentative clustering toward low without-CE fractions, alongside substantial scatter between models. In Section 3.2, we explore the physical assumptions responsible for these differences, demonstrating that the diversity in outcomes cannot generally be reduced to a single parameter variation, population synthesis code, or stellar-evolution track framework. The top axis indicates the corresponding CE fraction (defined as $1 - \text{fraction without CE}$). In a small number of cases (5 data points), this relation is approximate due to minor contributions from an “other” channel at Level 1; these contributions are negligible and do not affect the overall trends.

span $\sim [5, 1000] \text{ Gpc}^{-3} \text{ yr}^{-1}$. We return to this comparatively robust prediction in Section 4.1, including a discussion of the rare without-CE pathways.

3.1.2. Level 2: Detailed Subchannel Decomposition

Figures 6–8 present the Level 2 decomposition of formation pathways when available, revealing additional complexity beyond the simple with CE versus without-CE distinction.

Inside the without-CE branch, the classic only-stable-mass-transfer channel (SMT + SMT; stable mass transfer before SN1 and stable mass transfer after SN1) frequently contributes significantly, although its importance varies widely across simulations. Systems undergoing stable mass transfer in only one evolutionary phase (SMT + NON or NON + SMT) and systems without any mass transfer (NON + NON) also contribute non-negligibly in several models, while CHE appears as a minor but distinct channel where reported (Li et al. 2025a).

Within the CE branch, the classic CE pathway is typically the largest contributor, but significant diversity exists in the relative importance of single-core and double-core CE channels. Sequence-based combinations (e.g., CE+SMT or NON+CE) further illustrate the range of evolutionary histories captured across studies.

Overall, the Level 2 analysis demonstrates that the diversity observed at Level 1 is not driven by a clear single dominant subchannel, but instead reflects a broad distribution of evolutionary sequences across parameter space. This indicates that the large spread in Level 1 channel contributions is not driven by a single dominant alternative pathway, but rather by a redistribution across multiple evolutionary sequences, reflecting the high-dimensional nature of binary evolution parameter space and the complex pathways to form gravitational-wave sources.

3.2. Simulation Variations and Parameter Dependencies

The absence of clear clustering in the CE formation-channel fractions by code, stellar tracks, or individual parameter choices in Figures 3–5 already suggests that the origin of the variation is complex. Much of the diversity appears to arise either from differences between population-synthesis frameworks or from a small subset of influential parameters within

a given code. Within individual studies, only a few parameter variations typically drive large changes in the relative importance of with-CE and without-CE pathways, although the identity of these key parameters differs between studies. Moreover, models adopting nominally similar parameter values (e.g., identical α_{CE} or mass-transfer efficiency β) can still produce markedly different CE fractions across different codes. This demonstrates that the effect of a single parameter depends on its coupling to other physical assumptions, such as stellar structure, angular-momentum loss, and mass-transfer stability. As a result, similar parameter choices do not lead to unique formation-channel outcomes, complicating attempts to isolate the impact of individual assumptions across population-synthesis frameworks.

These considerations motivate the more detailed within-code comparison of parameter variations presented below. Because all studies report results at the Level 1 classification, we focus on the relative contributions of systems evolving with and without a CE phase, enabling a consistent comparison across models. We primarily focus on BBH and BHNS systems, which exhibit a broad range of CE contributions, while BNS formation remains consistently dominated by channels involving CE evolution (see Section 4).

Figures 9 (BBH) and 10 (BHNS) illustrate the Level 1 channel contributions across the compiled simulations and parameter variations by connecting, within each study, models that differ by only a single parameter. This one-at-a-time variation strategy is standard practice in population-synthesis studies and is widely used to probe the sensitivity of predictions to uncertain aspects of massive binary evolution (e.g., Belczynski et al. 2010b; Chruślińska et al. 2018; Broekgaarden et al. 2022c). More recent work, however, has begun to explore simultaneous variations in multiple parameters (e.g., Dorozzmai & Toonen 2024; Boesky et al. 2024a,b; Stevenson et al. 2022).

A perhaps counterintuitive result across these models is that a large increase in the relative contribution of the without-CE formation channel does not necessarily correspond to a lower total merger rate (or lower CE merger rate, see Figure 16). Intuitively, one might expect this to happen if in the simulations with higher without CE contributions, with-CE binaries in-

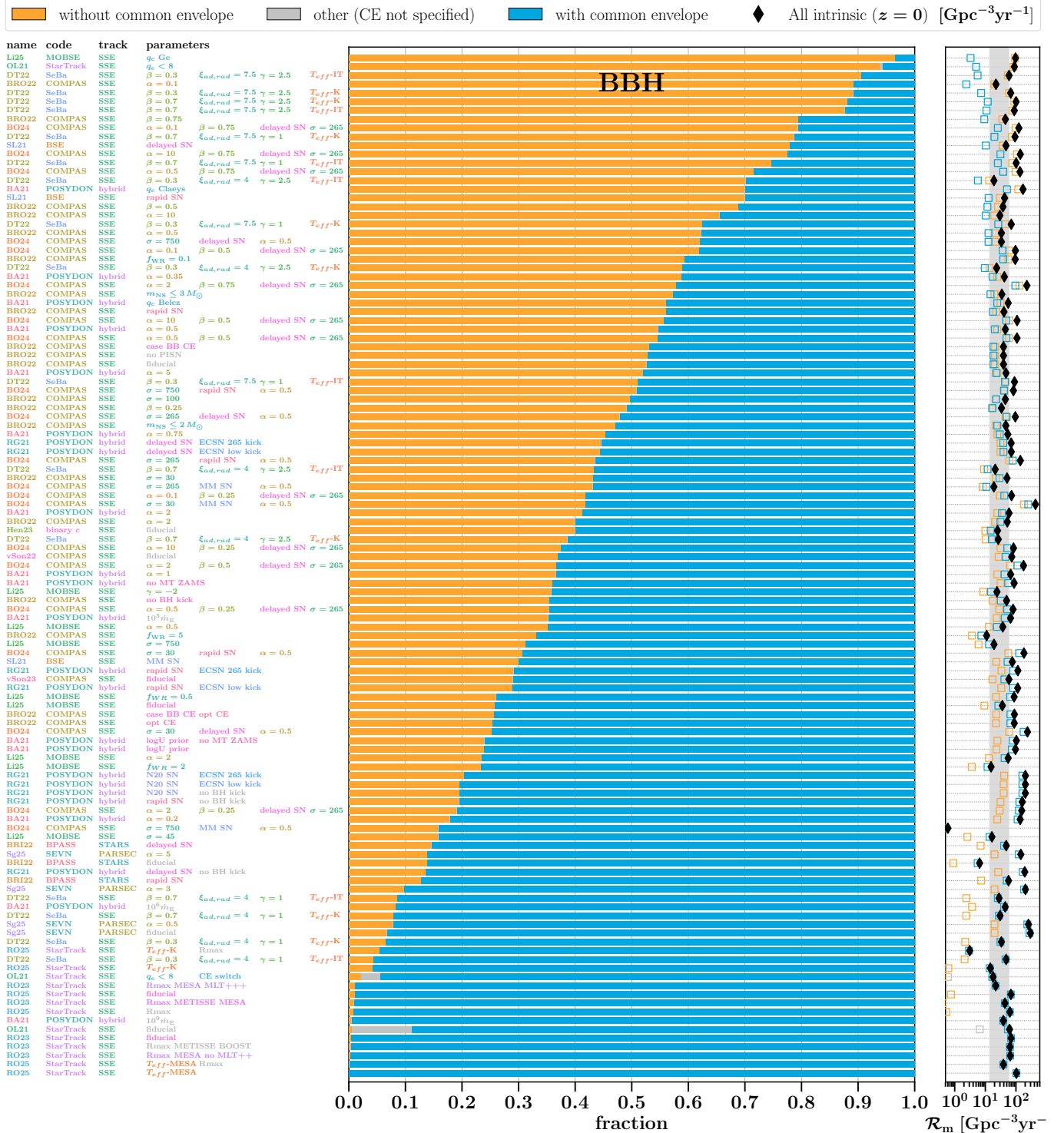


FIG. 3.— **Left panel:** Fractional contributions of Level 1 formation pathways (with and without CE) to the astrophysical BBH population across the compiled simulations. See Figure 1 for formation channels taxonomy. **Right panel:** corresponding intrinsic local merger rates (R_m), together with the 90% credible interval for the local BBH merger rate inferred from GWTC-5 (The LIGO Scientific Collaboration et al. 2026b, gray shaded region). Labels in the far left encode the study, population synthesis code, stellar-evolution tracks, and key model parameters varied relative to the fiducial model (when available), including the CE efficiency (α), mass-transfer efficiency (β), supernova kick dispersion (σ), and critical mass ratio for mass transfer stability (q_c). Tracks ‘hybrid’ indicate a hybrid combination of MESA and COSMIC in an early version of POSYDON. Details of the data compilation are provided in Appendix A, while a complete list of acronyms and parameter definitions is given in Appendix C. A online interactive table displaying all the values for each simulation and additional parameters can be found online and on Broekgaarden (2023a).

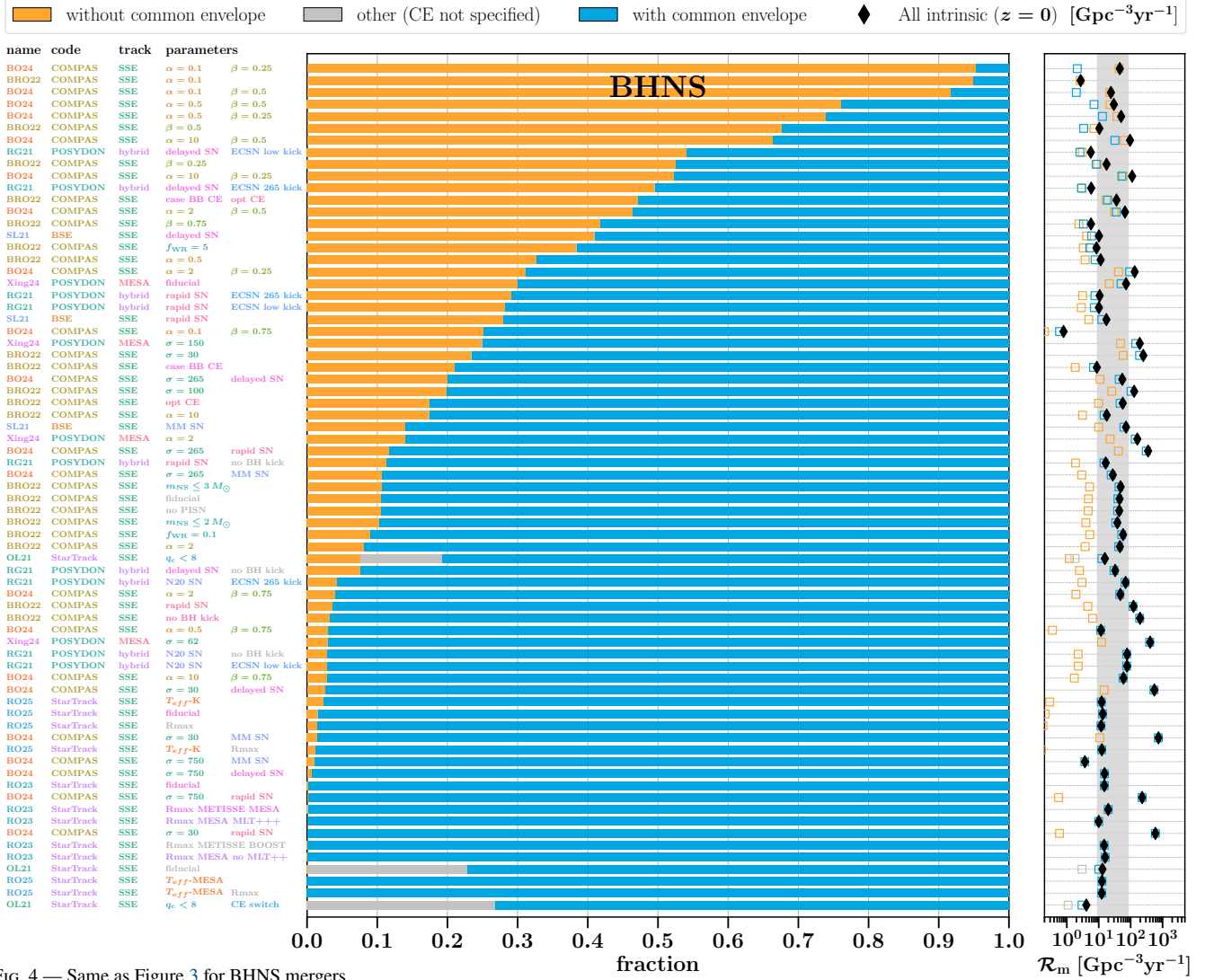


Fig. 4.— Same as Figure 3 for BHNS mergers.

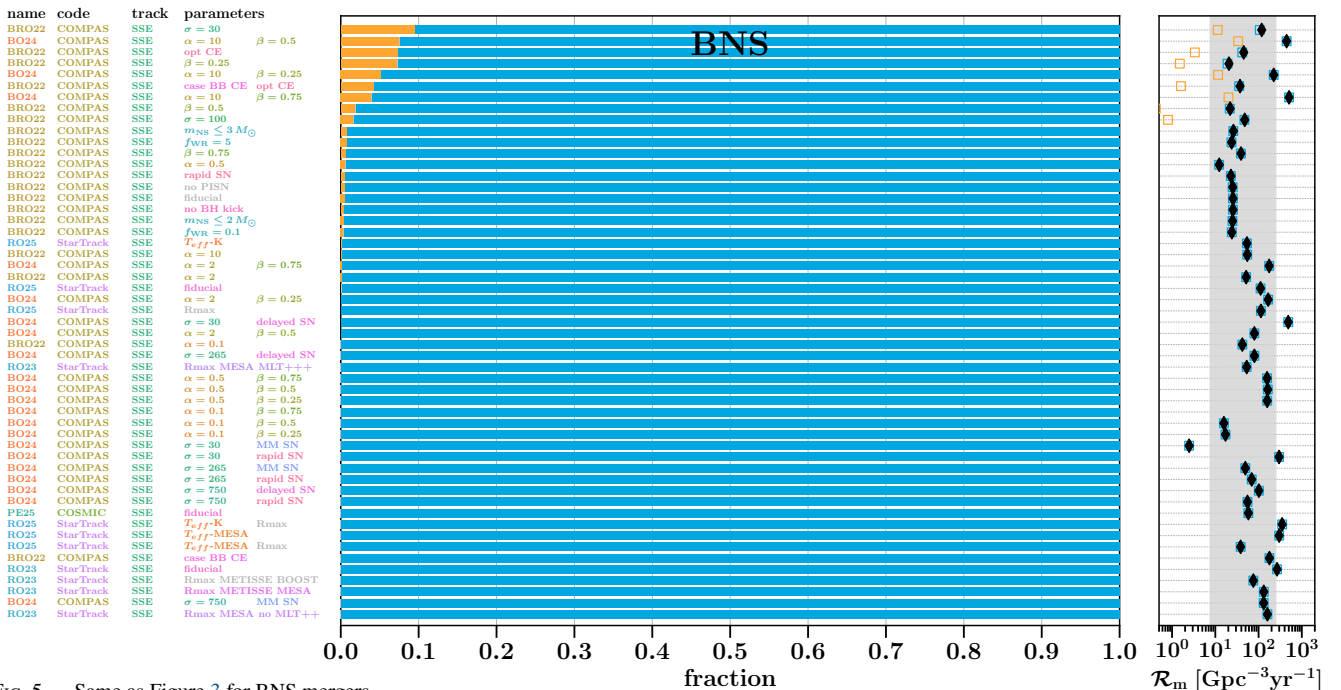


Fig. 5.— Same as Figure 3 for BNS mergers.

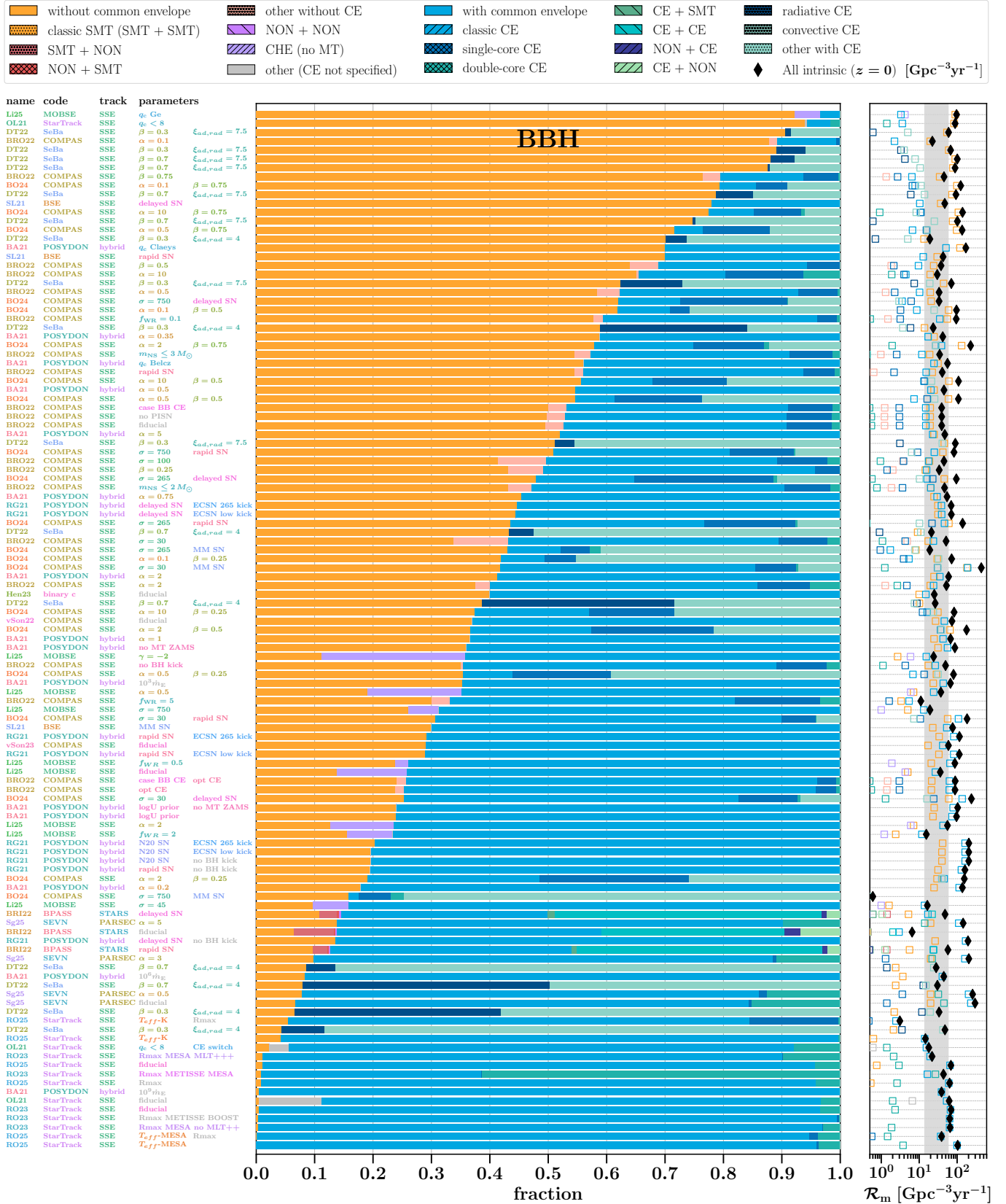


Fig. 6.— Detailed Level 2 decomposition of BBH formation channels across the compiled population-synthesis simulations. Each horizontal bar represents a simulation, showing the fractional contribution of individual evolutionary subchannels within the isolated binary evolution paradigm. Subchannels are grouped into systems that evolve without a common-envelope (CE) phase (e.g., sequences involving only stable mass transfer or no interaction) and systems that experience at least one CE episode. The without-CE branch includes channels such as classic stable mass transfer (SMT+SMT), systems with stable mass transfer in only one evolutionary phase (SMT+NON or NON+SMT), and systems that avoid binary interactions altogether (NON+NON). The CE branch includes the classic CE pathway, as well as variations such as single-core and double-core CE events and sequence-based combinations of CE and stable mass transfer. The right panel shows the intrinsic BBH merger rate predicted by each simulation. The figure illustrates that the diversity observed at the Level 1 classification arises from a wide range of underlying evolutionary sequences rather than from a single dominant subchannel. Different population-synthesis models populate different combinations of these pathways, highlighting the complex interplay between binary interactions, stellar evolution, and supernova physics in shaping the BBH population.

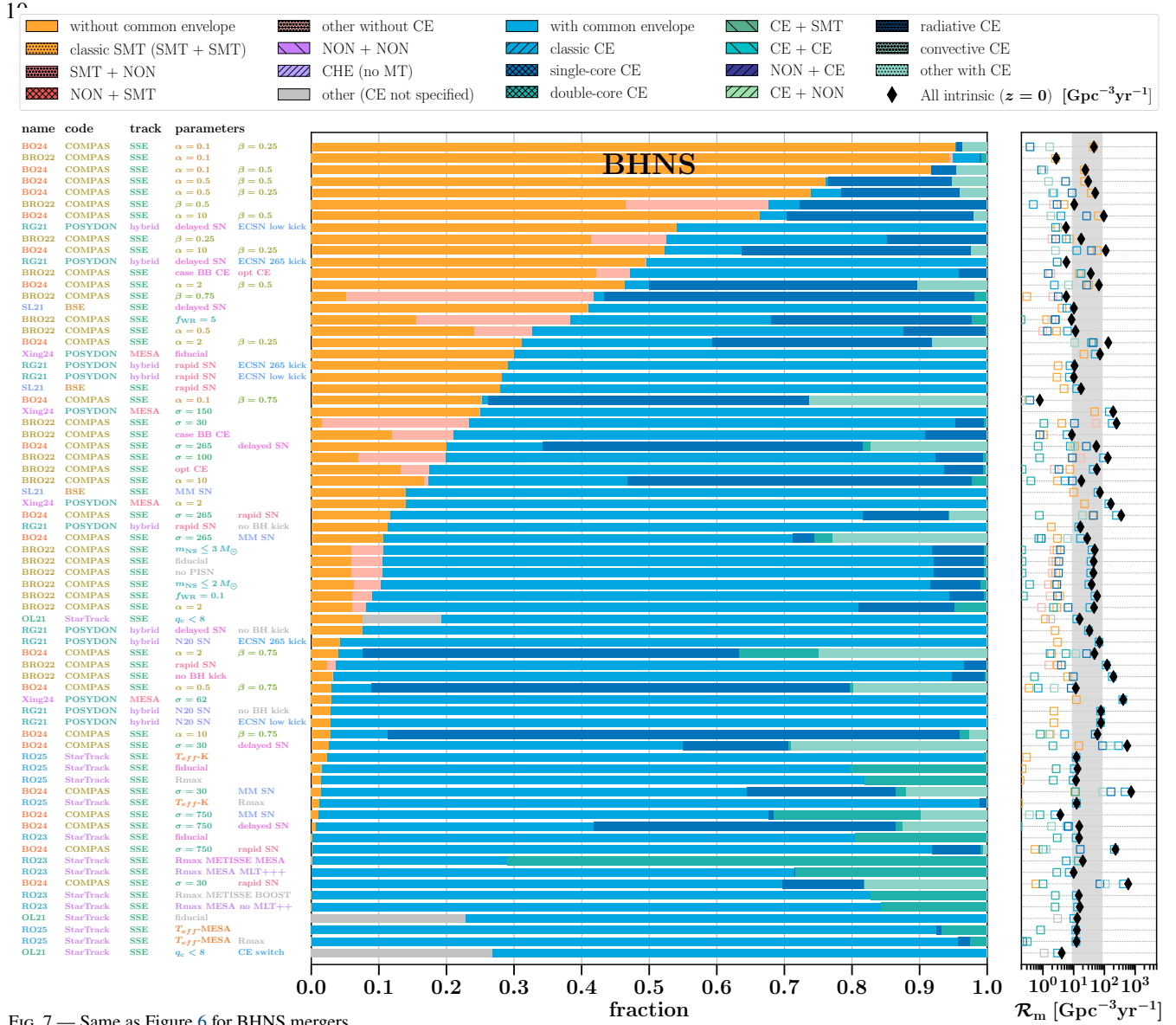


FIG. 7.— Same as Figure 6 for BHNS mergers.

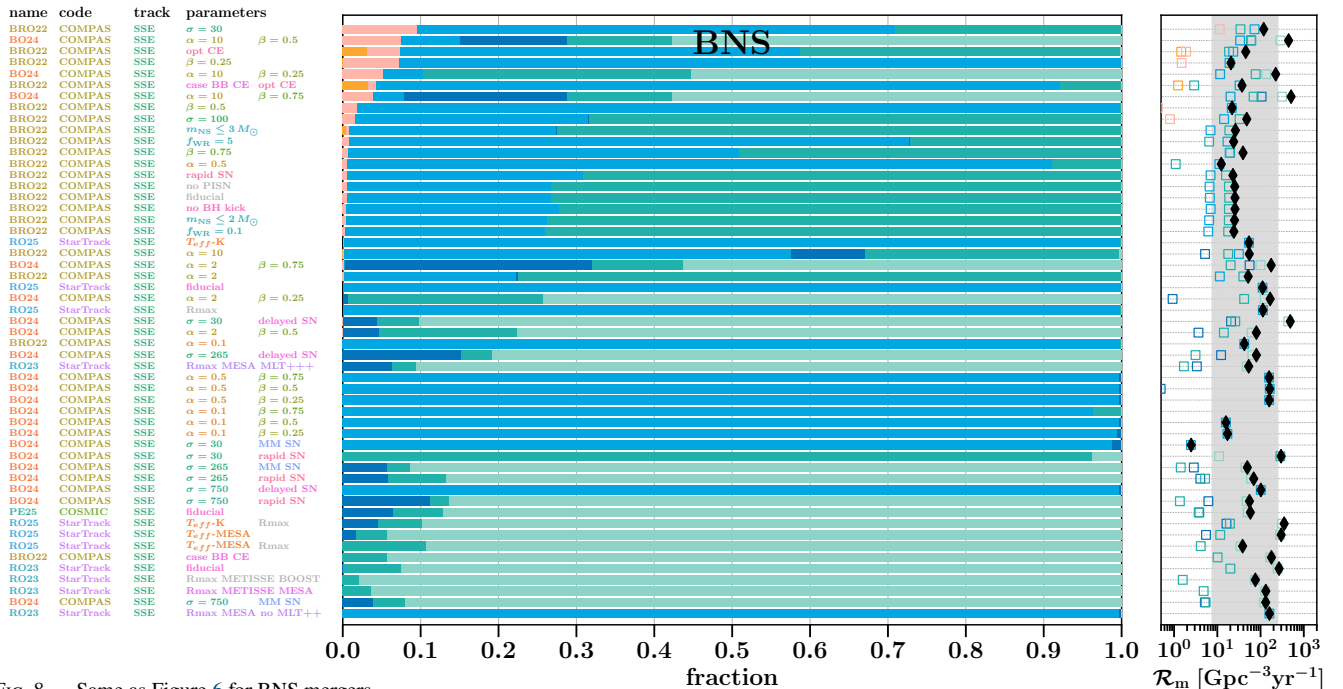


FIG. 8.— Same as Figure 6 for BNS mergers

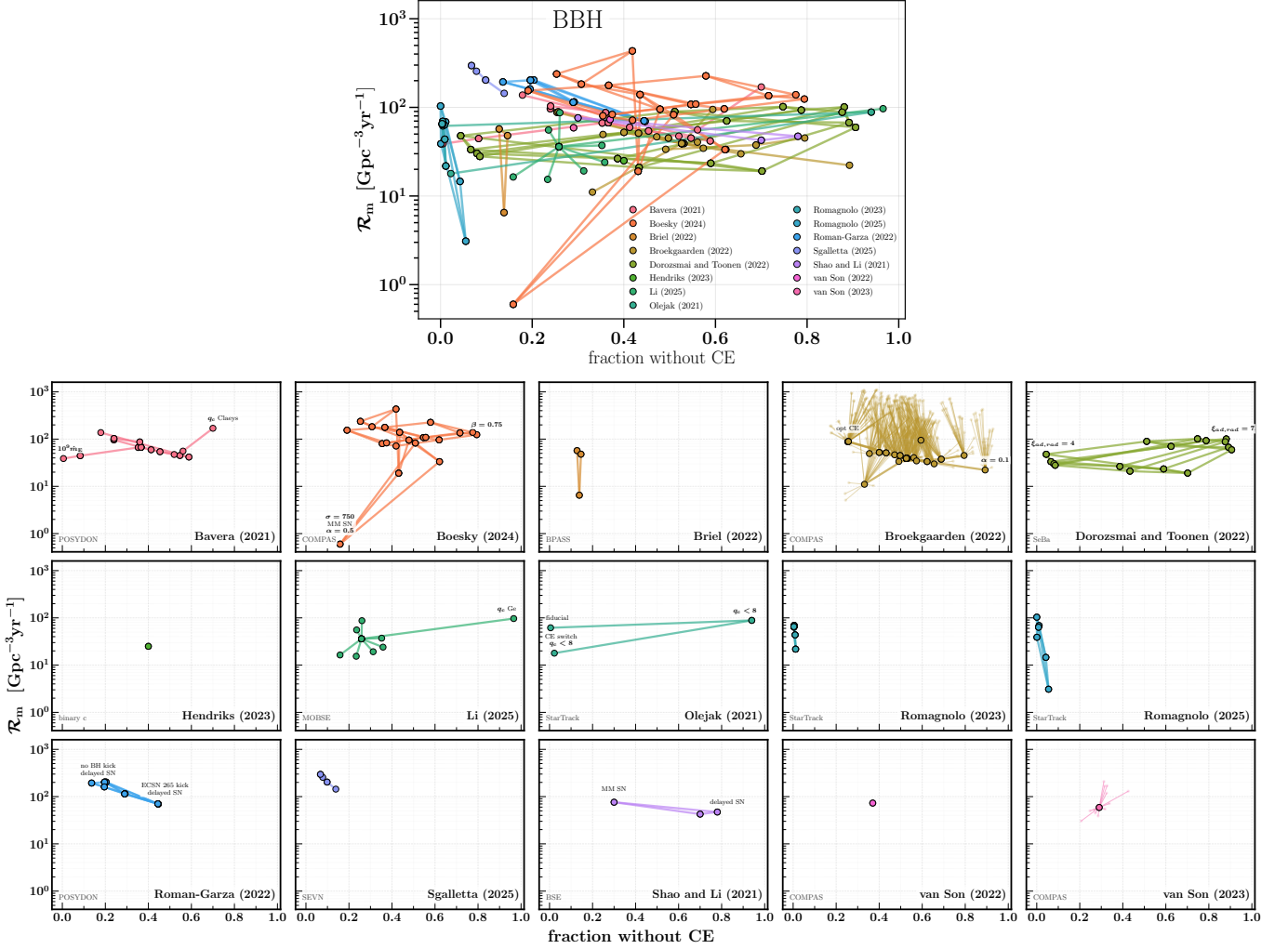


FIG. 9.— **Top:** Fraction of BBH mergers forming without a CE phase as a function of the total BBH intrinsic merger rate for the compiled population-synthesis simulations. Each point represents a single model, colored by study or stellar-evolution code. Lines connect models within the same study that vary a single parameter, illustrating the impact of controlled parameter changes. **Bottom:** Same data grouped by study in individual subpanels. Each panel highlights how variations of specific binary-evolution parameters within a given population-synthesis framework affect the predicted fraction of BBH mergers forming without CE evolution. For Broekgaarden et al. (2022c); van Son et al. (2023) additional lines are added for variations in the star formation history models. An interactive version of this figure can be accessed [online](#). The absolute with and without CE merger rates are shown in Appendix F. Some of the most extreme outlier models, corresponding to the lowest and/or highest without-CE fractions, are annotated with their dominant model assumptions where appropriate. See the labels in Figure 3 for the full parameter variations list.

stead evolve through without-CE pathways, or because avoiding CE often leads to wider post-interaction binaries and thus longer delay times, reducing the number of systems that merge within a Hubble time (see Section 4.2). As shown in Figures 9 and 10, several simulations exhibit simultaneous increases in both the without-CE fraction and the total merger rate (see also Appendix F for more detail). In particular for BBHs, some parameter variations produce a transition from predominantly with-CE to predominantly without-CE formation while maintaining comparable, or even higher, merger rates, indicating that the without-CE contribution itself can increase substantially rather than merely replacing suppressed CE systems. Overall, the lack of a clear monotonic relationship between formation-channel fractions and total merger rates highlights the difficulty of constraining formation pathways using merger rates alone. For BHNSs, some models do show lower merger rates at larger without-CE fractions, although numerous exceptions remain.

3.2.1. Key parameters driving formation-channel diversity

A striking feature of Figures 9 and 10 is that some simulations tend to cluster within relatively narrow regions of parameter space when considered within a single code framework, while the largest differences emerge between different population-synthesis codes or a limited subset of specific parameter variations (e.g. Li et al. 2025a; Romagnolo et al. 2023). This behavior highlights what we refer to as population synthesis ‘simulation silos’: variations within a given code, particularly single-parameter perturbations around a fiducial model, might probe only a limited region of the broader uncertainty space. In contrast, changes to the underlying physical framework (such as adopting a different population-synthesis code) or modifying a small number of particularly influential assumptions can lead to qualitatively different outcomes.

Across the BBH and BHNS simulations, the diversity in predicted formation-channel contributions can be described primarily in the context of several key physical assumptions that regulate (i) the stability of mass transfer and (ii) the efficiency of orbital tightening. This is highlighted in Figure 11 that quantifies the parameter impacts on the formation channel

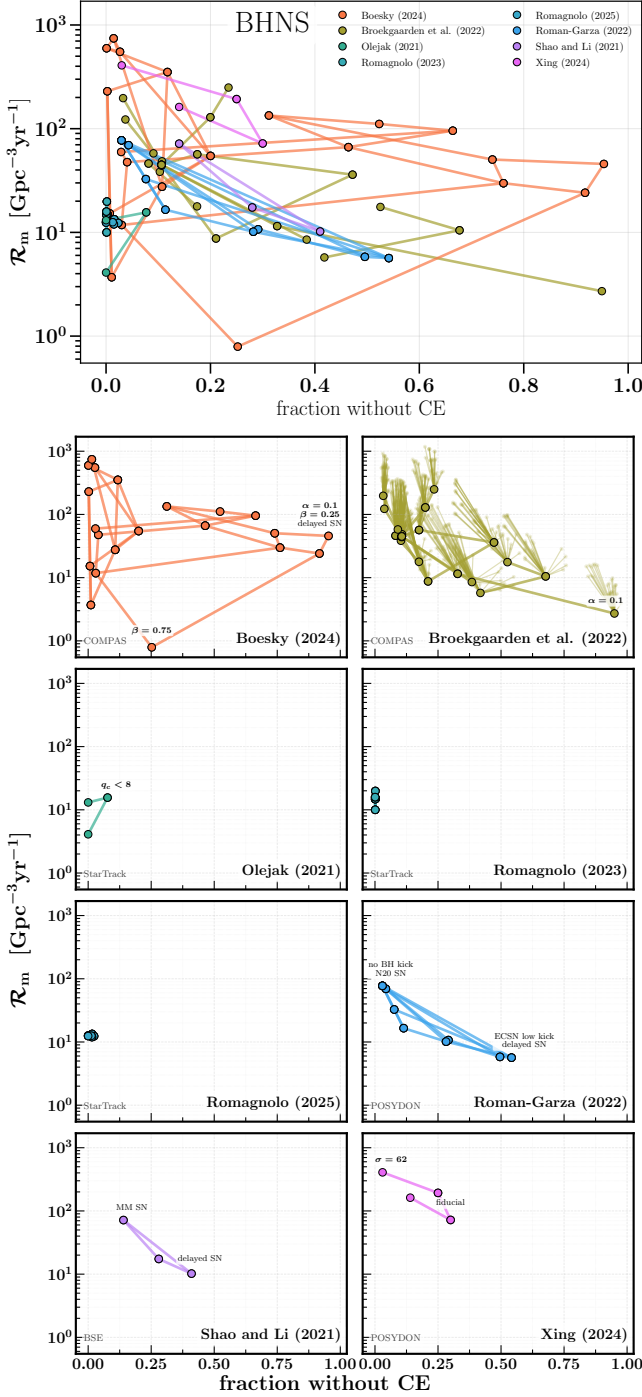


FIG. 10.— Same as Figure 9 for BHNS. For Broekgaarden et al. (2022c) additional lines are added for variations in the star formation history models. An interactive version of this figure can be accessed [online](#).

contributions across BBH and BHNS for parameters that affect it by at least an absolute change in the without-CE formation channel contribution of $\Delta\text{fraction} \gtrsim 0.1$ within a study, and Figures 12 (BBH) and 13 (BHNS) that quantify the CE contribution as a function of the parameter for a range of population synthesis parameters. Below we summarize the dominant parameters and their impact on BBH and BHNS populations across the simulations, noting that some aspects of this discussion are necessarily limited by the set of models included in our comparison.

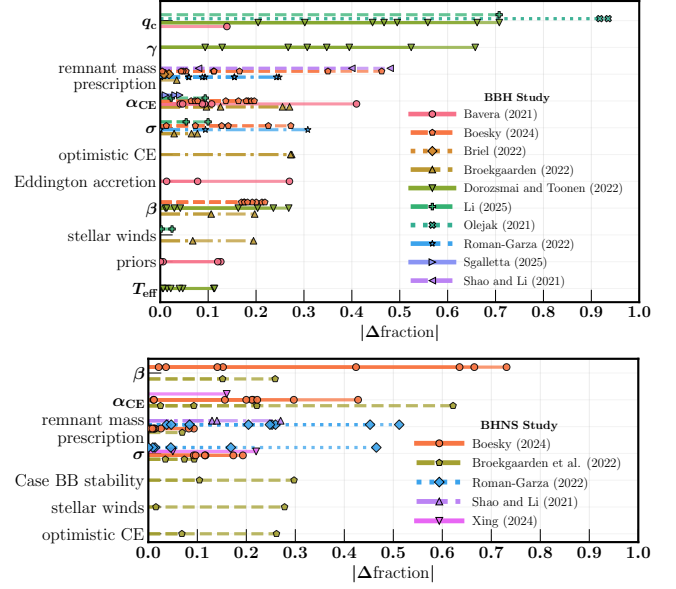


FIG. 11.— Change in the fraction of systems forming without a CE phase as a function of parameter-family variations for BBH (top) and BHNS (bottom) populations. Each scatter point shows the relative change in the without-CE fraction within a given study when varying a single model parameter, grouped by parameter family. Scatter points correspond to individual simulation variations within each family, while horizontal lines indicate the range of changes. Small vertical offsets (jitter) are applied to overlapping lines within the same parameter family for visual clarity and do not carry physical meaning. The figure only shows parameter variations that lead to changes of $|\Delta\text{fraction}| \geq 0.1$.

- **Mass-transfer stability criteria regulate whether systems avoid CE, with models adopting higher q_c or $\xi_{\text{ad,rad}}$ values leading to significantly higher without-CE contributions without necessarily lowering the total merger rate.** The assumed critical mass ratio for stability (q_{crit}), or equivalently the donor response parameter $\xi_{\text{ad,rad}}$ ¹, sets whether Roche-lobe overflow proceeds stably or leads to CE evolution². Models that adopt higher values for q_c or $\xi_{\text{ad,rad}}$ significantly expand the region of parameter space in which binaries avoid CE evolution.

For BBHs, this leads to some of the largest (outlier) variations in the without-CE fraction in Figure 9, with models adopting high stability thresholds (corresponding to $q_c \gtrsim 3-8$) finding that the majority of systems evolve through stable mass transfer in the simulations from Olejak et al. (2021); Dorozsmai & Toonen (2024); Li et al. (2025a), and overall, stability criteria impact the without CE BBH fraction by up to unity ($\sim 100\%$; Figure 11). Somewhat counterintuitively, such changes in the stability criterion do not necessarily reduce the total BBH rate, but often yield comparable, or even higher, BBH merger rates. This likely reflects that increased mass-transfer stability shifts parts of the parameter space that would previously have merged during a CE episode into systems that can instead survive and form merging BBHs through stable mass transfer (cf. Gallegos-Garcia et al. 2021). For BHNS systems, Ole-

¹ $\xi_{\text{ad,rad}} = (d \log R_d / d \log M)_{\text{ad}}$ is the mass radius exponent of giants with radiative envelopes (with R_d the donor radius, Soberman et al. 1997).

² There are notable exceptions to the q_{crit} formalism which explicitly model the donor's response to mass loss with detailed stellar models in population synthesis codes such as BPASS (Eldridge et al. 2017) and POSYDON (Fragos et al. 2023; Andrews et al. 2025)

jak et al. (2021) has explored this, finding a modest rise in the fraction forming without CE. Broekgaarden et al. (2022c) show that increasing the instability of case BB mass transfer can reduce the with-CE BHNS fraction by inducing mergers during this phase.

Another way to modify the effective stability boundary is to vary the assumptions for when stars develop convective envelopes, which directly affects the adopted critical mass ratio. For example, Dorozsmai & Toonen (2024); Romagnolo et al. (2025) vary the effective temperature (T_{eff}) as a proxy for the evolutionary stage at which a convective envelope forms. However, this variation has only a modest impact, changing the total BBH fraction forming without CE by at most $\sim 10\%$ (Figure 11).

- **Angular-momentum loss (γ) is a key parameter regulating the dominant BBH formation channel.** The efficiency with which mass loss removes angular momentum from the binary, often parameterized through γ -like prescriptions, has a first-order impact on the relative contribution of with- and without-CE BBH formation channels Marchant et al. (e.g. 2021b); Willcox et al. (e.g. 2023); Dorozsmai & Toonen (e.g. 2024). As shown in Figure 11, the γ variations of Dorozsmai & Toonen (2024) produce some of the largest shifts among all parameter studies: increasing γ from 1 to 2.5 substantially increases the without-CE contribution and, in some models, changes the population from predominantly CE-driven to without-CE dominated (Figure 12). In these models, γ represents the specific angular momentum carried away by mass lost during non-compact-object mass transfer, normalized to the binary’s specific angular momentum (cf. Portegies Zwart & Verbunt 1996). Physically, stronger angular-momentum loss leads to more efficient orbital shrinkage during stable mass transfer, allowing binaries to merge within a Hubble time without a CE phase, whereas weaker angular-momentum loss causes orbital widening and suppresses merging BBHs through the without-CE channel (Dorozsmai & Toonen 2024). This connects directly to the ongoing debate over whether stable mass transfer can simultaneously remain stable and shrink the orbit sufficiently to produce merging BBHs (Klencki et al. 2026; van Son et al. 2022).

- **Common-envelope efficiency (α_{CE}) values in the range ~ 0.5 – 2 typically produce the lowest without-CE contribution for BBH and BHNS mergers.** The CE efficiency parameter α determines how efficiently orbital energy is used to eject the envelope during CE evolution and therefore whether binaries survive the CE phase and emerge in sufficiently tight orbits to merge within a Hubble time. Figure 11 shows that, within the simulations included in our compilation, variations in α can change the without-CE fraction by up to ~ 0.4 for BBHs and ~ 0.6 for BHNSs, producing some of the strongest outliers in Figures 9 and 10.

Across most studies, both low and high α values reduce the with-CE BBH and BHNS rates and thereby increase the relative importance of without-CE pathways, with enhanced without-CE fractions consistently found outside the canonical range of ~ 0.5 – 2 (Figures 12 and 13; based on Bavera et al. 2021; Broekgaarden et al. 2022c;

Boesky et al. 2024a; Sgalletta et al. 2024; Li et al. 2025a; Xing et al. 2024a). This “sweet spot” arises because very low α values lead to more binaries merging during the CE phase as more orbital energy is required, while very high α values produce post-CE separations that are too wide for the systems to merge within a Hubble time (e.g. Bavera et al. 2021; Broekgaarden et al. 2022c; Kruckow et al. 2018). In other words, inefficient CE ejection (small α) requires binaries to deposit more orbital energy into the envelope, increasing the likelihood of stellar mergers during CE, whereas highly efficient CE ejection (large α) leaves binaries at wider separations and reduces the merger rate because fewer systems inspiral within a Hubble time.

A small exception to this trend is the $\alpha = 0.2$ model of Bavera et al. (2021), which exhibits an additional dip in the without-CE fraction due to an enhanced with-CE BBH merger rate at this specific CE efficiency. The authors find that, for this model, the post-CE orbital separations of BH–star systems peak in a regime that efficiently produces BBHs with short merger times, while still avoiding mergers during the CE phase. This corresponds to a second “sweet spot,” distinct from the broader $\alpha \sim 0.5$ – 2 regime, which instead reflects the tails of the post-CE separation distribution (see their Figure 3). Overall, this highlights that the dependence on α is complex and not strictly monotonic: although global trends favor reduced CE contributions at extreme α values, specific combinations of initial conditions and orbital evolution (likely correlated with the physical model assumptions for stellar expansion and mass transfer stability) can produce localized enhancements in the with-CE channel.

Given the uncertainty and ongoing debate surrounding the true underlying α values for massive-star CE evolution (e.g. Iaconi & De Marco 2019b; Ivanova et al. 2020; Hirai & Mandel 2022), it is therefore important for future work to explore broad, potentially correlated, variations in α when investigating its impact on BBH, BHNS, and BNS merger populations. We further note that comparisons between studies are complicated by differences in the adopted envelope-binding-energy parameter λ , which can vary substantially between population-synthesis frameworks depending on whether fixed values or stellar-structure-dependent fitting formulae are used (e.g. Xu & Li 2010a; Marchant et al. 2021b; Iaconi & De Marco 2019b; Ivanova et al. 2020; Sgalletta et al. 2026). Since binary evolution modeling such as the mapping between pre- and post-CE orbital separations depend primarily on the combination $\alpha\lambda$, variations in λ can significantly muddy direct comparisons between nominal α assumptions across different studies and codes. In this context, it is perhaps notable that many studies nevertheless consistently find a minimum in the without-CE contribution specifically around the same range of $\alpha \sim 0.5$ – 2 , despite substantial differences in the underlying CE implementations across population-synthesis frameworks.

- **Mass-transfer efficiency (β) can strongly alter the relative importance of with- and without-CE formation channels, particularly for BHNS mergers.** The mass-transfer efficiency parameter β , which describes the

fraction of transferred mass accreted during mass transfer between two non-compact stars, affects both the stability of Roche-lobe overflow and the orbital evolution of the binary. In the population-synthesis simulations considered here, the mass-transfer efficiency β is typically either modeled as a function of the thermal timescale of the accretor star (e.g. Team COMPAS: Riley et al. 2022), or assumed to take a fixed value throughout the evolution (e.g. Belczynski et al. 2008). In Figure 12 and 11 we only show the model variations where β is varied between fixed values. The impact of changing β is intrinsically complex because increasing β can influence binary evolution in several competing ways (Dorozsmai & Toonen 2024; Willcox et al. 2023).

On the one hand, more efficient accretion increases the mass of the companion accretor star, which can help binaries remain bound following the first supernova (e.g. Renzo et al. 2019; Renzo & Götzberg 2021; Broekgaarden et al. 2022c). Efficient accretion during the first mass-transfer phase can also generate highly unequal BH–star binaries (where the star can now be much more massive than the BH because it accreted efficiently earlier). Such systems might be particularly effective at shrinking their orbits during subsequent reverse stable mass transfer, thereby enhancing the formation of merging BBH and BHNS systems through stable pathways (e.g. Olejak et al. 2024; Klencki et al. 2026; Schürmann et al. 2025a).

On the other hand, more conservative mass transfer can also destabilize Roche-lobe overflow. Increased β causes the accretor to respond more strongly to the incoming mass and can lead to stronger orbital contraction during the transfer phase, effectively lowering the critical mass ratio for stable mass transfer and making runaway mass transfer more likely for a wider range of donor-accretor mass ratios (e.g. Dorozsmai & Toonen 2024; van Son et al. 2022). Moreover, higher β reduces the amount of mass and angular momentum lost from the system, which can limit orbital shrinkage in some evolutionary phases and prevent binaries from reaching sufficiently tight separations to merge within a Hubble time (e.g. Willcox et al. 2023; van Son et al. 2022). The relative importance of these effects depends sensitively on the evolutionary stage, donor structure, and the modeling assumptions for mass transfer and angular momentum loss.

As a result, β produces varying behavior in the without-CE contribution across BBH and BHNS populations. For BBHs, COMPAS based models from Broekgaarden et al. (2022c) and Boesky et al. (2024a) seem to find that higher β increases the without-CE contribution, while the SeBa based models from Dorozsmai & Toonen (2024) higher β leads to a similar or suppressed without-CE contribution (Figure 12). These results highlight that the impact of β cannot be understood purely through a single local stability criterion, but instead depends on how mass accretion reshapes the full multi-stage binary evolution.

For BHNS systems, the models in our compilation that vary β (Boesky et al. 2024a; Broekgaarden et al. 2022c) show a trend in which higher β decreases the without-CE contribution (Figure 13), changing the without-CE

BHNS fraction by up to ~ 0.75 (Figure 11). In these models, the trend is driven by a reduction in the absolute without-CE formation rate combined with a modest increase in the with-CE rate. Interestingly, these Boesky et al. (2024a); Broekgaarden et al. (2022c) models where the without-CE fraction decreases with increasing β are the models for which the BBH without-CE fraction increases with increasing β . This contrast highlights that identical parameter variations can shift the dominant formation pathway in opposite directions for BBH and BHNS mergers, reflecting differences in their typical mass ratios, compact-object formation histories, and subsequent binary-interaction phases.

• *The magnitude of natal kicks imparted to compact objects, parameterized by σ , plays an important role in setting the relative contribution of with- and without-CE formation channels for BBH and BHNS mergers.*

We find that the assumed distribution of compact-object natal kick magnitudes, parameterized with σ , can alter the without-CE contribution by up to ~ 0.3 for BBHs and ~ 0.5 for BHNSs (Figure 11). Natal kicks affect both binary survival and the post-supernova orbital evolution of the system. Higher kick magnitudes (larger σ values) increase the probability that binaries are disrupted during compact-object formation and therefore reduce the number of systems that remain bound. At the same time, natal kicks can significantly alter the post-supernova orbital properties by changing the orbital separation and inducing eccentricity, which can either promote or suppress mergers within a Hubble time depending on the kick magnitude, direction, and pre-supernova binary configuration.

For BBH mergers, models with low or zero BH kicks generally produce higher absolute merger rates for both with- and without-CE channels because fewer binaries are disrupted during compact-object formation. Since the natal-kick prescriptions in population synthesis models primarily affect low-mass BHs, this trend also reflects that the BBH merger-rate distribution peaks at relatively low BBH masses. Moderate kicks can preferentially enhance the without-CE contribution by tightening binaries that would otherwise remain too wide to merge within a Hubble time. As a result, several studies find somewhat larger without-CE BBH fractions for moderate or high kick dispersions ($\sigma \gtrsim 100$, km s⁻¹), with increases of ~ 0.2 – 0.4 in the relative without-CE contribution (Figure 12). This behavior reflects that many of the without-CE channels often lack the strong orbital tightening associated with a CE phase, making natal kicks particularly important for shrinking the orbit sufficiently to produce merging BBHs (e.g., Klencki et al. 2026; Chattaraj et al. 2026).

A related trend is seen for BHNS systems in Figure 13, where moderate natal kicks ($\sigma \sim 100$ – 300 , km s⁻¹) can enhance the without-CE contribution by increasing the absolute without-CE BHNS merger rate in studies such as Román-Garza et al. (2021a); Xing et al. (2025) and Boesky et al. (2024a). However, other simulations instead find little effect or even a decrease in the without-CE contribution for similar kick assumptions (e.g. Broekgaarden et al. 2022c; Román-Garza et al. 2021b). These differences highlight the complex inter-

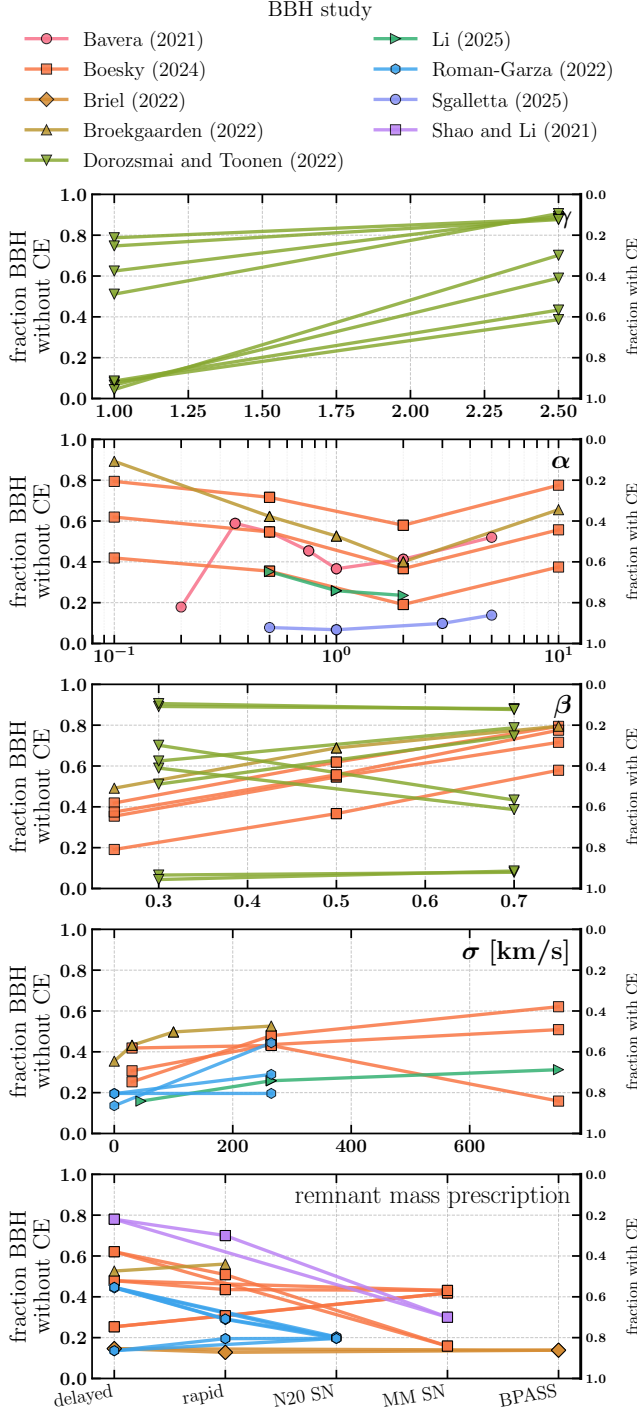


FIG. 12.— Dependence of the fraction of BBH mergers forming without a CE phase on commonly varied binary-evolution parameters. Each panel shows simulations in which a single parameter is varied while other assumptions are kept fixed within the same population-synthesis study. Shown parameters include the angular momentum loss parameter γ , CE efficiency parameter α_{CE} , the mass-transfer efficiency β , supernova natal kick magnitude parameter σ , and the adopted supernova remnant prescription. Each point represents a simulation, with lines connecting models belonging to the same simulation grid to highlight controlled (single) parameter variations. For the remnant-mass prescription panel, the x-axis represents discrete model assumptions rather than a continuous parameter range. Consequently, individual parameter variations may connect non-adjacent x-axis positions, leading to triangular connection patterns within a given study. Multiple lines for a given study indicate simultaneous variation of additional parameters (i.e., multi-dimensional grids).

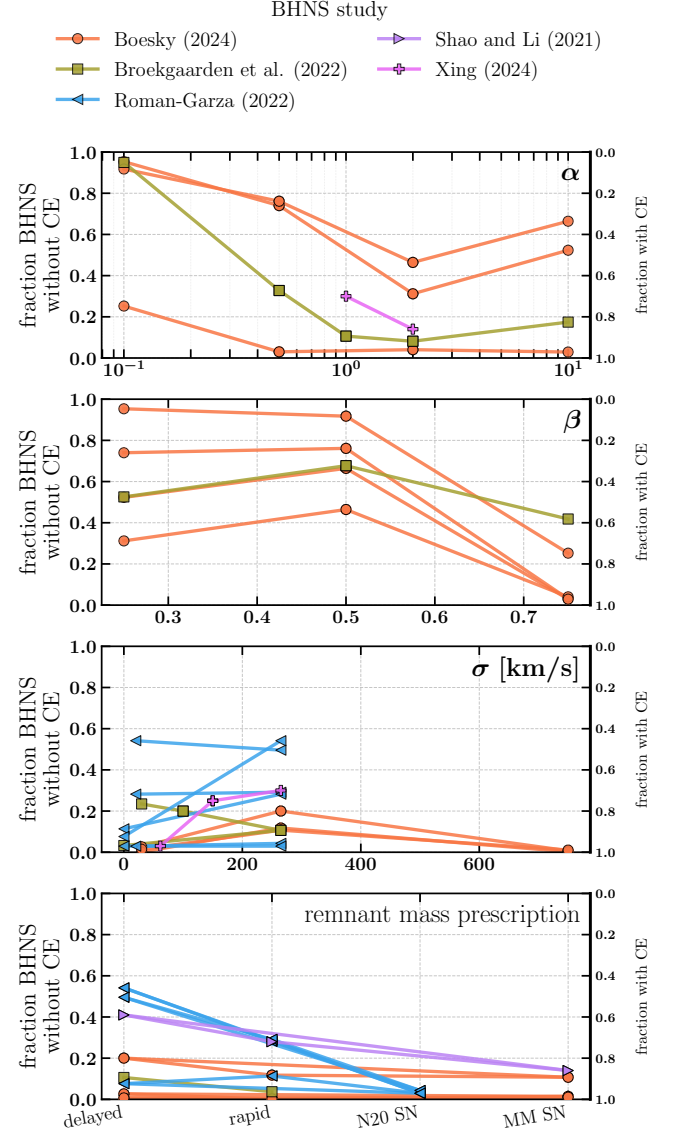


FIG. 13.— Same as Figure 12 for BHNS mergers. The panel with γ is not shown as no BHNS simulations include variations in γ . For the σ panel, the quoted σ values are not always defined identically across studies. In some models the natal-kick dispersion is applied to both NSs and BHs, while in others it is applied only to NS kicks, only a subset of NS kicks (e.g. only electron-capture supernovae) or only to BH kicks.

play between natal kicks and other binary-evolution assumptions, including mass transfer, orbital separations, and compact-object formation physics, in determining the dominant BBH and BHNS formation pathways.

- **Remnant-mass prescriptions strongly affect the CE contribution to BBH and BHNS formation.** The adopted compact object remnant mass prescription can play a major role in determining the dominant formation pathway for BBH and BHNS mergers, changing the without-CE contribution by up to ~ 0.5 for both populations (Figure 11).

For BBH mergers, Figure 12 shows that changing the remnant-mass prescription can alter the without-CE contribution by ~ 10 –50% in several studies (Román-Garza et al. 2021b; Broekgaarden et al. 2022c; Boesky et al. 2024a; Shao & Li 2021a), with no clear consistent

correlation between remnant mass prescription and CE formation channel contribution. For [Briel et al. \(2023\)](#), [Broekgaarden et al. \(2022c\)](#), and the $\sigma = 265 \text{ km s}^{-1}$ models of [Boesky et al. \(2024a\)](#) find that the remnant-mass prescription changes the without-CE fraction by only $\lesssim 5\%$, highlighting the strong interplay between remnant prescriptions and other model assumptions, particularly natal kicks.

For BHNS mergers, Figure 13 shows that most studies find the largest without-CE contributions for models adopting the delayed remnant-mass prescription of [Fryer et al. \(2012\)](#), with the exception of [Román-Garza et al. \(2021b\)](#), who instead find a slightly larger without-CE contribution for the rapid prescription. While some studies report only weak sensitivity to the remnant-mass prescription ($\lesssim 5\%$ changes in the without-CE fraction; e.g., [Boesky et al. 2024a](#); [Broekgaarden et al. 2022c](#)), others find variations as large as $\sim 50\%$.

- **Variations in the assumed star formation history primarily affect the normalization of the merger rate.** Finally, changes in the adopted star formation history mainly affect the normalization of the merger rate, often by up to an order of magnitude, while leaving the relative formation-channel contributions largely unchanged. This is seen for the models of ([Broekgaarden et al. 2022c](#)) and [van Son et al. \(2023\)](#) in Figures 12 and 13. This suggests that the star formation history primarily reweights the underlying population across cosmic time and metallicity, rather than fundamentally changing the balance between CE and without-CE formation pathways (cf. [Levina et al. 2026](#); [Willcox et al. 2025](#); [de Sá et al. 2024](#)).

All in all, taken together, these results emphasize that the expected balance between CE and without-CE formation pathways remains highly sensitive to the underlying assumptions of stellar and binary evolution models and parameter impacts can be highly correlated. Consequently, caution is warranted when interpreting trends in the relative contribution of formation-channels from a limited set of simulations or from a single population-synthesis code.

4. DISCUSSION

Our compilation reveals a central challenge in interpreting gravitational-wave sources: merger-rate predictions alone do not uniquely constrain the evolutionary pathways of compact-object binaries. Across population-synthesis studies, models that reproduce similar intrinsic merger rates can correspond to fundamentally different formation histories, reflecting a strong degeneracy between rates and formation channels. We find that this degeneracy is primarily driven by systematic differences between population-synthesis frameworks: “*simulation silos*”, rather than parameter variations within individual models. At the same time, the degree of uncertainty varies across compact-object classes, with BBH and BHNS systems showing large diversity in formation pathways, while BNS mergers display strong agreement and are almost universally associated with CE evolution. In the following, we interpret these results in terms of the underlying binary-interaction physics and discuss how future simulation work in combination with future observations can break these degeneracies.

4.1. A Robust Result: BNS Formation Requires CE

In contrast to the large diversity observed for BBH and BHNS systems, the formation of BNS mergers is strikingly consistent across models. All studies find that BNS systems form mainly through evolutionary pathways that include at least one common-envelope (CE) phase, with only a negligible fraction ($\lesssim 10\%$) forming without CE ([Boesky et al. 2024a](#); [Broekgaarden et al. 2022c](#); [Romagnolo et al. 2023](#); [Pellouin et al. 2025](#)). Although our compilation in Figure 5 includes a limited number of studies—primarily because many works do not report intrinsic merger rates—this conclusion is robust. The dominance of CE-mediated formation channels for BNS mergers is consistently reported across binary population-synthesis studies, including those beyond our compilation such as studies that only report formation channel contributions for simulations at a fixed metallicity (see Table D for an overview).

To highlight this, we show in Figure 14 the overview of reported formation channel contributions as a function of formation efficiency across a range of metallicities for BNS (and BHNS and BBHs). Which indeed shows that studies find consistently a dominant contribution of channels with CE across metallicity and for the large range of models (including [Belczynski et al. 2002](#); [Andrews et al. 2015](#); [Chruślińska et al. 2018](#); [Dominik et al. 2012](#); [Vigna-Gómez et al. 2018](#); [Tanaka et al. 2023](#); [Mestichelli et al. 2025](#); [van Son et al. 2024](#); [Iorio et al. 2023b](#); [Broekgaarden et al. 2022c](#); [van Son et al. 2024](#); [Iorio et al. 2023b](#); [Chattaraj et al. 2026](#)).

This result of CE for BNS mergers can be understood from simple binary-evolution considerations. At some stage, BNS progenitors must evolve through a neutron-star–star binary configuration (e.g. [Tauris et al. 2015a](#); [Tauris et al. 2017](#)). Forming a merging system requires a sufficiently tight orbit prior to the second supernova. However, the typical mass ratios in such systems (e.g., $\sim 1.4 M_{\odot}$ neutron star with a $\gtrsim 7 M_{\odot}$ companion) strongly favor dynamically unstable mass transfer, leading to a CE phase (e.g. [Gallegos-Garcia et al. 2023](#); [Chattaraj et al. 2026](#)). Avoiding CE would require a narrow and fine-tuned region of parameter space involving low-mass companions or unusually massive neutron stars, making such pathways impossible or extremely rare ([van Son et al. 2024](#); [Gallegos-Garcia et al. 2023](#); [Chattaraj et al. 2026](#)).

Only a few CE-free pathways have been identified for the formation of merging BNSs. In [Gallegos-Garcia et al. \(2023\)](#), all merging BNSs with a $1.4 M_{\odot}$ NS companion undergo a CE phase, while a small subset of systems with a $2 M_{\odot}$ NS companion can merge after only stable mass transfer, provided the NS receives a sufficiently large natal kick. Because these calculations begin from pre-existing NS–star binaries, rather than from zero-age binaries, the formation efficiency of the full evolutionary pathway remains difficult to assess. Similarly, [van Son et al. \(2024\)](#) find that BNS mergers are formed almost exclusively through the CE channel; only their no-natal-kick model shows a non-zero, but negligible, contribution from the only-stable-mass-transfer channel, and no representative evolutionary sequence is given for these systems. This is consistent with the arguments of [Chattaraj et al. \(2026\)](#), who find that the with-CE channel overwhelmingly dominates BNS formation at solar metallicity ($\gtrsim 95\%$), but also show that $\sim 5\%$ of BNS systems can form without any mass transfer at all (which would fall under without-CE) while the channel with stable mass transfer only is intrinsically inefficient, contributing only a very small fraction ($\lesssim 1\%$) of merging BNS systems. Together, these results further support the conclusion that merging BNS formation is overwhelmingly CE dom-

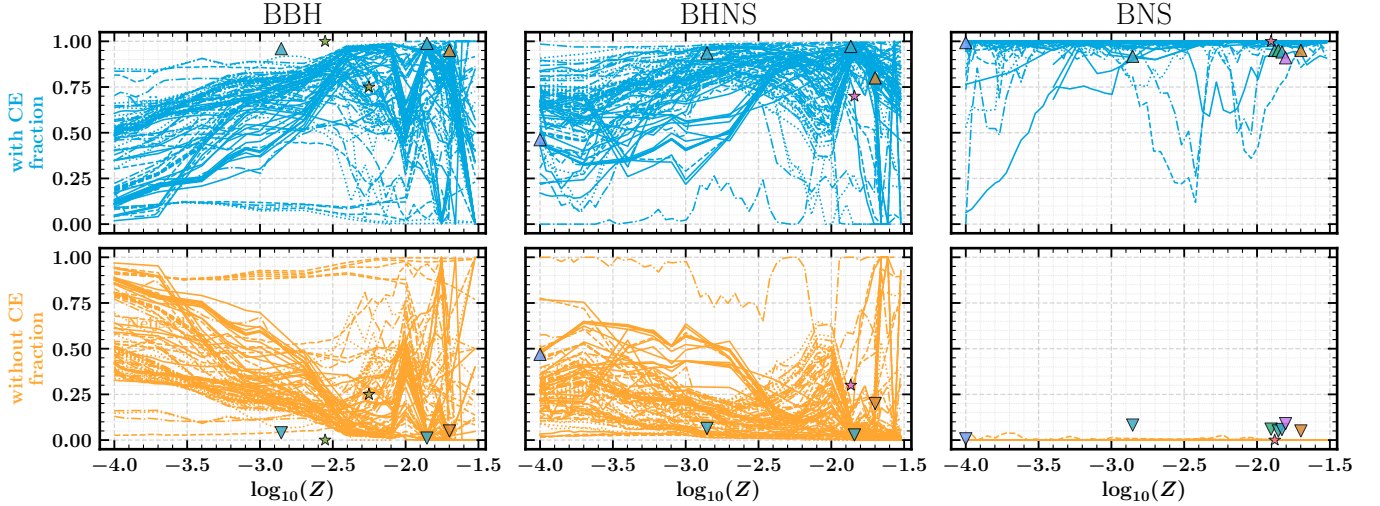


FIG. 14.— Formation-channel fractions as a function of metallicity for compact-object binaries across multiple population-synthesis studies and literature constraints. Models are shown from Iorio et al. (2023b), Broekgaarden et al. (2022c), van Son et al. (2024), and Neijssel et al. (2019) (a single BBH model), while colored markers indicate literature constraints at specific metallicities compiled in Table 1; arrows denote upper or lower limits and stars indicate approximate values. Columns correspond to the compact-object merger type (BBH, BHNS, and BNS), while the top and bottom rows show the fractions of systems forming with and without a CE phase, respectively. The van Son et al. (2024) BBH models include contributions from chemically homogeneous evolution (CHE), although these do not significantly alter the overall trends shown here. Some Broekgaarden et al. (2022c) models additionally contain a non-negligible “other” channel for which the presence or absence of a CE phase is not specified (leading to some with CE and without CE contributions adding up to less than 1). Low-number statistics typically introduces increased stochastic scatter at $\log_{10}(Z) \gtrsim -2$ for BBH and BHNS models. All metallicities are converted to $\log_{10}(Z)$, and complementary fractions are inferred where needed assuming $f_{\text{with CE}} + f_{\text{without CE}} = 1$. The figure highlights the large diversity in predicted BBH and BHNS formation-channel fractions across metallicity and population-synthesis frameworks. In contrast, all studies consistently predict that BNS formation is strongly dominated by channels involving a CE phase, with typically $\lesssim 10\%$ of systems forming without CE evolution across the explored metallicity range. For BBH and BHNS mergers, the relative contribution of without-CE formation channels generally increases toward the lowest metallicities ($\log_{10}(Z) \lesssim -2.5$), particularly for BBH systems, with the exception of . See Appendix D for additional details.

inated. The studies in Figure 5 that report non-negligible BNS contributions from without-CE channels (Boesky et al. 2024a; Broekgaarden et al. 2022c) do not provide detailed descriptions of the corresponding formation pathways. Both studies also assume that Case BB mass transfer, a phase thought to play a key role in BNS formation (e.g. Andrews et al. 2015; Vigna-Gómez et al. 2018; Chattaraj et al. 2026), always proceeds stably, which may overestimate the fraction of BNS systems forming without a CE phase. Thus, while without-CE BNS formation may occur in finely tuned regions of parameter space, current population-synthesis studies generally support the conclusion that merging BNS formation is overwhelmingly CE-dominated.

This robust requirement of CE evolution for BNS formation provides an important anchor point: while the relative importance of CE and stable mass-transfer channels remains highly uncertain for BBH and BHNS systems, CE evolution appears to be an essential ingredient for producing merging BNS systems. This suggests that BNS mergers may serve as a particularly clean calibration channel for CE physics. Because their formation pathways almost universally require a CE phase, the observed BNS merger rate directly traces the efficiency and success of CE evolution. In particular, a deficit of BNS mergers relative to BBH or BHNS populations (e.g. The LIGO Scientific Collaboration et al. 2025d; Fishbach et al. 2026) could indicate inefficient envelope ejection or a high failure rate of CE interactions. This indicates that relative merger rates between BNS, BHNS, and BBH populations may provide a complementary constraint on CE physics, helping to break degeneracies that cannot be resolved using merger rates of individual populations alone.

4.2. Breaking the Degeneracy: Observational Signatures of Formation Channels

The degeneracy between merger rates and formation pathways implies that rates alone cannot uniquely constrain binary evolution. Robust inference therefore requires the joint interpretation of multiple observables, as different formation channels imprint correlated signatures in masses, mass ratios, delay times, and spins. We discuss below how different observables may distinguish between formation channels with and without a CE phase, for which qualitative trends for with and without CE channels from the simulation literature are highlighted in Tables 2–4.

Masses — A robust trend across studies is that formation channels without a CE phase—i.e., those dominated by stable mass transfer (SMT)—preferentially produce the most massive BBH systems (Neijssel et al. 2019; Belczynski et al. 2022; van Son et al. 2022; van Son et al. 2022; Hendriks et al. 2023; Li et al. 2025a; Willcox et al. 2025). This reflects the ability of stable mass transfer to grow the accretor mass while avoiding premature mergers. In contrast, the contribution of CE-driven channels to the mass distribution is less consistent across studies. Some find that CE pathways dominate the lower-mass population (Li et al. 2025a; Hendriks et al. 2023), while others attribute low-mass BBH systems primarily to without CE channels (e.g. van Son et al. 2023), or find that both with CE and without-CE channels contribute comparably (e.g. Willcox et al. 2025).

Mass Ratios — The mass-ratio distribution provides a complementary diagnostic, though with more complex and sometimes non-monotonic behavior. Without-CE channels can produce relatively equal-mass systems ($q \sim 0.8$ – 1) in the absence of strong kicks (Briel et al. 2026), but can also yield intermediate mass ratios ($q \sim 0.4$ – 0.7) or contribute to the most unequal BBH mass ratios (van Son et al. 2022; Briel et al. 2026; Olejak et al. 2024; Banerjee & Olejak 2024; Zevin et al. 2020).

depending on the model assumptions including efficiency of mass accretion, BH kicks, and angular-momentum loss. By contrast, CE channels are found to produce broader distributions and/or sometimes found to produce the equal mass BBH mergers (e.g. [Oh et al. 2023](#); [Zevin et al. 2020](#); [van Son et al. 2022](#)). These trends emphasize that the mass-ratio distribution is not a simple discriminator, but instead encodes the interplay between the with and without CE channels and the underlying model assumptions.

Delay times. — Delay-time distributions might provide one of the clearest separations between formation channels. Without-CE channels generically produce long delay times ($\gtrsim 0.1$ – 1 Gyr), reflecting the limited orbital tightening achievable through stable mass transfer, particularly when angular-momentum loss is inefficient ([Gallegos-Garcia et al. 2021](#); [van Son et al. 2022](#); [Olejak et al. 2024](#); [Oh et al. 2023](#); [Briel et al. 2026](#); [Klencki et al. 2026](#)). In contrast, CE evolution can shrink binaries efficiently, producing typically shorter delay times ($\lesssim 1$ Gyr) and dominating rapidly merging populations ([Neijssel et al. 2019](#); [van Son et al. 2022](#)), though this might strongly depend on the assumed CE efficiency (e.g. [Bavera et al. 2021](#)). For BHNS, the without-CE channel might lead to shorter periods and delay times at solar metallicity [Xing et al. \(2024b\)](#). However, this distinction is itself sensitive to angular-momentum loss prescriptions: more efficient angular-momentum loss during stable mass transfer can partially compensate for the lack of a CE phase and produce shorter delay times ([Olejak et al. 2024](#)) whilst different assumptions in the CE efficiency can drastically change the BBH delay time from the with-CE channel ([Boesky et al. 2024a](#)). This highlights that delay times are not purely a channel diagnostic, but also probe the underlying physics of orbital angular-momentum evolution.

Spins. — Spin distributions provide an additional, partially independent probe of binary evolution. SMT channels typically produce low or modest spins, as post-interaction binaries remain relatively wide and tidal spin-up is inefficient ([Klencki et al. 2026](#)). In contrast, CE channels can lead to higher spins for the second-born compact object due to tighter post-CE orbits and enhanced tidal interactions ([Bavera et al. 2021](#)).

At the same time, SMT channels can produce a range of effective spins depending on accretion efficiency and angular-momentum transport, and may contribute to both low-spin and moderate-spin populations ([Olejak et al. 2024](#); [Briel et al. 2026](#)). As with masses and delay times, the spin distribution reflects not only the formation channel, but also the underlying assumptions about mass transfer and angular-momentum transport.

White dwarf populations — Another angle might be to also investigate the impact on the formation channels of binaries containing a white dwarf (e.g., [Toonen et al. 2012, 2018](#); [Roy & Kalita 2026](#)). While this study focuses on BBH, BHNS and BNS systems, compact object binaries that contain a WD can provide a direct pathway to calibrate the key physical processes governing binary evolution and to break the degeneracies inherent in formation-channel inference from gravitational-wave data alone. Moreover, the abundance of white dwarf binaries identified throughout the Milky Way from large astronomical surveys (e.g., [Rebassa-Mansergas et al. 2010](#); [Ren et al. 2014](#); [Rebassa-Mansergas et al. 2021](#); [Grondin et al. 2024](#); [Nayak et al. 2024](#); [Li et al. 2025b](#)) makes them powerful observational probes of CE evolution through their ability to directly

measure post-CE outcomes.

For example, the range of orbital separations observed in white dwarf binaries offers important constraints on CE evolution. Observed short-period white dwarf binaries (e.g., [Nebot Gómez-Morán et al. 2011](#); [Hernandez et al. 2021, 2022](#); [Li & Zhang 2024](#)) can be explained by efficient orbital energy deposition and strong inspiral, whereas wider post-CE systems (e.g., [Yamaguchi et al. 2024a,b](#)) imply that additional energy reservoirs, such as recombination energy, convection, or accretion-driven outflows, may play a significant role in unbinding the envelope. As such, the white dwarf binary population serves as a powerful observational constraint on the underlying physical mechanisms of CE evolution.

Multi-messenger and intermediate populations. — A key avenue to break degeneracies is to connect compact-object mergers to their progenitor populations. Systems such as high-mass X-ray binaries, stripped stars, Wolf–Rayet binaries, and supernova progenitors directly probe the phases of mass transfer and envelope ejection that set the relative importance of CE for forming tight compact object binaries (e.g., [Vinciguerra et al. 2020](#); [Gallegos-Garcia et al. 2022](#); [Rocha et al. 2024](#); [Zapartas et al. 2025](#)). In particular, these intermediate populations can constrain the physical parameters identified in Section C; including mass-transfer stability criteria (q_{crit}), CE efficiency (α_{CE}), and angular-momentum loss during non-conservative transfer, which ultimately control both the formation-channel fractions and their observable signatures. Future work should therefore move beyond merger-only constraints and adopt a multi-phase, multi-messenger approach, linking gravitational-wave observations to electromagnetic populations such as Be/X-ray binaries, *Gaia* astrometric binaries ([Chawla et al. 2022](#)), Wolf–Rayet stars ([Korb et al. 2025](#)), and supernova progenitors.

A complementary and relatively untapped avenue to break these degeneracies is provided by electromagnetic transients that directly probe phases of unstable mass transfer, envelope ejection, and stellar coalescence. In particular, luminous red novae (LRNe) are now robustly associated with stellar mergers and common-envelope (CE) events (e.g., [Soker & Tylenda 2006](#); [Tylenda et al. 2011](#)), tracing the onset of dynamical instability and the subsequent inspiral in real time ([Ivanova et al. 2013c](#); [Pejcha et al. 2016](#); [MacLeod et al. 2017](#); [Blagorodnova et al. 2021](#); [Twum et al. 2026](#)). Observations and modeling show that LRNe arise from envelope ejection following unstable mass transfer, often preceded by sustained mass loss through the outer Lagrange point, and encode ejecta masses, velocities, and energetics that directly reflect the efficiency of envelope removal ([Matsumoto & Metzger 2022](#)). As such, LRNe provide one of the few direct empirical probes of CE physics, constraining the conditions under which binaries merge versus survive envelope ejection, the same processes that dominate the uncertainty in formation-channel predictions.

Fast blue optical transients (FBOTs; [Drout et al. 2014](#)), and in particular the luminous FBOT (LFBOT) subclass (e.g., [AT2018cow](#); [Prentice et al. 2018](#); [Margutti et al. 2019](#); [Ho et al. 2019](#); [Perley et al. 2019](#)), may probe a complementary regime involving compact objects embedded within stellar envelopes (e.g., [Soker et al. 2019](#)). A promising scenario is that LFBOTs arise from delayed mergers between a black hole or neutron star and the helium core of a companion, either following a classical CE phase ([Metzger 2022](#)), or after an extended phase of stable mass transfer that terminates in a delayed dynamical instability ([Klencki & Metzger 2025](#)). In this

picture, the multiwavelength transient is powered by highly super-Eddington accretion onto the compact object and interaction with the circumstellar medium shaped by the early mass-transfer phase (Klencki & Metzger 2025; Aspegren & Kasen 2026). Importantly, these events trace systems that lie near the boundary between stable and unstable mass transfer, i.e., those governed by the critical mass-ratio and angular-momentum-loss prescriptions that drive the diversity between CE and without-CE channels. LFBOTs may therefore represent “failed” gravitational-wave progenitors, directly sampling binaries that would otherwise contribute to the compact-object merger population if they avoided late-stage instability (Klencki & Metzger 2025).

Finally, longer-lived electromagnetic signatures such as ultra-luminous X-ray sources (ULXs; Kaaret et al. 2017; Pavlovskii et al. 2017) and associated energetic synchrotron radio nebulae (“hyper-nebulae”), provide a time-integrated probe of hyper-Eddington mass transfer onto compact objects. During phases of extreme accretion, powerful jets and disk winds inflate extended nebulae of relativistic particles, which can remain observable for $\gtrsim 10^3$ – 10^6 yr and trace the cumulative energy injection from mass transfer episodes (Sridhar & Metzger 2022). Population models show that these systems arise across both stable and unstable mass-transfer channels, spanning a wide range of accretion rates and evolutionary states (Uno et al.). As a result, ULX and hypernebulae provide a statistical probe of the prevalence, duration, and energetics of mass-transfer phases, complementing the instantaneous snapshots provided by LRNe and LFBOTs. Taken together, these electromagnetic counterparts offer a multi-phase, multi-timescale view of binary interaction physics, from pre-instability mass loss to final merger or survival, and thus provide a future pathway to break the degeneracies between CE and without-CE formation channels.

A complementary constraint on binary evolution comes from the observed rates and electromagnetic counterparts of neutron star mergers themselves. Kilonovae—thermal transients powered by the radioactive decay of r-process elements synthesized in the ejecta—provide a direct probe of the merger rate and ejecta properties of compact-object binaries (e.g., Metzger & Berger 2011). However, despite their importance, the intrinsic kilonova rate remains poorly constrained, with only a small number of confirmed detections and a lack of robust blind discoveries in wide-field surveys. Similarly, short gamma-ray bursts (SGRBs), widely interpreted as beamed emission from neutron star mergers, offer an independent tracer of the merger population (e.g., Berger 2014). Comparisons between SGRB rates and gravitational-wave-inferred binary neutron star (BNS) merger rates suggest broad consistency, but also highlight significant uncertainties related to jet opening angles and observational selection effects (e.g., Fong et al. 2015). Recent work indicates that the true BNS merger rate may be lower than previously inferred from early gravitational-wave detections, with implications for the efficiency of r-process production and the population of merging double neutron stars (e.g., Fishbach et al. 2026). Together, these multi-messenger constraints provide a critical normalization of the BNS merger rate and, by extension, of the efficiency of the evolutionary pathways that produce them. In particular, given the robust theoretical result that BNS mergers predominantly form through channels involving common-envelope evolution, the observed rates of kilonovae and SGRBs offer one of the most direct empirical probes

of common-envelope physics. Future improvements in blind kilonova searches and joint GW–EM detection statistics will therefore provide a key avenue to break degeneracies between formation channels by anchoring population-synthesis predictions to the absolute rate of neutron star mergers.

Observational constraints from gravitational-waves on formation-channel fractions.— A growing number of studies have attempted to directly constrain the relative contributions of CE and SMT channels using gravitational-wave observations, typically by combining population-synthesis predictions with hierarchical Bayesian inference. Early work demonstrated that even with $O(100)$ detections, the relative fraction of binaries formed through different channels can be constrained by jointly modeling the mass, spin, and redshift distributions (Zevin et al. 2017). More recent analyses have extended this framework to explicitly compare isolated-binary sub-channels, including CE, SMT, and chemically homogeneous evolution (CHE), showing that only channels with sufficiently distinct physical imprints can be robustly distinguished with current data (e.g., van Son et al. 2022; Colloms et al. 2025; Alvarez-Lopez et al. 2025). In particular, studies incorporating GWTC catalogs find emerging, albeit still model-dependent, evidence favoring a mixture of with CE and without CE formation, with the inferred branching ratios sensitive to assumptions about mass-transfer efficiency and CE physics (Bavera et al. 2021, 2022; Zevin et al. 2020; Stevenson et al. 2022).

However, these efforts also highlight a key limitation: CE and SMT channels often share overlapping regions of parameter space in mass, spin, and delay time, making them difficult to disentangle without multi-dimensional inference and flexible population models (Zevin et al. 2017; Colloms et al. 2025). As a result, current constraints on CE versus SMT fractions remain broad and degenerate with uncertain binary-physics prescriptions, particularly angular-momentum loss and mass-transfer stability. Nevertheless, as the number of detected mergers grows and inference frameworks become increasingly agnostic and high-dimensional, gravitational-wave observations are beginning to place meaningful constraints on the relative importance of CE and SMT channels, providing the first empirical calibration of these key phases of massive binary evolution.

4.3. What constitutes a Formation Channel?

A fundamental ambiguity in the literature is that there is no universally agreed-upon definition of what constitutes a distinct formation channel. Different studies adopt markedly different classification schemes, often tailored to the specific physical processes or questions under investigation.

For example, some works define (sub)channels based on the sequence of binary interactions—tracking whether common-envelope (CE), stable mass transfer (SMT), or no mass transfer occurs before or after the first supernova (Briel et al. 2021; Oh et al. 2023). Others distinguish channels based on the nature of the CE phase itself, such as single-core versus double-core CE (e.g. Belczynski et al. 2002; Vigna-Gómez et al. 2018), or the evolutionary state (i.e., structure or stellar type) of the donor at the onset of the CE (e.g. Ruiter et al. 2019; Vigna-Gómez et al. 2020; Chattaraj et al. 2026; Dorozsmai & Toonen 2024).

Additional approaches emphasize the number of mass-transfer episodes, including repeated CE or SMT phases—an aspect that is often neglected but can be critical for binary tightening (e.g. Dominik et al. 2013; Stevance et al. 2023). For binary neutron star formation in particular, several stud-

ies highlight the role of specific physical ingredients, such as case BB mass transfer, ultra-stripped supernovae (Tauris et al. 2015b; Tauris et al. 2015a; Tauris et al. 2017; Vigna-Gómez et al. 2018; Andrews et al. 2015; Xing et al. 2024a), electron-capture supernovae (Kruckow et al. 2018; Tanaka et al. 2023; Vigna-Gómez et al. 2018), or accretion-induced collapse (Dominik et al. 2012; Ruiter et al. 2019; Chruślińska et al. 2018; Pellouin et al. 2025). Other formation channel distinctions that have been pointed out include systems that initiate mass transfer only after the first supernova (Willcox et al. 2025; Mestichelli et al. 2025), quasi-chemically homogeneous evolution (Eldridge et al. 2019), pathways leading to very short delay times (Andrews et al. 2020), or systems that survive via favorable (“lucky”) natal kicks (Broekgaarden et al. 2022c).

Taken together, these examples illustrate that what is labeled a “formation channel” is not uniquely defined, but instead reflects a choice of which physical processes are considered most informative for organizing the evolutionary pathways and answering the key open question: “how do merging BNS, BHNS, and BBHs form?”. In practice, the level of granularity often depends on the scientific question: whether one aims to isolate key physical mechanisms (e.g., CE physics, supernova prescriptions), trace evolutionary sequences, or connect to observable populations.

It is therefore perhaps unsurprising that, despite compact-object mergers being intrinsically rare outcomes, the literature identifies a wide diversity of pathways leading to BBH, BHNS, and BNS systems. At the same time, this diversity is itself a key result: it highlights that different physical assumptions and modeling choices can open or suppress distinct routes to merger. Consequently, even the seemingly simple question—“how does a BBH, BHNS, or BNS system form?”—does not admit a unique answer within current binary evolution frameworks.

4.4. Mass Transfer and Binary Interactions as the Dominant Uncertainty

Our results reinforce the emerging view that mass transfer, particularly its stability, is a dominant uncertainty in modeling the formation of merging compact-object binaries.

Mass-transfer stability. — Mass-transfer stability is a key uncertainty, particularly for the SMT channel. van den Heuvel et al. (2017b) first pointed out that the greater stability of mass transfer from radiative donors compared to convective giants opens the possibility for an SMT channel in systems with BH accretors. Subsequent studies have demonstrated that systems evolving to form GW sources through the SMT channel often lie close to the boundary between stable and unstable mass transfer evolution, further emphasizing the importance of the location of this stability boundary (e.g. Marchant et al. 2021b; Gallegos-Garcia et al. 2021; Picco et al. 2024; Klencki et al. 2026).

Determining mass transfer stability remains an area of active research. Numerous studies have shown that it depends sensitively on the stellar structure, evolutionary state, stellar mass, and metallicity of the donor, rather than following a simple prescription based on stellar type or fixed critical mass ratios (Pavlovskii & Ivanova 2015; Pavlovskii et al. 2017; Ge et al. 2015, 2020a, 2024; Marchant et al. 2021b; Klencki et al. 2020, 2021a; Renzo et al. 2023; Temmink et al. 2023, 2025). Detailed stellar-evolution calculations demonstrate that the critical mass ratio for stability varies significantly with the internal structure of the donor star, particularly for radiative-envelope

stars (Ge et al. 2015, 2020b; Pavlovskii et al. 2017; Klencki et al. 2026). This challenges the simplified prescriptions commonly adopted in rapid population-synthesis codes (Hurley et al. 2002), which may either overestimate or underestimate the prevalence of SMT depending on the adopted assumptions (Gallegos-Garcia et al. 2021; Zhao et al. 2024; Schürmann et al. 2025b). More generally, this suggests that progress in the theory of mass-transfer stability requires not only more sophisticated 3D simulations of gas dynamics in binaries (e.g. Ryu et al. 2025) and of the outer envelope layers of stars (e.g. Jiang 2023), but also improved understanding of stellar interiors, including the response of mass-gainer stars (Renzo & Göteborg 2021; Renzo et al. 2023; Schneider et al. 2024; Xu et al. 2025; Klencki et al. 2026). Observationally, valuable constraints may soon come from massive post-interaction binaries, such as the recently discovered populations of stripped-star systems (Drout et al. 2023; Göteborg et al. 2023; Villaseñor et al. 2023; Ramachandran et al. 2023, 2024; Marchant & Bodensteiner 2024).

Variations in stability criteria can shift systems between SMT and CE channels, thereby strongly affecting the relative contributions of formation pathways Dorozsmai & Toonen (2024). In extreme cases, adopting highly permissive stability criteria can lead to populations dominated by without-CE evolution (Li et al. 2023; Olejak et al. 2021).

Notably, mass-transfer stability depends not only on the donor star, but also on the accretor, particularly when the accretor is a star rather than a compact object. A growing number of studies suggests that, for stellar accretors, the response of the accretor to gaining a substantial amount of mass on a short timescale may itself strongly determine the boundary between stable and unstable mass transfer (Henneco et al. 2024; Lau et al. 2024; Schürmann & Langer 2024). This could imply substantially different response before and after the formation of the first compact object, an effect that is only beginning to be taken into account in rapid population-synthesis models (Schürmann et al. 2025b; Klencki & Metzger 2025).

Orbital evolution and angular-momentum loss. — In addition to the stellar response, mass-transfer stability also depends sensitively on the orbital evolution driven by the transfer process itself. One of the primary uncertainties is how much mass escapes the binary and how much angular momentum it carries away. Many prescriptions for non-conservative mass loss assume isotropic re-emission (specific angular momentum of the donor and/or accretor), while additional mass-loss channels such as L2 outflows can significantly enhance angular-momentum loss and lead to more efficient orbital tightening (van Son et al. 2022; Willcox et al. 2025; Lu et al. 2023; Picco et al. 2024; Olejak et al. 2024; Gallegos-Garcia et al. 2024). In some cases, these outflows may remain partially bound to the system, forming a circumbinary disk which can significantly alter the long-term orbital evolution of the inner binary (e.g., Siwek et al. 2023; Valli et al. 2024; Wei et al. 2024). These uncertainties directly impact whether SMT channels can produce binaries sufficiently compact to merge within a Hubble time, as well as their predicted properties such as mass ratios (e.g. Olejak et al. 2024).

At the same time, although orbital evolution and mass-transfer stability are often treated as separate uncertain ingredients, they are physically coupled. More rapid orbital shrinkage, for example due to enhanced angular-momentum loss through L2 outflows, generally decreases the stability of mass transfer by accelerating the response of the Roche lobe

relative to that of the donor (e.g. [Picco et al. 2024](#); [Willcox et al. 2025](#); [Ge et al. 2024](#); [Klencki et al. 2026](#)).³ This introduces a feedback loop: increased angular-momentum loss during SMT can help systems shrink to separations small enough to merge, but it also reduces the region of parameter space in which the interaction is stable in the first place.

This coupling may make the uncertainty associated with the SMT channel smaller than one might infer by varying angular-momentum loss and mass transfer stability independently. A useful reference point is the population of compact-object + massive-star binaries, which is an intermediate evolutionary stage expected in both SMT and CE channels (e.g. [Picco et al. 2024](#); [Korb et al. 2025](#)). [Klencki et al. \(2026\)](#) find that varying the angular-momentum loss strongly affects which specific BH + massive-star systems are the expected progenitors of the SMT channel, but only weakly affects the overall size of the parameter space in which the SMT channel operates. This behavior arises from the coupling between orbital shrinkage and stability, which introduces a fundamental limit on orbital tightening in SMT-driven evolution.

This opens a promising avenue for calibrating the contribution of the SMT channel to the observed GW-source population. While only a handful of BH + massive-star systems is known to date, this number is predicted to grow substantially in the coming years thanks to large-scale astrometric, spectroscopic, and time-domain surveys ([El-Badry 2024](#); [Shenar et al. 2024](#); [Abrams et al. 2025](#)) yielding a well-characterized population of tens to hundreds of BH systems ([Langer et al. 2020](#); [Janssens et al. 2022](#)). If models can predict what fraction of these systems later form merging compact objects through SMT, than the observed population provides a direct normalization point for the SMT channel contribution.

Common-envelope efficiency and binding energy. — In addition to mass-transfer stability, the efficiency of CE ejection remains a major uncertainty. Variations in the CE efficiency parameter α_{CE} can significantly affect both merger rates and formation-channel contributions ([Bavera et al. 2021](#); [Tanaka et al. 2023](#); [Boesky et al. 2024a](#)). Importantly, the effects of CE efficiency can be degenerate with other parameters, complicating attempts to constrain it observationally (e.g. [Boesky et al. 2024a](#); [Sgalletta et al. 2026](#)). Taken together, these uncertainties demonstrate that the balance between with-CE and without-CE channels is highly sensitive to poorly constrained aspects of binary interaction physics, making it difficult to robustly infer formation pathways from current models.

4.5. Hydrodynamic Constraints on Common-Envelope Prescriptions

One important point that emerges from comparing these formation channels is that many commonly adopted common-envelope (CE) prescriptions should not be viewed simply as interchangeable parameter choices, but as approximations to a fundamentally unresolved hydrodynamic problem. Quantities such as the CE efficiency parameter α_{CE} , the envelope binding-energy parameter λ , critical mass-ratio stability criteria, and prescriptions tied to convective or radiative envelope structure are widely used throughout population-synthesis calculations. However, these quantities ultimately act as effective descriptions of multidimensional processes involving inspiral-driven

energy deposition, angular-momentum transport, shocks, envelope restructuring, recombination physics, and the dynamical response of the stellar interior.

Over the past decade, hydrodynamic simulations have significantly improved our understanding of CE evolution, particularly for highly extended RGB, AGB, and red-supergiant donors ([Ricker & Taam 2012](#); [Passy et al. 2012](#); [Ohlmann et al. 2016](#); [Iaconi et al. 2017](#); [Iaconi & De Marco 2018](#); [Sand et al. 2020](#); [Chamandy et al. 2020](#)). Despite large differences in numerical methods, resolution, and donor structure, these simulations reveal several remarkably consistent features. In most calculations, the interaction begins with a rapid plunge-in phase in which orbital energy and angular momentum are transferred to the envelope through shocks, tidal torques, and hydrodynamic drag. Large-scale spiral shocks and asymmetric outflows develop quickly, leading to substantial envelope expansion. However, many simulations find that the inspiral slows considerably after the initial plunge, often entering a prolonged phase of slower orbital evolution characterized by circulation flows, repeated passages through shocked gas, and continued redistribution of energy and angular momentum over many dynamical timescales ([Ivanova & Nandez 2016](#); [Iaconi et al. 2017](#); [Chamandy et al. 2020](#)).

A major lesson emerging from these studies is that the outcome of the interaction is not determined solely by the orbital energy released during the initial inspiral, but by the subsequent hydrodynamic evolution of the disturbed envelope. In many simulations, the initial plunge-in alone is insufficient to fully unbind the envelope, and the longer-term redistribution of energy and momentum through shocks, turbulent flows, fallback, envelope inflation, and recombination-assisted expansion becomes essential in determining the final ejecta mass and orbital separation ([Ivanova et al. 2013b, 2015](#); [Lau et al. 2022](#); [Everson et al. 2025](#)). Hydrodynamic calculations also increasingly emphasize the importance of donor structure in regulating CE outcomes. Successful envelope ejection and compact binary survival appear to occur preferentially for highly evolved donors with extended weakly bound envelopes and steep density gradients between the core and envelope ([MacLeod et al. 2015](#); [Law-Smith et al. 2020](#); [Marchant et al. 2021a](#)). In these systems, the outer envelope binding energy per unit mass can become comparable to the available recombination-energy reservoir, allowing recombination-assisted outflows to contribute significantly during the late inspiral and expansion phases ([Ivanova et al. 2015](#); [Lau et al. 2022](#)).

Importantly, these studies also show that the envelope binding energy is not static during the interaction. As orbital energy and angular momentum are deposited into the envelope, the donor dynamically restructures, altering the density profile, entropy stratification, and effective binding energy of the remaining gas as the inspiral proceeds. This restructuring is difficult to capture in rapid prescriptions that rely on pre-computed stellar profiles or fixed λ parameterizations. Consequently, differences between analytic stellar fits, interpolated stellar tracks, hybrid MESA-based approaches, and fully resolved stellar models do not merely correspond to calibration differences, but often reflect qualitatively different assumptions regarding stellar structure and the hydrodynamic evolution of the envelope itself. In practice, different prescriptions for the envelope structure and core-envelope boundary can produce binding energies that differ by factors of several and in some cases by nearly an order of magnitude, leading to substantially different predictions for CE survival and merger

³ Possible ways of accounting for this effect in rapid population synthesis include the ζ formalism ([Willcox et al. 2025](#)), interpolating grids of detailed calculations ([Ge et al. 2020a](#)), or adopting a critical minimum separation rather than a critical mass-ratio prescription ([Klencki et al. 2026](#)).

outcomes. Prescriptions based on fixed λ values, tabulated λ calculations (e.g., Xu & Li 2010b), enthalpy-based formalisms (Ivanova & Chaichenets 2011), recombination-inclusive energy budgets (Webbink 2008; Ivanova et al. 2015; Lau et al. 2022), and hydrodynamic inspiral calculations therefore encode distinct physical assumptions regarding what constitutes available energy for envelope ejection and what conditions define successful binary survival.

Another important uncertainty concerns the final inspiral or “parking” conditions near the donor core. In many population-synthesis implementations, the stellar core is effectively treated as a point mass and the inspiral termination is determined through simplified energetic arguments. Hydrodynamic studies, however, suggest that the final separation depends sensitively on the detailed structure of the core-envelope transition region and on how the inspiraling companion couples to the surrounding gas near the core (Law-Smith et al. 2020; Cohen & Soker 2023). Entropy gradients, pressure support, and unresolved gas dynamics near the inner envelope can therefore materially influence whether inspiral stalls, survives, or proceeds to merger.

Taken together, these results suggest that the diversity of formation-channel predictions across population-synthesis studies reflects not only differences in binary-evolution parameters, but also differing implicit assumptions about the hydrodynamic evolution of the CE interaction itself.

4.6. Limitations of the Current Compilation

Our compilation focuses on intrinsic formation-channel predictions from isolated binary evolution and therefore excludes several classes of studies that could provide complementary constraints. First, we do not include studies that report results with detection-weighted populations, or without channel-resolved rates. While such studies provide valuable insights, they are difficult to incorporate into a homogeneous comparison due to differences in assumptions about metallicity, selection effects, and detector sensitivity. Second, certain formation channels, such as chemically homogeneous evolution or residual (other) channels, are treated inconsistently across studies. A more systematic inclusion of these channels will be necessary for a complete mapping between theoretical predictions and observations. Third, we rely on published intrinsic merger rates rather than re-analyzing simulation outputs directly. A fully self-consistent re-analysis of publicly available datasets, using uniform assumptions for cosmic star formation history and channel classification, would provide a more robust comparison but is beyond the scope of this work. Finally, the classification of subchannels varies significantly across studies, with different works adopting distinct definitions based on interaction sequences, CE types, or evolutionary phases.

4.7. Recommendations for Population-Synthesis Studies

Based on this meta-analysis, we highlight several practical recommendations for future population-synthesis studies:

- *Adopt transparent (and, if possible, consistent) formation-channel definitions.* Clearly define formation channels and avoid ambiguous terminology such as “only stable mass transfer,” particularly in cases where systems may not undergo mass transfer at all. Explicitly state the selection criteria used to classify systems into different channels.
- *Ensure reproducibility through open data and post-processing pipelines.* Make simulation outputs and

post-processing pipelines publicly available to enable re-analysis under consistent assumptions. This is especially critical for formation-channel studies, where differences in classification schemes can significantly affect inferred contributions.

- *Systematically explore key parameter uncertainties.* Prioritize variations in parameters that strongly impact formation-channel contributions, particularly mass-transfer stability criteria and processes governing angular-momentum loss (e.g. γ). This includes the mass-transfer efficiency (β), common-envelope efficiency (α_{CE}), supernova prescriptions, supernova kick magnitudes (e.g. σ), and envelope binding-energy assumptions.
- *Expand formation-channel analyses across populations.* Extend detailed formation-channel studies to a broader set of simulations, especially for BHNS and BNS populations, where systematic comparisons remain limited. Future work should additionally extend formation-channel analyses beyond merging compact-object binaries to other astrophysical populations, including intermediate evolutionary stages of gravitational-wave sources. This will help map how different physical assumptions impact different compact-object populations, including whether certain parameters affect BBH, BHNS, and BNS systems differently or alter their relative formation efficiencies and channel contributions in distinct ways.
- *Move beyond merger rates* Investigate the observable imprints of different formation channels beyond rates such as their impact on mass, spin, and redshift distributions, to enable comparisons with current and future gravitational-wave observations and help quantify and characterize distinguishable observable features for these channels.

Overall, our results support a growing consensus that the key question is no longer whether CE or only stable mass transfer channels dominate, but rather how the interplay between these channels—and the uncertainties in mass transfer physics—shapes the observable population of gravitational-wave sources.

5. CONCLUSIONS

In this work, we compiled and systematically compared formation-channel predictions from more than one hundred population-synthesis simulations to assess the role of common-envelope (CE) evolution in forming gravitational-wave sources within the isolated binary evolution paradigm. Our main conclusions are:

- ***BBHs and BHNS can form both mostly with or without CE and merger rates alone do not constrain formation pathways.*** Figures 3 and 4 and the global overview in Figure 2 demonstrate that BBH and BHNS populations span nearly the full allowed range from CE-dominated to without-CE-dominated formation, while often predicting comparable merger rates. This reveals a fundamental degeneracy: qualitatively different evolutionary pathways can reproduce similar observable merger-rate populations, implying that rates alone cannot uniquely constrain the physics of massive binary evolution.

- **BNS formation robustly requires CE evolution, unlike BBH and BHNS.** In contrast to BBH and BHNS systems, Figure 5 shows that BNS mergers are consistently dominated ($\gtrsim 90$ – 100%) by channels involving at least one CE phase across all compiled simulations. This is further confirmed in Figure 14 that shows BNS formation across metallicity has negligible contribution from without CE channels. This qualitative difference implies that CE physics may play a critical role in setting the relative rates of BNS versus BBH/BHNS mergers, and that constraints on CE efficiency from observations could directly impact population ratios and their relative rates.
- **A small number of key physical assumptions dominate the uncertainty in formation-channel predictions.** Figures 9–13 show that parameter variations within a given population-synthesis framework typically explore a relatively limited region of formation-channel parameter space, while the largest differences often arise between different population-synthesis codes. This highlights the presence of “simulation silos,” where differences in underlying physical prescriptions can dominate over within-code parameter variations in determining the relative contributions of with- and without-CE formation channels. At the same time, Figure 11 demonstrates that a small set of key assumptions can substantially alter the channel fractions, often by $\gtrsim 30\%$. In particular, assumptions governing mass-transfer stability (e.g., q_{crit}) and angular-momentum loss during non-conservative mass transfer have the largest impact on the relative contributions of CE and without-CE pathways for BBHs. The mass-transfer efficiency β , CE efficiency α , supernova remnant-mass prescriptions, and natal kicks also significantly affect both BBH and BHNS channel contributions.
- **Common-envelope efficiency, mass-transfer efficiency, and SN natal kick parameters introduce non-linear and competing effects.** Figures 12 and 13 further show that parameters such as the CE efficiency (α_{CE}) and mass-transfer efficiency (β), and SN kick magnitude (σ) have non-monotonic effects on formation-channel fractions. These parameters regulate both binary survival and orbital tightening, leading to complex and sometimes counterintuitive trends in the relative importance of CE versus SMT pathways.
- **Formation-channel diversity arises from a redistribution across multiple evolutionary pathways.** The Level 2 decomposition (Appendix E) shows that the large spread in Level 1 channel contributions is not driven by a single dominant alternative pathway, but rather by a redistribution across multiple evolutionary sequences. This reflects the high-dimensional nature of binary evolution and the complex interplay of physical processes governing compact-object formation.
- **Breaking degeneracies requires multi-dimensional and multi-messenger constraints.** Given the strong degeneracy between formation pathways and merger rates, additional observables—such as masses, mass ratios, spins, and delay times—are required to distinguish between competing evolutionary scenarios. Furthermore, connecting gravitational-wave sources to intermediate

populations (e.g., X-ray binaries, stripped stars, and supernova progenitors) offers a critical pathway to directly constrain the underlying physics of binary interactions.

- **Future progress requires systematic cross-code comparisons and exploration of key uncertainties.** Our results highlight the need for coordinated efforts to map the uncertainty space of massive binary evolution. In particular, future work should (i) systematically quantify formation-channel contributions, (ii) explore parameter variations affecting mass-transfer stability and angular-momentum loss (e.g., q_{crit} , α_{CE} , β , γ), and (iii) develop frameworks that enable direct comparison across population-synthesis codes. Such efforts will be essential for robustly connecting gravitational-wave observations to the physical processes shaping massive stars across cosmic time.

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This work made use publicly available code for creating the BBH, BHNS, and BNS formation efficiencies in Figure 14 from van Son (2024) for results from van Son et al. (2024), the code and corresponding data Iorio et al. (2023a) to reproduce the formation efficiencies from Iorio et al. (2023b), and Broekgaarden (2023b) for the formation efficiencies based on Broekgaarden et al. (2022c,a) for which also the software from was used.

DATA AVAILABILITY

All data and code to reproduce all results and figures in this work are publicly available at [Broekgaarden \(2023a\)](#).

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APPENDIX

APPENDIX A: DATA RETRIEVAL

This appendix describes the process used to collect the formation-channel contributions and merger rates presented in this study. We compile data from recent population synthesis studies of isolated binary evolution that report intrinsic merger rates and explicitly quantify the relative contributions of different formation channels, with particular attention to the occurrence of CE phases in the formation of BBH, BHNS, and BNS systems. We restrict our sample to studies published within the past decade that (i) provide cosmologically integrated intrinsic merger rates and (ii) report a breakdown of these rates by distinct formation channels. We do not include studies that only present formation-channel fractions at fixed metallicity, or that focus exclusively on detectable (observer-frame) rates. Incorporating such results would require re-computing cosmological rate integrations under a consistent set of assumptions (most notably regarding detector sensitivity and selection effects), which is beyond the scope of this work but constitutes an important direction for future study. A list of excluded studies is provided in the following Appendix section. For studies that do not tabulate rates for the full set of channels in our taxonomy, we adopt a uniform bookkeeping convention: any unreported channel is assigned a rate of zero for that study. This choice enables a consistent cross-study compilation while explicitly reflecting the information content

of the original publication; it may therefore be interpreted as a lower limit on the contribution from unreported channels. All included datasets are listed alphabetically below. Throughout this appendix, we adopt the formation-channel classification scheme introduced in the main text (see Figure 1). At the first level, we distinguish between systems that do and do not undergo a CE phase during their evolution. Where available, we further adopt the more detailed sub-classifications reported by the original authors (level 2 classification). We list the studies alphabetically, and in brackets list the acronyms used in the figures.

GWTC-4 inferred local merger rates — The LIGO Scientific Collaboration et al. (2025d) infer the local merger rates of BBH, BHNS, and BNS systems from the gravitational-wave detections in GWTC-4 The LIGO Scientific Collaboration et al. (2025b) under a range of population models. We adopt their reported “merged” rate constraints from Table 2 as a baseline, which already include both statistical and model uncertainties. For BBHs, this corresponds to a central estimate of $\sim 26 \text{ Gpc}^{-3} \text{ yr}^{-1}$ with a quoted range of $14\text{--}26 \text{ Gpc}^{-3} \text{ yr}^{-1}$. To account for additional systematic uncertainty from the assumed redshift evolution, we further include the constraints from Section 6.4 of The LIGO Scientific Collaboration et al. (2025d) at $z \sim 0.2$, which give $29^{+8.5}_{-6.5} \text{ Gpc}^{-3} \text{ yr}^{-1}$ for a power-

law redshift model and $38^{+19}_{-10} \text{ Gpc}^{-3} \text{ yr}^{-1}$ for a B-spline model. Combining these estimates, we adopt a conservative overall BBH rate range of $14\text{--}58 \text{ Gpc}^{-3} \text{ yr}^{-1}$.

For BHNS and BNS systems, we adopt the ‘merged’ rate constraints directly from Table 2, yielding ranges of $9.1\text{--}84 \text{ Gpc}^{-3} \text{ yr}^{-1}$ for BHNS and $7.6\text{--}250 \text{ Gpc}^{-3} \text{ yr}^{-1}$ for BNS mergers. These ranges are used throughout as observational benchmarks for comparison to the population-synthesis predictions.

Bavera et al. (2021) [Ba21] — *Bavera et al. (2021)* compute BBH merger rates by coupling detailed binary stellar evolution simulations performed with MESA to the rapid population synthesis code COSMIC via an early version of the POSYDON framework. Their study explores a range of population synthesis assumptions varying CE parameters, mass transfer and accretion parameters, and initial conditions and reports BBH rates subdivided into the classic CE and SMT formation channels, which correspond to the ‘with CE’ and ‘without CE’ categories in Fig 1, respectively. We extract total merger rates from Tables 1, 3, and 4 of *Bavera et al. (2021)*. From Table 1, which explores variations in the common-envelope efficiency parameter α_{CE} , we adopt the seven CE rate estimates listed in the first seven rows and pair each with the constant SMT rate given in the eighth row in order to compute the relative CE and SMT contributions. The lower six rows of Table 1 provide three additional total BBH rate model estimates (CE+SMT combined). Tables 3 and 4 contribute three and two additional models for which the authors report total BBH rate estimates, respectively. In each case, we exclude the fiducial models listed in the first row of these tables, as they duplicate the fiducial model already included from Table 1. In total, this yields 15 distinct BBH merger-rate estimates and their corresponding CE and SMT contributions.

Boesky et al. (2024b,a) [Bo24] — *Boesky et al. (2024a,a)* use the COMPAS population synthesis code to compute 21 model variations: 12 exploring different values of the CE efficiency parameter (α) and binding energy parameter (β), and 9 varying the remnant mass prescription and the supernovae (SN) natal-kick magnitude distribution. We extract formation channel contributions from Figure 12 of *Boesky et al. (2024a)* (at redshift $z \sim 0$) and from the authors through private communication. The published figure includes a non-negligible contribution for some models from an “other” formation channel, without explicitly stating whether these systems experience a CE phase. After inspecting the underlying model data, we verified that all systems classified as “other” do undergo a CE episode (but, for example, do not undergo SMT prior to the first SN hence the ‘other’ classification). Accordingly, we assign these systems to the “with CE” category in the Level 1 classification, and label them as “other with CE” in the Level 2 classification.

Briel et al. (2023) [BRI22] — *Briel et al. (2023)* employ the BPASS population synthesis framework to investigate the formation of BBH mergers, with a particular focus on distinguishing systems formed through the CE and SMT channels. Formation-channel contributions for their fiducial model are presented in Figure 7 of their study. For the Level 1 classification, we assign all systems that experience at least one CE episode (i.e., those labeled as undergoing CEE in Figure 7) to the “with CE” category, and classify the remaining systems as “without CE”. For the Level 2 classification, we adopt the

authors’ more detailed taxonomy, which distinguishes systems according to whether CE, SMT, or no mass transfer occurs in the primary and/or secondary component. We obtain the total BBH merger rate for the fiducial model from Table 1 of *Briel et al. (2023)*, which reports $\mathcal{R}_{\text{BH-BH}} = 6.5 \text{ Gpc}^{-3} \text{ yr}^{-1}$. The authors also explore variations in the supernova-engine prescription, adopting the “DELAYED” and “RAPID” remnant mass prescription models of *Fryer et al. (2012)*. As these results are not explicitly tabulated in the published article, we obtained the corresponding merger rates and formation-channel contributions via private communication. The resulting total BBH merger rates are $\mathcal{R}_{\text{BH-BH}} = 57.1 \text{ Gpc}^{-3} \text{ yr}^{-1}$ for the “RAPID” model and $\mathcal{R}_{\text{BH-BH}} = 48.0 \text{ Gpc}^{-3} \text{ yr}^{-1}$ for the “DELAYED” model. All data derived from these models are included in our accompanying data release.

Broekgaarden et al. (2021) — *Broekgaarden et al. (2021)* employ an earlier version of the COMPAS population synthesis framework and explore a range of star-formation history and stellar evolution assumptions, yielding 560 distinct binary population-synthesis realizations. For our analysis, we use the BBH, BHNS, and BNS merger rates from their 20 primary population-synthesis model variations. We extract the corresponding formation-channel fractions and total merger rates assuming the star-formation history model $\text{xyz} = 312$, which best matches the gravitational-wave observed merger rates (*Broekgaarden et al. 2022a*). These data are obtained from the authors’ publicly available repository.⁴ Unless stated otherwise, we foremost include the $\text{xyz} = 312$ model, as the other cosmic star formation model variations probe uncertainties in the star-formation history rather than in binary evolution. For the detailed comparisons shown in Figure 9, however, we include the full set of model variations including variations in the cosmic star formation history model. It can be seen that the choice of star formation history model impact the formation channel contribution slightly (changing it by 20%) and especially impacts the total (BBH) rate. The authors classify binaries into several formation channels. Systems that undergo a CE phase are subdivided into the *classic CE* channel, characterized by stable mass transfer prior to the formation of the first compact object followed by a CE phase after the SN (i.e. the ‘SMT < SN < CE’ channel) and two additional CE-related channels: the *single-core CE* and *double-core CE* channels. These distinguish whether one or both stars, respectively, have evolved off the main sequence (and thus developed a core-envelope structure) before the first compact object forms. The stable mass-transfer channel comprises systems that undergo stable mass transfer both before and after the formation of the first compact object, analogous to the ‘SMT < SN < SMT’ channel. Systems that experience stable mass transfer only before or only after compact-object formation are grouped into the *other* category, corresponding to ‘other no CE’ in Fig. 1. We adopt these definitions for our Level 2 classification. For the Level 1 categorization, we combine the three CE-related channels (classic, single-core, and double-core CE) into a single “with CE” group and classify all remaining systems as “without CE”.

Dorozsmai & Toonen (2024) — *Dorozsmai & Toonen (2024)* use the SEBA population-synthesis code to study the formation channels, properties, and merger rates of BBH systems,

⁴ https://github.com/FloorBroekgaarden/DCO_FormationChannels

exploring the impact of variations in several uncertain physical parameters, including the mass-transfer stability criterion, mass-transfer efficiency, specific angular-momentum loss, and stellar-wind strength. The authors consider three principal formation channels: a stable mass-transfer channel and two subtypes of the common-envelope channel, distinguished by the evolutionary state of the donor star during the second mass-transfer episode. In subtype 5b, the donor has a radiative envelope (rCEE), whereas in subtype 5c the donor has a deep convective envelope (cCEE). We extract the corresponding formation-channel rates from the legends of their figures. For our Level 2 classification, we treat channels 5b and 5c as distinct CE subchannels, but for the Level 1 classification we combine them into a single “with CE” category.

Hendriks et al. (2023) — *Hendriks et al. (2023)* use the `BINARY_C` population-synthesis code to model BBH mergers and to investigate how different prescriptions for pair-instability supernovae affect the resulting merger population. From their Figure 4 and the accompanying text, we extract the total BBH merger rate and the relative contributions of the different formation channels for the fiducial model. The reported total BBH merger rate, obtained by combining the stable mass transfer (SMT) and common-envelope (CE) channels, is approximately $25 \text{ Gpc}^{-3} \text{ yr}^{-1}$, with a contribution of about $10 \text{ Gpc}^{-3} \text{ yr}^{-1}$ from the SMT channel. We therefore classify their SMT channel as “without CE” and assign the remaining contribution to the “with CE” category at Level 1.

Li et al. (2025a) — *Li et al. (2025a)* study the formation channels, mass distributions, and merger rates of BBH systems using the binary population-synthesis code `MOBSE`, explicitly incorporating chemically homogeneous evolution (CHE) alongside the more traditional CE and SMT isolated-binary evolution pathways. The authors explore a grid of population-synthesis models in which key uncertain physical parameters are varied, including the angular-momentum loss prescription during non-conservative mass transfer, the CE ejection efficiency α_{CE} , the Wolf-Rayet wind mass-loss multiplier, the natal kick velocity dispersion, and the mass-transfer stability criterion. Across all models, BBH mergers are classified into three mutually exclusive formation channels: (i) a CHE channel, in which both stars evolve chemically homogeneously without undergoing any Roche-lobe overflow; (ii) a common-envelope (CE) channel, in which at least one dynamically unstable mass-transfer episode leads to envelope ejection; and (iii) a stable mass-transfer (SMT) channel, in which both interaction phases proceed stably without a CE phase. We extract the intrinsic merger rates at redshift $z = 0.2$ for each formation channel directly from the legends of Figure 4 in *Li et al. (2025a)*, which report channel-specific merger rates (in units of $\text{Gpc}^{-3} \text{ yr}^{-1}$) for each model in the parameter study. For our analysis, we treat CHE, CE, and SMT as three distinct Level 1 formation channels; no further subdivision of the CE or SMT channels is applied. The total merger rate for each model is taken as the sum of the three reported channel contributions, consistent with the authors’ definitions⁵.

Olejak et al. (2021) — *Olejak et al. (2021)* use the `STARTRACK` population-synthesis code to investigate the formation chan-

nels of BBH mergers within the isolated binary evolution scenario. They consider three `STARTRACK` models: M380.B (the fiducial model), M480.B (which adopts a revised criterion for CE development), and M481.B (which further modifies this criterion via an additional stability flag). Model M480.B introduces a revised CE onset criterion that requires four simultaneous conditions to be satisfied for a CE phase to occur. In particular, this prescription allows SMT to proceed for donors within a specified radius range even for systems with strongly unequal mass ratios at the onset of mass transfer (up to $q < 8$), based on the grid of donor-radius ranges presented by *Pavlovskii et al. (2017)*. The authors extend this grid with a fitted prescription applicable to a broader range of masses and compact-object accretors. Model M481.B is identical to M480.B, but includes an additional “MT switch” flag that forces systems undergoing stable RLOF to enter a CE phase if the donor radius exceeds twice its Roche-lobe radius. An overview of these assumptions is provided in Figure 9 of *Olejak et al. (2021)*. We extract formation-channel contributions for the three models from Table 4 (BBH) and Table 5 (BHNS) of *Olejak et al. (2021)*. For BBHs, we interpret channel 1 in Table 4 as the ‘SMT<SN<SMT’ channel (SMT before and after the formation of the first compact object). Channels 2, 3, and 4 are grouped into the classic CE channel (‘SMT<SN<CE’), as they all involve SMT prior to the first supernova (SN1) followed by a CE phase. Channel 5 corresponds to the double-core CE channel (DCCE). Channel 6 is assigned to the “other” category, as no detailed evolutionary pathway is specified. For BHNS systems, we group channels 1, 3, and 4 into the classic CE channel (‘SMT<SN<CE’), since they involve SMT prior to SN1 followed by a CE phase. Channel 2 is classified as the only SMT channel (‘SMT<SN<SMT’). Channel 5 is assigned to the “other” category due to the lack of further specification. For the Level 1 classification, we assign all channels involving at least one CE episode to the “with CE” category, and all channels involving only stable mass transfer to the “without CE” category.

Pellouin et al. (2025) — *Pellouin et al. (2025)* use `COSMIC` to study the contribution of different isolated binary evolution formation channels to the formation of BNS. They consider three main channels. First, their ‘standard channel’ corresponds to the classic CE channel (SMT<SN<CE in Figure 1). Second, their *equal mass-ratio at ZAMS channel* is extremely similar to the double-core CE channel in Figure 1 as the stars are so similar in mass on ZAMS that they evolve simultaneously and become a giant at the same time leading to a double-core CE. Third, they consider an accretion-induced collapse (AIC) channel, in which one neutron star forms from a white dwarf that collapses after mass transfer. We extract the BNS formation-channel rates from the $z \approx 0$ results shown in their Figure 10 and convert them to units of $\text{Gpc}^{-3} \text{ yr}^{-1}$. This yields a total local BNS merger rate of approximately $298 \text{ Gpc}^{-3} \text{ yr}^{-1}$, with contributions of $287 \text{ Gpc}^{-3} \text{ yr}^{-1}$ from the double-core CE channel and $11 \text{ Gpc}^{-3} \text{ yr}^{-1}$ from the AIC channel. Following the authors’ statement that they do not find any BNS forming without at least one CE phase, we classify the AIC channel as an “other” channel *with CE* in our taxonomy. A 100% of their BNS form with at least one CE, meaning that the contribution from channels without a CE is zero. The authors find that the classical CE channel (their “standard” channel) contributes negligibly to the *local* BNS merger rate. This is because, in their simulations, BNS formed through the standard CE channel do not form efficiently at higher metal-

⁵ Note that due to small rounding errors, the total BBH merger rates reported in each figure in *Li et al. (2025a)* is not the sum of the subchannel rates. We take the sum of the subchannel rates for that reason as the total BBH rate to be consistent throughout.

licities, as they tend to produce systems with inspiral times longer than a Hubble time. Different assumptions about mass transfer and CE evolution could therefore substantially modify this result.

Romagnolo et al. (2023) — *Romagnolo et al. (2023)* use StarTrack to investigate how imposing an upper limit on stellar radial expansion affects the formation of BNS, BHNS, and BBH mergers. We obtain their BBH, BHNS, and BNS merger rates and associated formation-channel fractions via private communication, as the channel breakdown is not reported in the published paper. The total merger rates differ slightly (few percent) from those listed in their Table 3 due to statistical sampling noise in the authors’ merger-rate estimation. For the formation channels, *Romagnolo et al. (2023)* largely adopt the same classification scheme as *Broekgaarden et al. (2022a)*, with a few notable differences. In particular, their “only stable mass transfer” channel includes systems with no mass transfer either before or after the first supernova (though this is a negligible fraction). We therefore classify this entire category as a “no CE” channel, without subdividing into OSMT subchannels. They further distinguish a classic channel, a double-core CE channel, and a single-core CE channel. Although their single-core CE definition does not formally require the CE to occur prior to the first supernova, in practice this is true for the vast majority of systems, as post-SN1 systems are otherwise likely to disrupt. We therefore map their CE channels onto our “single-core” and “double-core” subchannels. Finally, due to their channel definitions, no systems are assigned to an “other” category in their classification.

Romagnolo et al. (2025) — *Romagnolo et al. (2025)* use StarTrack to investigate how different CE survival criteria, based on the presence of a convective envelope, affect the formation of compact-object mergers. We adopt their reported BBH, BHNS, and BNS merger rates and formation channels classification from private communication, as the latter are not explicitly presented in the paper. The total merger rates used here differ slightly from those in their Table 2 due to statistical sampling noise in the authors’ merger-rate calculations. The models in this study explore alternative prescriptions for unsuccessful (“pessimistic”) CE evolution. The fiducial model assumes the standard SSE-based prescription in which Hertzsprung-gap (HG) donors are not allowed to survive a CE phase. Two alternative prescriptions replace this stellar-type criterion with envelope-structure criteria derived from detailed stellar-evolution calculations. The first, $M_{\alpha, \text{ML } 1.5}$, uses fits from *Klencki et al. (2021b)* to determine whether the donor envelope is convective. The second, $M_{\alpha, \text{ML } 1.82, \text{MLpp}}$, employs new fits based on MESA models provided by the authors. For each of these prescriptions, the authors additionally consider a variant in which the maximum stellar radius in SSE is artificially capped (RMAX). We retrieve and classify formation channels for all these models following the same procedure described in the previous subsection for *Romagnolo et al. (2023)*.

Román-Garza et al. (2021a) — *Román-Garza et al. (2021a)* compute BBH and BHNS merger rates by coupling detailed binary stellar evolution simulations performed with MESA to the rapid population synthesis code COSMIC via an early version of the POSYDON framework. They compute rates for nine population-synthesis model variants that differ in their supernova remnant-mass prescriptions and natal-kick velocity assumptions. We extract their BBH and BHNS merger rates

from their Table 3, adopting the combined “CE+SMT” rate as the total merger rate and the individual “CE” and “SMT” contributions as proxies for the classic CE channel (level I) and the stable mass-transfer channel (level II), respectively. The authors do not further subdivide these channels.

Sgalletta et al. (2025) — *Sgalletta et al. (2025)* use SEVN to compute BBH merger rates, with particular emphasis on the impact from different cosmic star formation history assumptions. They use the same formation channels as considered in *Broekgaarden et al. (2022a, 2021)* based on only stable mass transfer, the classic CE channel, and the single-core CE and double-core CE channels. We retrieve their formation channel rates from digitalizing Figure 6, which shows the contributions of the formation channels to the BBH merger rate. We take the rate at redshift $z \sim 0$ using a plot digitizer. Note that in the Figure it is clear that the classic CE channel dominates the formation of BBH mergers in all model variations. Interestingly, they find that the OSMT channel can especially contribute (and become dominant) at high redshifts $z \gtrsim 4$ for a few of the submodels ($\alpha = 3$ and $\alpha = 5$).

Shao & Li (2021b) — use an updated version of BSE to investigate the impact of different supernova prescriptions on the formation and merger rates of compact-object binaries, including BBH and BHNS systems. We extract the intrinsic merger rates for BBH and BHNS systems from their Table 1 for the three supernova models considered (rapid, delayed, and stochastic prescriptions). The corresponding formation channel breakdown into stable mass transfer and common-envelope pathways is obtained via private communication, as these are not explicitly tabulated in the paper.

van Son et al. (2022) — *van Son et al. (2022)* use COMPAS to compute BBH merger rates, with particular emphasis on the relative contributions of the only stable mass-transfer (OSMT) and common-envelope (CE) formation channels. We extract the formation-channel fractions from the right-hand panel of their Figure 4, adopting the values in the lowest-redshift bin. Although this figure presents the *detectable* merger rate, it assumes a perfect detector and therefore corresponds to the underlying astrophysical rate at low redshift. This yields an OSMT fraction of $\simeq 0.37$ and a CE fraction of $\simeq 0.63$ in the lowest-redshift bin. We adopt their total local merger rate of $R_{\text{BBH}} \simeq 73 \text{ Gpc}^{-3} \text{ yr}^{-1}$ at $z \approx 0$ from the main text (Section 7.3).

van Son et al. (2023) — *van Son et al. (2023)* use COMPAS to study how different metallicity-dependent star-formation histories affect the BBH merger rate in both the only stable mass-transfer (OSMT) and common-envelope (CE) channels. We extract their BBH merger rates and channel contributions from their Figures 3 and 4, summing the stable channel and CE channel components to obtain the total rate. Unless stated otherwise, we adopt the fiducial model, as the model variations primarily probe uncertainties in the star-formation history rather than in binary evolution. For the detailed comparisons shown in Figure 9, however, we include the full set of model variations including variations in the cosmic star formation history model.

Xing et al. (2024a) — *Xing et al. (2024a)* use the POSYDON binary population synthesis framework to study the formation of BHNS mergers across a broad range of metallicities. We extract the total BHNS merger rates from Table 1, using the reported $\mathcal{R}_{\text{NSBH}}$ values from the published version

of the paper.⁶ This yields four models: a fiducial model, a variant with $\alpha_{\text{CE}} = 2$, and two variants with modified core-collapse kick dispersions σ_{cc} . The authors additionally explore versions of each model with the maximum neutron-star mass reduced from the default $2.5 M_{\odot}$ to $2.0 M_{\odot}$. However, formation-channel contributions for these submodels are not tabulated, and inspection of Figure 3 indicates that this variation has a negligible effect on the relative channel fractions. We therefore exclude these submodels from our analysis.

APPENDIX C: ADDITIONAL STUDIES NOT INCLUDED

In this section, we briefly summarize additional simulations from the literature that report formation channel contributions within the isolated binary evolution paradigm. These studies are not included in our main comparative analysis because they do not meet the criteria required for a consistent cross-study comparison. In particular, many of these works either consider only a single metallicity, do not provide intrinsic merger rates, or do not provide the quantitative breakdowns of formation channels necessary to map onto our unified taxonomy. Nevertheless, we include a brief discussion here for completeness and to highlight their qualitative contributions to the field. See also Section 4 for more details.

Andrews et al. (2015) — *Andrews et al. (2015)* perform a large population-synthesis study of BNS using *StarTrack*, comparing their model predictions to the observed Galactic BNS population in the orbital period-eccentricity plane. They find that all (short-period) BNS systems form through binary evolution channels that include a CE phase following the first SN. They identify three dominant formation channels, distinguished by (i) whether the system undergoes stable Case BB mass transfer after the CE phase and (ii) whether the second neutron star forms via an iron core-collapse or an electron-capture SN. Channels that include Case BB mass transfer produce systematically tighter binaries, while channels involving electron-capture supernovae, and hence low natal kicks, give rise to systems with lower eccentricities. At (approximately) solar metallicity, they find that channels involving Case BB mass transfer dominate the formation of BNS systems. Their results further highlight that electron-capture supernovae are required to reproduce the observed population of low-eccentricity systems, and they highlight the coupled role of post-CE mass transfer and natal kick physics in shaping the observable DNS population. We do not include this study in our formation-channel comparison because it does not report formation-channel fractions for intrinsic merger rates and focuses instead on a fixed metallicity.

Banerjee & Olejak (2024) — *Banerjee & Olejak (2024)* use the rapid population-synthesis code *BSE* to model the properties of BBH mergers formed through isolated binary evolution, with a focus on the spin-mass ratio correlation. They explore both CE and SMT pathways by constructing models that are dominated by one channel or the other by changing the critical mass ratio (cf. *Olejak et al. 2021*). In particular, their default models yield $\sim 90\%$ of BBH mergers through CE evolution, while models adopting extreme mass-transfer stability assumptions (e.g., $q_{\text{crit}} = 8$) instead produce $\sim 90\%$ of systems through SMT. We do not include this study in our formation-channel comparison because it does not report formation-channel fractions for intrinsic merger rates based on self-consistent model

populations. Instead the analysis focuses on normalized distributions at fixed metallicities rather than intrinsic rates.

Belczynski et al. (2002) — *Belczynski et al. (2002)* use the *StarTrack* population-synthesis code to model the formation of BBH, BHNS, and BNS systems at solar metallicity $Z = 0.02$. They present representative formation pathways in their Table 3, classifying channels based on the number and type of common-envelope (CE) episodes, including single-core CE, double-core CE, and CE with significant accretion during the CE (‘hyper-critical accretion’). In these models, merging systems predominantly form through channels involving at least one CE phase, highlighting the importance of CE evolution in producing sufficiently tight binaries. We do not include this study in our formation-channel overview because it does not report systematic formation-channel fractions for intrinsic merger rates across its model grid.

Belczynski et al. (2022) — *Belczynski et al. (2022)* investigate BBH formation using the *StarTrack* population-synthesis code, exploring variations in the initial mass function and supernova engine prescriptions (delayed and rapid). They focus on the relative contributions of SMT and CE channels, showing that SMT-formed systems tend to produce slightly more massive BBH mergers than those formed through CE evolution. They further demonstrate that assumptions about the onset and development of CE evolution significantly affect both the dominant formation pathway (i.e., CE versus Roche-lobe overflow) and the resulting BBH mass distribution. In particular, they find that BBH formation at low metallicities ($Z \lesssim 0.4 Z_{\odot}$) is dominated by the CE channel (with a factor $\sim 3\times$ higher formation efficiencies), while at higher metallicities $Z \gtrsim 0.4 Z_{\odot}$ the contributions from SMT and CE become comparable. We do not include this study in our formation-channel overview because the absence of a quantitative decomposition into channel contributions as a function of merger rate prevents a direct comparison with the framework adopted in this work. We note that the models considered are the same as those presented in *Olejak et al. (2021)* which are included.

Briel et al. (2026) — *Briel et al. (2026)* present a full ZAMS-to-merger population study of BBH formation through the SMT channel using *POSYDON* with detailed binary grids across eight metallicities. They discuss details of the SMT channel, but do not present intrinsic merger rates, which is why we do not include this study in our key results. They find that SMT BBH mergers arise predominantly from short-period progenitors ($P_{\text{ZAMS}} \lesssim 10$ days) in which *both* the initial STAR+STAR interaction and the later STAR+BH interaction occur while the donor is on the main sequence (i.e., Case A in both phases), with only a limited contribution from wider Case B/C systems. In their fiducial (zero natal-kick) models, SMT does not produce BBH mergers above $Z \gtrsim 0.2 Z_{\odot}$ because wind-driven mass loss widens the orbit, preventing merger within a Hubble time. The resulting SMT population favors near-unity mass ratios (via mass-ratio reversal) and shows a bimodal χ_{eff} distribution with peaks at $\chi_{\text{eff}} = 0$ and $\chi_{\text{eff}} \simeq 0.1\text{--}0.15$; importantly, the $\chi_{\text{eff}} = 0$ peak appears only at low metallicity ($Z < 0.01 Z_{\odot}$), while at higher Z only the $\chi_{\text{eff}} \sim 0.15$ peak remains. They also find characteristically long delay times: even at $Z = 10^{-4} Z_{\odot}$ the minimum delay time is $\sim 0.2\text{ Gyr}$, and the delay-time distribution shifts to even longer times with increasing metallicity, implying an intrinsic SMT BBH merger-rate density that rises toward the present day ($z = 0$). Finally, introducing a mass-scaled natal-kick prescription

⁶ Note that their earlier arXiv version reports slightly different NSBH merger rates.

leaves the main SMT population largely intact but can add eccentricity at second collapse, enabling an additional low-mass, unequal-mass-ratio sub-population (and allowing SMT mergers at $Z > 0.2, Z_{\odot}$) that would otherwise be too wide to merge within a Hubble time.

Chattaraj et al. (2026) — *Chattaraj et al. (2026)* use detailed binary-evolution models with POSYDON to study the formation of BNS systems at solar metallicity, focusing on the role of CE evolution and its dependence on donor structure and evolutionary stage. In particular, they find that BNS almost always form through two distinct CE subchannels split by Case B and Case C mass transfer that lead to different outcomes in terms of whether binaries merge within a Hubble time, and explore how assumptions about envelope ejection efficiency and supernova physics affect BNS merger rates and properties. They find that $\geq 95\%$ of all BNS systems form through these two CE subchannels and that such systems always undergo a stable case BB mass transfer phase and an ultra-stripped SN after the CE phase. Up to 5% of BNS form mostly through a non-interacting channel, and more rare ($\lesssim 1\%$) a channel with only stable mass transfer, and the double-core CE channel ($\lesssim 2\%$). We do not include their results in our formation-channel overview because the study does not report formation-channel fractions and only has intrinsic merger rates based on a single (solar) metallicity, making it difficult to consistently compare with the studies included in our compilation.

Chruślińska et al. (2018) — *Chruślińska et al. (2018)* use StarTrack to investigate the formation channels of BNSs, focusing primarily on individual metallicity environments. For solar metallicity, they find that the dominant formation pathway is the “classic” CE channel, which typically includes a subsequent phase of stable mass transfer onto the compact object (i.e., Case BB mass transfer). This model is an order of magnitude more common than any other subchannels they identify. They also identify subdominant channels, including formation through accretion-induced collapse, where stable mass transfer prior to the first supernova leads to the formation of a white dwarf that subsequently collapses into a neutron star. This pathway can contribute at most 10% of all BNS but most of them (90%) require CE with HG donor before the formation of the second NS. So combined, the CE channel likely forms at least $\geq 95\%$, if not more, of their BNS at solar metallicity, which we add to Table 1. Additionally, they discuss alternative channels involving two CE phases after the first supernova, as well as channels with CE evolution prior to the first supernova, although the latter are found to be rare and at least an order of magnitude less efficient than the classic channel. We do not include this study in our main comparative analysis because the results are primarily presented for individual metallicities (e.g., solar versus low metallicity), and the work does not provide a systematic breakdown of formation channel contributions integrated over cosmic star-formation and metallicity evolution. In particular, the relative contributions of different channels to the cosmological BNS merger rate are not quantified in a way that can be consistently mapped onto our unified taxonomy.

Dall’Amico et al. (2025) — *Dall’Amico et al. (2025)* investigate the formation of BBH, BHNS, and BNS mergers through accretion-induced chemically homogeneous evolution (CHE). They find that, in standard binary evolution channels, a CE phase is typically essential to shrink the orbit and enable mergers, although this is not quantified within their framework. In

contrast, models including accretion-induced CHE produce binaries that often avoid CE evolution altogether, as the secondary stars remain compact and do not fill their Roche lobes, thereby suppressing mass transfer phases. However, the majority of these CHE systems have inspiral times exceeding a Hubble time, resulting in a strongly reduced number of merging double compact objects. We do not include this study in our formation-channel comparison because it does not provide a systematic breakdown of channel contributions within a simulation to the merging population that can be consistently mapped onto our unified taxonomy.

Dominik et al. (2012, 2013) — *Dominik et al. (2013, 2013)* present an extensive suite of population-synthesis simulations using StarTrack, exploring a wide range of binary-evolution assumptions. They identify representative formation pathways (Tables 4 and 5 of *Dominik et al. 2012*) at fixed metallicities, highlighting characteristic evolutionary sequences for BBH, BHNS, and BNS systems. Across their models, merging systems overwhelmingly originate from channels that involve at least one CE phase. Quantitatively, at solar metallicity, at least 94.2% (BNS), 97.2% (BHNS), and 98.9% (BBH) of merging systems form through channels that include a CE phase, with the remaining fraction categorized as “other” (i.e., not assigned to a specific formation pathway). At subsolar metallicity (0.1, $Z_{\odot}$), the CE fraction remains similarly high: $\geq 91.8\%$ (BNS), 93.6% (BHNS), and 96% (BBH). We add these fractions to Table 1. The identified channels span a range of CE configurations, including CE episodes prior to the first supernova, double-core CE, pathways involving multiple CE phases, and involving accretion-induced collapse (AIC). Notably, none of the reported channels correspond to purely stable mass transfer sequences or to systems that avoid mass transfer altogether (although these could contribute to ‘other’). We do not include these studies in our formation-channel overview because they do not report quantitative formation-channel fractions for intrinsic merger rates across their model grid.

Iorio et al. (2023b) — *Iorio et al. (2023b)* explore the formation and properties of BBH, BHNS, and BNS mergers using an extensive suite of SEVN simulations spanning a wide range of physical assumptions. They present a detailed breakdown of formation-channel contributions to the formation efficiency as a function of metallicity (their Figures 14 and 16), adopting a Level 2 classification that distinguishes between classic CE (SMT+CE), OSMT (SMT+SMT), single-core and double-core CE, and an “other” category.

We do not include these results in our main overview, as the authors do not report formation-channel contributions in terms of intrinsic merger rates. Moreover, their publicly available post-processing scripts are designed to reproduce figures directly rather than to enable re-analysis of the underlying data, which complicates a consistent extraction of channel fractions within our framework. Instead, we incorporate their metallicity-dependent formation-channel trends using the publicly available data from *Iorio et al. (2023a)* in Figure 14.

Across their broad model variations—including changes in α_{CE} , chemically homogeneous evolution, supernova natal kicks, remnant-mass prescriptions, and envelope binding energies—they find a robust and consistent result: BNS mergers form exclusively through channels involving at least one CE phase. In contrast, BHNS and BBH systems exhibit a strong dependence on model assumptions and metallicity, with contributions from both CE and non-CE channels.

Kruckow et al. (2018) — *Kruckow et al. (2018)* used the COMBINE framework to study the formation of gravitational-wave sources from isolated binary evolution. We exclude their results from our cosmological rate comparison because their analysis is limited to a single metallicity ($Z = 0.0088$) and includes all compact-object binaries, including systems with delay times exceeding a Hubble time. Importantly, they state that their formation-channel distribution of merging systems differs significantly from that of the full formation population for which they present quantified formation channel breakdowns in their study. While their Appendix C indicates that BBHs predominantly form without mass transfer (channel D) in the overall population, the subset of merging systems is instead dominated by channels involving stable mass transfer (channel A) and CE evolution (channel C). For merging BHNS systems, channels involving stable mass transfer (channel A) and classical CE evolution (channels B/C) appear to dominate. In the case of binary neutron star mergers, the classical CE channel (B/C) is the primary formation pathway. The authors do not provide precise quantitative breakdowns for the merging population, and we therefore treat these conclusions as qualitative.

Mesticelli et al. (2025) — *Mesticelli et al. (2025)* model the formation of BHNS and BNS mergers from metal-free Population III and early Population II progenitors using the SEVN binary population-synthesis code. They compute merger-rate densities by convolving their merger catalogs with metallicity-dependent cosmic star-formation histories using CosmoRate. Their analysis focuses exclusively on low-metallicity environments, adopting $Z \sim 10^{-11}$ for Population III and $Z \sim 10^{-4}$ for Population II star models.

The authors define formation channels (their Table 2) based on whether systems undergo stable mass transfer and/or CE evolution relative to the formation of the first compact remnant: Ch. 0 (no interaction), Ch. 1 (SMT+CE), Ch. 2 (SMT+SMT), Ch. 3 (single-core CE), Ch. 4 (double-core CE), and Ch. 5 (interaction only after the first remnant; CE/SMT not specified). Because the study is restricted to fixed low-metallicity simulations and does not provide a quantitative breakdown of formation channels for the intrinsic merger population, we do not include it in our main results. However, the authors report channel contributions to the merger formation efficiency η (Tables 4 and 6), which we include in Appendix D. For consistency with our classification, we group Channels 1, 3, and 4 as ‘with CE’, Channels 0 and 2 as ‘without CE’, and Channel 5 as ‘other’ (CE not specified). Under this mapping, BHNS systems show a substantial contribution from both pathways: 46.2% (36%) form through channels with CE and 46.9% (55.9%) without CE for Population II (Population III), respectively, with the remaining 6.7% (8%) in the “other” category. In contrast, BNS formation is overwhelmingly dominated by CE evolution: 99.3% (91.9%) of Population II (Population III) systems form through channels involving at least one CE phase, with the remainder attributed to the “other” category, which likely also includes CE evolution. No contribution from “without CE” channels is reported for BNS systems.

Neijssel et al. (2019) — *Neijssel et al. (2019)* use COMPAS to model the rates and properties of detectable BBH, BHNS, and BNS populations under different cosmic star-formation histories. They find that the detectable BBH population is strongly dominated by channels involving only stable mass

transfer (i.e., without a CE phase), contributing nearly 80% of mergers (their Figure 12), compared to 15.5% from the classic CE channel, 0.3% from double-core CE, and the remainder from other pathways.

We do not include this work in our main overview, as formation channel fractions are reported only for the detectable population rather than intrinsic merger rates. In addition, more recent COMPAS studies with comparable fiducial assumptions provide directly comparable intrinsic channel breakdowns and are therefore prioritized in our compilation.

However, *Neijssel et al. (2019)* also present formation channel contributions to the BBH formation efficiency as a function of metallicity (their Figure 1), which we include in Figure 14 using a web plot digitalizer. From these results, they infer that $\sim 30\%$ – 50% of BBH systems form through channels without CE, with the remainder forming through CE-involving pathways. This balance shows only a weak dependence on metallicity, with a modest decrease in the contribution from without-CE channels toward higher metallicity.

Olejak et al. (2024) — *Olejak et al. (2024)* use StarTrack to investigate the spin and mass ratio properties of BBH population from the CE and SMT channels. We do not include their study as they do not report BBH merger rates or formation channel contributions. Their models are similar to the ones used in *Olejak et al. (2021)* which are included in our main overview. The default model in *Olejak et al. (2024)* finds a dominant contribution from the SMT channel. For the same reason we also do not include (*Olejak & Belczynski 2021*).

Oh et al. (2023) — *Oh et al. (2023)* use the rapid population-synthesis code COSMIC to investigate the evolutionary pathways leading to asymmetric BBH and BHNS mergers, with a particular focus on the role of natal kicks and the timing of mass-transfer phases (e.g., whether systems undergo CE before or after the first supernova). They identify several formation channels and show that highly asymmetric systems are preferentially produced through channels involving CE before the first supernova. We do not include this study in our formation-channel comparison because it does not report formation-channel fractions for intrinsic merger rates, and therefore cannot be consistently incorporated into our quantitative channel accounting. In addition, the simulations are performed at only three discrete metallicities, which limits direct comparison with studies that explore a broader metallicity range or integrate over cosmic star formation.

Riley et al. (2021) — *Riley et al. (2021)* implement CHE in the rapid population-synthesis code COMPAS and, under a consistent set of assumptions, compute BBH merger rates including the redshift-dependent merger-rate density and detectability for CHE and “conventional” isolated binaries, finding CHE could contribute up to $\sim 70\%$ of isolated-binary BBH detections. Their CHE systems do not undergo mass transfer that could harden the binary except for RLOF at ZAMS, which is assumed to be an over-contact phase leading to mass equilibration and CHE. We do not include their rates in our formation-channel overview because, while they model both dynamically stable mass transfer and common-envelope evolution in the non-CHE channel, they do not report a quantitative breakdown of what fraction of their BBH progenitors experienced CE versus only stable MT, which is the split we need for our channel accounting.

Stevance et al. (2023)— *Stevance et al. (2023)* use BPASS to explore isolated binary evolution pathways leading specifically to BNS systems analogous to GW170817. Because the study focuses on reproducing a single event rather than providing a comprehensive formation-channel breakdown for the full BNS population, we do not include it in our main analysis. They find that all GW170817-like systems in their models undergo at least one CE phase, with many experiencing two or even three CE episodes (see their Figure 2). The identified pathways include variants of the classic CE channel (stable mass transfer prior to the first supernova followed by a CE phase), but the dominant contribution arises from systems that undergo a CE phase both before and after the formation of the first compact object, with or without additional stable mass transfer. We note that BPASS adopts a different treatment of CE evolution compared to most population-synthesis codes and is based on the STARS stellar evolution tracks, which may contribute to differences in inferred channel contributions. Interestingly, the authors still find that all formation channels for GW170817 include at least one CE phase consistent with other work.

Tanaka et al. (2023)— *Tanaka et al. (2023)* use the `binary_c` population-synthesis code to study BNS merger formation across a wide range of model assumptions, varying the CE efficiency parameter α , the core-collapse kick dispersion σ_{cc} , and metallicity. Their simulations are limited to three discrete metallicities and do not provide a detailed breakdown of formation-channel contributions to the intrinsic merger rates, and we therefore do not include this study in our main analysis. Nonetheless, the authors report that a large majority of BNS systems—between 81% and 100%—undergo at least one CE phase involving a neutron star during their evolution (but they don’t specify whether this is for specific metallicity grids or for the entire population). They also identify a small contribution from accretion-induced collapse, with up to 0.7% of all BNS systems and up to 1.7% of merging BNS systems forming one of the neutron stars via a white dwarf accretion-induced collapse channel. These fractions are reported for individual simulations and are not weighted to represent the intrinsic merging population.

Ugolini (2025)— *Ugolini (2025)* use SEVN to evolve a large grid of isolated binary simulations of BBH systems, exploring how variations in stellar evolution and binary interaction physics (including CE efficiency, supernova prescriptions, and the initial mass function) shape the resulting black hole mass distribution. In particular, they demonstrate that both stable mass transfer and CE evolution can contribute to features in the BBH mass spectrum, such as the $\sim 35 M_{\odot}$ peak, with their relative importance depending sensitively on the assumed CE efficiency. The CE channel seems to produce the low mass BH peak below $\lesssim 10 M_{\odot}$, but this is based on samples from the simulation (not intrinsic rate). We do not include this study in our comparative analysis because it is a preliminary work and does not calculate intrinsic BBH merger rates required for our comparison.

van Son et al. (2022)— *van Son et al. (2022)* use the rapid population-synthesis code COMPAS to investigate the formation of BBH and BHNS mergers through the stable mass transfer (SMT) channel, with a particular focus on understanding the origin of features in the low-mass end of the black hole mass distribution ($\lesssim 10 M_{\odot}$). They derive an analytical expression for the minimum primary black hole mass set by stability criteria during binary evolution and explore how variations in

key physical assumptions (e.g., mass-transfer efficiency, core mass fraction, and supernova prescriptions) impact the resulting merger rates and mass distributions. Their models are explicitly constructed to include only systems that undergo exclusively stable mass transfer, thereby isolating the SMT channel and excluding binaries that experience a CE phase or other evolutionary pathways. We do not include this study in our formation-channel comparison because it does not provide formation-channel fractions or total intrinsic merger rates across multiple channels. Instead, the reported rates correspond solely to the SMT channel by design, and therefore cannot be directly compared to studies that model the relative contributions of different formation pathways within a self-consistent population.

van Son et al. (2024)— *van Son et al. (2024)* present an extensive suite of COMPAS simulations to investigate the formation efficiency of BBH, BHNS, and BNS mergers as a function of metallicity. They report formation-channel rates in terms of formation efficiencies (their Figure 4), which we reproduce using the publicly available data from (*van Son 2024*) and include in our Figure 14. Their results show that BNS systems form almost exclusively through channels involving a CE phase across all metallicities, BHNS systems are predominantly formed via CE channels, and BBH systems can receive a substantial contribution from channels without CE, including CHE. We do not include this study in our main tables and figures, as it does not provide a consistent breakdown of formation channels in terms of intrinsic merger rates.

Vigna-Gómez et al. (2018)— *Vigna-Gómez et al. (2018)* use COMPAS simulations at a single (solar-like) metallicity to investigate the rates and properties of BNS mergers. In their fiducial model ($Z = 0.0142$), they identify two dominant formation pathways: the classic channel, accounting for $\sim 70\%$ of systems, and the double-core CE channel, contributing $\sim 21\%$. The remaining $\sim 9\%$ of systems form through a range of alternative pathways (likely still involving CE phases). Their total contribution from channels with CE is thus at least $\gtrsim 91\%$ at $Z = 0.0142$. Because the analysis is restricted to a single metallicity and does not incorporate cosmological weighting, we do not include these results in our comparison of intrinsic merger-rate predictions.

Vigna-Gómez et al. (2020)— *Vigna-Gómez et al. (2020)* use COMPAS to investigate the formation channels and CE properties of binaries that produce BNS mergers at solar metallicity ($Z = 0.0142$). They focus in particular on how different double neutron star subpopulations—classified by the donor type during the CE phase—map onto distinct evolutionary pathways, such as the classic CE channel and the double-core CE channel. In their fiducial model, $\sim 69\%$ of BNS systems form through the classic channel and $\sim 14\%$ through the double-core CE channel (see their Section 3.1). The remaining $\sim 17\%$ arise from variations of these dominant pathways, typically involving modifications to the sequence of mass-transfer episodes or the omission of specific phases. While not always explicitly classified, these alternative channels likely also involve at least one CE phase. Taken together, this implies that at least $\gtrsim 83\%$ of BNS systems in their simulations form through channels involving CE evolution. Because the study is restricted to a single metallicity and does not provide cosmologically weighted merger rates, we do not include it in our main analysis.

Xing et al. (2024b) — *Xing et al. (2024b)* use detailed binary-evolution models to investigate the formation of compact-object binaries, with a particular focus on the role of mass transfer and CE evolution in shaping their properties. They identify characteristic evolutionary pathways and examine how different mass-transfer assumptions influence the resulting populations. For their solar-metallicity simulation, they find that $\sim 70\%$ of BHNS mergers form through channels involving a CE phase (primarily the classic CE channel), while the remaining $\sim 30\%$ form through exclusively stable mass transfer, with most of these systems undergoing two stable mass-transfer episodes; one before SN1 and one after SN1. They also find distinct (effective) spin properties and period distributions between the CE and SMT populations, which we summarize in Section 4. We do not include this study in our formation-channel comparison, as it is restricted to a single (solar) metallicity and does not provide channel fractions for intrinsic merger rates across a broader model grid. This limits a consistent comparison with studies that explore metallicity-dependent trends and report quantitative channel contributions. We instead include the companion study *Xing et al. (2024a)*, which extends this analysis to multiple metallicities and provides a more suitable basis for comparison within our framework.

Zevin et al. (2020) — *Zevin et al. (2020)* use the rapid population-synthesis code COSMIC to investigate the formation of GW190814-like compact binaries. They identify two main evolutionary pathways: Channel A, which proceeds predominantly through stable mass transfer, and Channel B, which resembles the classic CE channel. These pathways highlight the role of supernova physics and natal kicks in producing highly asymmetric merging systems. They find that at low metallicities both channels contribute comparably to the formation of GW190814-like BBH systems, whereas at higher metallicities ($Z \gtrsim Z_{\odot}/30$) the only stable mass transfer channel (Channel A) becomes dominant, accounting for $\sim 73\%$ of such systems. We do not include this study in our formation-channel comparison because the authors focus on GW190814-like BBH rates but do not present a formation channel breakdown for the intrinsic BBH merger rate.

White Dwarf Binaries — *Roy & Kalita (2026)* use COMPAS to investigate the formation of double white dwarf (DWD) systems, classifying channels based on whether binaries evolve exclusively through stable mass transfer or experience at least one CE phase. Their study is restricted to a single (solar) metallicity and does not compute cosmological merger-rate densities, and we therefore do not include it in our main analysis. The authors identify a pronounced bimodality in the final orbital separations: systems formed through purely stable mass transfer predominantly produce wide DWDs, whereas those undergoing a CE phase lead to tight systems. This dichotomy results in a gap in the DWD separation distribution between these two populations. They note that further work is required to connect these predictions to observable populations and to assess sensitivities to key model assumptions. In particular, the relative importance of the two channels depends strongly on the adopted CE efficiency parameter α as well as on the binding energy parameter λ . For their fiducial choice, stable mass transfer dominates the DWD formation pathways, while adopting $\alpha\lambda = 0.5$ leads to a CE-dominated population at solar metallicity.

Toonen et al. (2012, 2018); Roy & Kalita (2026) Studies

investigating DWD and WD-NS systems exist, but typically many of these calculate galactic populations representative of our Milky Way or potential LISA sources and do not include a large metallicity distribution (sometimes) or cosmological formation rates. Most studies typically also do not break up the studies by RLOF or CE channel. This includes *Toonen et al. (2012, 2018)* which includes a detailed breakdown of common formation pathways for DWD and NS-WD systems and type 1A supernovae. Or (*Li et al. 2023*)

APPENDIX D: POPULATION-SYNTHESIS PARAMETERS, ACRONYMS, AND ASSUMPTIONS

This appendix summarizes the key population-synthesis parameters, assumptions, and acronyms used throughout this work and in the compiled literature sample. Our meta-analysis combines results from a wide range of studies that adopt different physical prescriptions, parameterizations, and nomenclature. To enable a consistent comparison, we standardize terminology and group related assumptions (e.g., mass-transfer stability, angular-momentum loss, and supernova physics) under common labels used in our figures. For each parameter, we provide a brief description of its physical meaning, indicate the symbol used in this work, and list representative studies that vary it.

α (*common-envelope efficiency*) — The CE efficiency parameter α sets how effectively orbital energy is used to unbind the donor envelope during a CE phase. Higher α values correspond to more efficient envelope ejection and typically wider post-CE separations, while low α promotes tighter binaries or mergers during the CE phase. This parameter is denoted by α in our figures. It is widely varied across studies, including *Broekgaarden et al. (2022b); Boesky et al. (2024a); Briel et al. (2023); Bavera et al. (2021); Xing et al. (2024a); Li et al. (2025a)*, and *Sgalletta et al. (2025)*.

β (*mass-transfer efficiency*) — The mass-transfer efficiency β describes the fraction of transferred mass that is accreted by the companion during Roche-lobe overflow. It controls both the mass-ratio evolution and the amount of angular momentum lost from the system, thereby affecting the stability of mass transfer and orbital evolution. This parameter is denoted by β in our figures. It is varied in, for example, *Broekgaarden et al. (2022b); Boesky et al. (2024a)*, and *Dorozsmai & Toonen (2024)*. In these studies it is only used for mass transfer onto a star (for compact objects Eddington limited accretion is assumed).

γ (*angular-momentum loss parameter*) — The angular-momentum loss parameter γ sets the specific angular momentum carried away by mass lost from the binary, but its physical meaning depends on the adopted prescription. In the γ -formalism (e.g., *Dorozsmai & Toonen (2024)*), γ directly parameterizes the specific angular momentum of the ejected material relative to the binary orbit, with larger γ corresponding to more efficient orbital angular-momentum loss and thus stronger orbital shrinkage during non-conservative mass transfer. In contrast, other implementations (e.g., *Li et al. (2025a)*) adopt alternative prescriptions in which γ can take negative or varying values depending on the assumed mass-loss channel, effectively modifying whether mass loss extracts or redistributes angular momentum within the system. As a result, variations in γ can qualitatively change the orbital evolution during stable mass transfer, determining whether binaries tighten sufficiently to merge within a Hubble

time or instead widen and avoid merger. This parameter is denoted by γ in our figures and is explicitly varied in Dorozsmai & Toonen (2024) and Li et al. (2025a).

q_{crit} and ξ (mass-transfer stability criteria)— The stability of mass transfer is commonly parameterized either in terms of a critical mass ratio q_{crit} or through response coefficients such as the adiabatic mass–radius exponent $\xi \equiv d \ln R / d \ln M$. In the q_{crit} formalism, mass transfer becomes dynamically unstable when the donor-to-accretor mass ratio exceeds a threshold value, leading to a common-envelope phase. In contrast, ξ -based prescriptions compare the donor’s radial response to mass loss with the response of the Roche lobe, providing a more physically motivated, structure-dependent stability criterion. Different studies adopt different implementations: some assume fixed values of q_{crit} , while others use prescriptions based on detailed stellar models (e.g., Claeys et al. 2014; Ge et al. 2020a) or directly compute ξ as in Dorozsmai & Toonen (2024). These choices can shift systems between stable and unstable mass-transfer regimes and therefore strongly affect the relative importance of formation channels with and without common-envelope evolution. This parameter is denoted by q_{crit} or ξ in our figures (and grouped with stability criteria more generally) and is varied in studies such as Bavera et al. (2021), Olejak et al. (2021), Li et al. (2025a), and Dorozsmai & Toonen (2024). The different assumptions are either a fixed value for q_{crit} or ξ , or q_{crit} based on (Claeys et al. 2014) or Ge et al. (2020a).

T_{eff} (convective-envelope proxy)— The effective temperature T_{eff} is used in several studies as a proxy for the evolutionary stage at which a star develops a convective envelope, which in turn affects the stability of mass transfer. Variations in this threshold can shift systems between stable and unstable mass-transfer regimes. This parameter is denoted by T_{eff} in our figures and different assumptions for T_{eff} are explored in Dorozsmai & Toonen (2024), and Romagnolo et al. (2025): $T_{\text{eff}} - \text{K}$ for models based on (Klencki et al. 2021a), $T_{\text{eff}} - \text{MESA}$ for MESA based grids developed in Romagnolo et al. (2025), or $T_{\text{eff}} - \text{B}$ for the commonly used convective boundary of $T_{\text{eff}} = 10^{3.73} \text{ K}$ from Belczynski et al. (2008).

Supernova model: delayed, rapid, MM SN, N20 SN— Supernova (SN) prescriptions determine remnant masses and natal kicks, affecting binary survival and post-SN orbital properties. Common variations include ‘rapid’ Fryer et al. (2012), ‘delayed’ Fryer et al. (2012), the stochastic explosion model from Mandel & Müller (2020) which we denote as ‘MM SN’, the model which we denote with ‘N20 SN’, and the BPASS model. These assumptions are labeled as SN model variations in our figures and are explored in Broekgaarden et al. (2022b), Briel et al. (2023), Boesky et al. (2024a), and Shao & Li (2021a).

σ (natal kick dispersion)— The parameter σ sets the dispersion of the natal kick velocity distribution imparted to compact objects at formation. Larger kicks increase binary disruption probabilities and alter merger rates and formation channels. This parameter is denoted by σ in our figures and is varied in Broekgaarden et al. (2022b), Boesky et al. (2024a), ?, and Li et al. (2025a).

f_{WR} (Wolf–Rayet wind scaling)— The parameter f_{WR} scales the mass-loss rates of Wolf–Rayet stars, influencing stellar masses prior to collapse and thus compact-object masses and orbital evolution. This parameter is denoted by f_{WR} in our figures and is varied in Broekgaarden et al. (2022b) and Li et al. (2025a).

$M_{\text{NS,max}}$ (maximum neutron-star mass)— The assumed maximum neutron-star mass determines whether compact remnants are classified as neutron stars or black holes, thereby affecting the relative rates of BNS, BHNS, and BBH systems. This parameter is denoted by $M_{\text{NS,max}}$ in our figures and is varied in Broekgaarden et al. (2022b).

Pair-instability supernova (PISN) treatment— Assumptions about pair-instability and pulsational pair-instability supernovae affect the upper mass spectrum of black holes and the formation efficiency of massive BBHs. These assumptions are varied in Broekgaarden et al. (2022b).

$\dot{m}_{\text{Edd}}$ (Eddington-limited accretion)— The Eddington accretion rate $\dot{m}_{\text{Edd}}$ sets the maximum rate at which compact objects can accrete matter under the balance between radiation pressure and gravity. In binary evolution models, this limit effectively caps the accretion onto the compact object during Roche-lobe overflow, such that any excess mass is lost from the system. This introduces an additional source of non-conservative mass transfer, even in cases where the mass-transfer efficiency β is assumed to be high. Different studies adopt different treatments of this limit, including strict Eddington-limited accretion or allowing for super-Eddington accretion with multiplicative factors (e.g., $10^3 - 10^6 \times \dot{M}_{\text{Edd}}$). These choices affect both the mass growth of compact objects and the amount of mass and angular momentum lost from the system, thereby influencing orbital evolution and the stability of mass transfer. This parameter is denoted by $\dot{M}_{\text{Edd}}$ in our figures and is varied in studies such as Bavera et al. (2021).

R_{max} (maximum stellar radius / radial expansion limit)— The parameter R_{max} sets the maximum radius that a star can reach during its evolution in Romagnolo et al. (2023), effectively controlling the extent of radial expansion phases (e.g., giant branches). This directly impacts the likelihood and timing of Roche-lobe overflow, and therefore whether binaries interact, undergo stable mass transfer, or enter a common-envelope phase. Romagnolo et al. (2023) adopt different implementations for R_{max} , including limits derived from different stellar-evolution models or treatments of envelope inflation and mixing, see Romagnolo et al. (2023) for details.

Opt CE (Optimistic common-envelope assumption)— The optimistic CE assumption refers to models in which binaries are allowed to survive CE phases that would otherwise be assumed to lead to mergers under more conservative criteria. In particular, this includes allowing systems undergoing CE initiated by Hertzsprung-gap donors to successfully eject the envelope and form a bound post-CE binary, despite the lack of a well-defined core-envelope structure in such stars. This assumption increases the number of systems that survive CE evolution, thereby boosting the overall merger rate and enhancing the contribution of formation channels involving CE phases. In contrast, more conservative treatments assume that such systems merge during the CE phase and do not contribute to the merging compact-object population. This assumption is denoted as ‘opt CE’ in our figures and is explored in studies such as Broekgaarden et al. (2022b).

$\log U$ (initial orbital-separation prior)— In Bavera et al. (2021), the authors present formation channels for models varying the initial distribution for the orbital separation. By default this is X . But they change this to uniform in the logarithm of the separation, motivated by observations of massive binaries. This

TABLE 1
FORMATION CHANNEL CONTRIBUTIONS

BNS formation channels

Study	Z	CE statement	other notes
Mestichelli et al. (2025)	10^{-11}	$\geq 91.9\%$ with CE	rest not specified
Mestichelli et al. (2025)	10^{-4}	$\geq 99.3\%$ with CE	rest not specified
Dominik et al. (2012)	$0.1 Z_{\odot}$	$\geq 91.8\%$ with CE	rest not specified
Gallegos-Garcia et al. (2023)	$0.1 Z_{\odot}$	$\sim 100\%$ with CE	rare stable-MT-only BNS mergers for $M_{\text{NS}} = 2 M_{\odot}$ and large natal kicks
Belczynski et al. (2002)	0.02	$\geq 95\%$ with CE	DCCE before SN1
Andrews et al. (2015)	$Z_{\odot}$	$\sim 100\%$ with CE	case BB / electron-capture SN
Chruślińska et al. (2018)	$Z_{\odot}$	≥ 95 with CE	
Chattaraj et al. (2026)	$Z_{\odot}$	$\geq 95\%$ with CE	CE always with caseBB MT and USSN, case B/C MT subchannels
Chattaraj et al. (2026)	$Z_{\odot}$	$\leq 6\%$ without CE	5% can form without MT at all, 1% OSMT
Dominik et al. (2012)	$Z_{\odot}$	$\geq 94.2\%$ with CE	
Vigna-Gómez et al. (2018)	$Z_{\odot}$	$\geq 91\%$ with CE	remaining 9% CE not specified
Gallegos-Garcia et al. (2023)	$Z_{\odot}$	$\sim 100\%$ with CE	stable-MT-only BNS mergers absent for fiducial $1.4 M_{\odot}$ NSs

BHNS formation channels

Study	Z	with CE	other important fc
Mestichelli et al. (2025)	10^{-11}	≥ 36 (55.9)% with (without) CE	(rest not specified)
Mestichelli et al. (2025)	10^{-4}	≥ 46.2 (46.9)% with (without) CE	(rest not specified)
Dominik et al. (2012)	$0.1 Z_{\odot}$	$\geq 93.6\%$ with CE	
Belczynski et al. (2002)	0.02	$\geq 80\%$ with a CE	SCCE before SN1
Dominik et al. (2012)	$Z_{\odot}$	$\geq 97.2\%$ with CE	
Xing et al. (2024b)	$Z_{\odot}$	~ 70 (30)% with (without) CE	

BBH formation channels

Study	Z	with CE	other important fc
Belczynski et al. (2022)	$\leq 0.4 / Z_{\odot}$	$\sim 75\%$ with CE	CE $3\times$ higher than SMT channel
Dominik et al. (2012)	$0.1 Z_{\odot}$	$\geq 96\%$ with CE	
Belczynski et al. (2022)	$\geq 0.4 / Z_{\odot}$	without CE contribution higher	without CE channel dominates over CE channel at higher Z
Briel et al. (2026)	$\geq 0.2 Z_{\odot}$	0% without CE	without SN natal-kicks SMT does not produce BBH mergers
Belczynski et al. (2002)	0.02	$\geq 95\%$ with a CE	SCCE before SN1
Dominik et al. (2012)	$Z_{\odot}$	$\geq 98.9\%$ with CE	

choice sets the relative fraction of systems formed at wide versus close separations and therefore strongly affects the likelihood of binary interaction, including Roche-lobe overflow and common-envelope evolution. This parameter is denoted as “log U prior” in our figures and is varied in Bavera et al. (2021).

APPENDIX E: FORMATION CHANNEL CONTRIBUTION TO FORMATION EFFICIENCY

In Figure 14 we show for additional studies the formation channel contribution to the BBH, BHNS, and BNS formation efficiency. Some studies provide this (often instead of BBH, BHNS, BNS intrinsic merger rates) or only formation channels at a specific metallicity which we provide in an overview there.

Overall: agreement that BNS always form with a CE, no matter what the metallicity. Large variety in BBH, BHNS contribution of the without CE channel, but some agreement that the SMT channel for BBH might be more important at low metallicity. Same but smaller effect for BHNS where there is a slight increase around low metallicities in the without CE contribution, and a small higher contribution around solar metallicity.

APPENDIX F: OBSERVABLE FEATURES TABLES

To facilitate comparison across the literature, we summarize in Tables 2–?? the main gravitational-wave observables associated with each study and the physical parameter(s) that appear to have the largest impact on the predicted formation-channel contributions. These entries are intentionally qualitative and study-level: they are meant to highlight the dominant trends discussed in each work rather than provide an exhaustive summary of every individual simulation variant. This is particularly useful here because our compilation includes more than one hundred simulations across many population-synthesis studies.

APPENDIX G: ABSOLUTE MERGER RATES PARAMETER VARIATIONS

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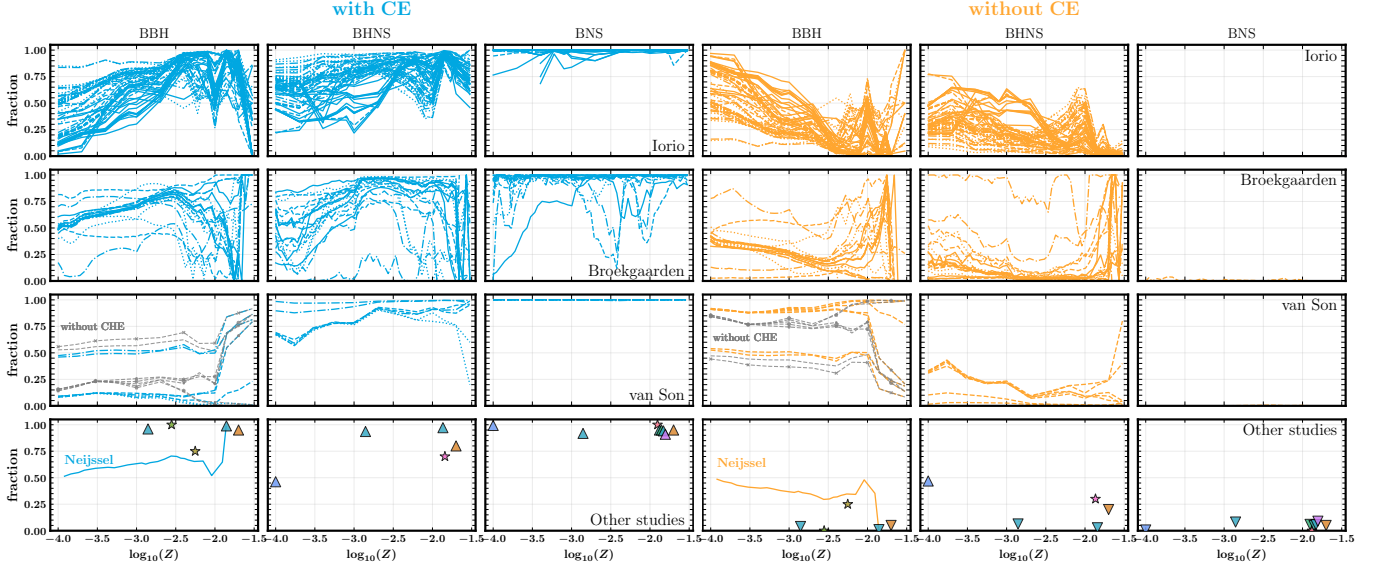


FIG. 15.— Formation-channel fractions as a function of metallicity for compact-object binaries across multiple population-synthesis studies and literature constraints. Rows corresponds (top to bottom) to Iorio et al. (2023b), Broekgaarden et al. (2022c), van Son et al. (2024), and a compilation of additional literature constraints. Columns show double compact object types (BBH, BHNS, BNS), while the left and right blocks correspond to systems formed with and without a CE phase, respectively. For van Son et al. (2024) BBH models, gray curves show fractions excluding chemically homogeneous evolution (CHE). Some Broekgaarden et al. (2022c) models include a non-negligible “other” channel (CE not specified). Low-number statistics introduce noise near $Z_{\odot}$ for BBH and BHNS models. In the bottom row, black curves show results from Neijssel et al. (2019), while colored markers indicate literature constraints at specific metallicities (Table 1); arrows denote limits and stars approximate values. All points are converted to $\log_{10}(Z)$, and complementary fractions are inferred where needed assuming $f_{\text{with CE}} + f_{\text{without CE}} = 1$. The figure highlights the wide diversity of BBH and BHNS predictions and the consistent result that **BNS formation is dominated by CE channels**, with $\lesssim 10\%$ forming without CE across models and metallicities. See Appendix D for details.

TABLE 2
QUALITATIVE STATEMENTS IN THE LITERATURE ABOUT THE OBSERVATIONAL IMPRINT OF THE SMT CHANNEL. HM: HIGH MASS, LM: LOW MASS

Study	Observable	Main statement about SMT channel
Li et al. (2025a)	Mass distribution	SMT produces somewhat higher-mass BBHs than CE
Neijssel et al. (2019)	Mass distribution	SMT contributes substantially to HM BBHs
Willcox et al. (2025)	Mass distribution	SMT contributes to HM+HM BBHs; both CEE and SMT to LM+LM BBHs (Chirp mass)
van Son et al. (2022)	Mass distribution	SMT contributes substantially to higher-mass BBHs
van Son et al. (2022)	Mass distribution	SMT produces a low-mass cutoff ($\sim 2.5\text{--}9 M_{\odot}$) and a peak around $\sim 8\text{--}10 M_{\odot}$ set by MT stability
Hendriks et al. (2023)	Mass distribution	SMT preferentially produces the HM BBHs
Belczynski et al. (2022)	Mass distribution	SMT produces substantially more massive BBHs than CE channels
Klencki et al. (2026)	Mass distribution	SMT preferentially produces more massive BBHs
Xing et al. (2024a)	Mass distribution	SMT might be important for forming BHNS containing BHs $\gtrsim 11 M_{\odot}$
Briel et al. (2026)	Mass Ratio	SMT BBH mergers formed without BH kick have, independent of metallicity, mass ratios distribution
Briel et al. (2026)	Mass Ratio	SMT BBH mergers formed with BH kick might have more unequal mass ratios
Banerjee & Olejak (2024)	Mass Ratio	The SMT channel combined with the delayed remnant mass model produces the most unequal-mass BBHs
Olejak et al. (2024)	Mass ratio	SMT can produce BBHs with lower mass ratios $q \in [0.4, 0.7]$ and relatively high effective spins (e.g. $\chi_{\text{eff}} \gtrsim 0.1$)
Zevin et al. (2020)	Mass Ratio	Both Classic CE channel and OSMT channels might contribute to the formation of unequal mass ratios
Xing et al. (2024b)	Mass Ratio and Period	Most SMT BHNS have periods $\lesssim 50$ d and wide spread of mass ratios (at $Z_{\odot}$)
Zevin et al. (2020)	Mass Ratio	at $Z \gtrsim Z_{\odot}/30$ stable channel dominates formation of GW190814-like (unequal q) BBH systems
Klencki et al. (2026)	Delay time	SMT preferentially produces long delay times because orbital tightening is limited
van Son et al. (2022)	Delay time	SMT preferentially produces long BBH delay times because orbital tightening is limited
van Son et al. (2022)	Delay time	SMT BBH and BHNS delay times depend on orbital tightening during stable mass transfer and can be long
Briel et al. (2026)	Delay time	SMT channel produces long delay times $\gtrsim 0.2$ Gyr
Olejak et al. (2024)	Delay time	SMT delay times depend sensitively on the degree of orbital tightening during mass transfer
Oh et al. (2023)	Delay time	SMT channel cannot produce BBH and BHNS with short enough delay times, except for low metallicities
Klencki et al. (2026)	Spin	SMT generally leads to low BH spins because post-SMT binaries remain relatively wide
Briel et al. (2026)	Spin	SMT BBH mergers have peaks at $\chi_{\text{eff}} = 0$ and $\chi_{\text{eff}} = 0.15$.
Olejak et al. (2024)	Spin	SMT can produce low or moderate BH spins depending on the efficiency of angular-momentum transfer
Xing et al. (2024b)	Spin	SMT BHNS can have large ($\gtrsim \pi/2$ rad) spin tilt angles and effective spins spanning $-0.2\text{--}0.2$, due to SMT
Briel et al. (2026)	SMT progenitors	SMT BBH mergers arise predominantly from short-period case A mass transfer progenitors (P_{ZAMS})

TABLE 3
QUALITATIVE STATEMENTS IN THE LITERATURE ABOUT THE OBSERVATIONAL IMPRINT OF THE WITH CE CHANNEL.

Study	Observable	Main statement
Li et al. (2025a)	Mass distribution	CE dominates the lower-mass peak of the BBH mass distribution
Hendriks et al. (2023)	Mass distribution	CE channel struggles to produce the highest-mass BBHs, with massive systems preferentially forming through SMT
Belczynski et al. (2022)	Mass distribution	CE channel produces slightly lower-mass BBHs compared to SMT
Willcox et al. (2025)	Mass distribution	CEE contributes to HM+LM BBHs, and both CEE and SMT to LM+LM
Oh et al. (2023)	Mass Ratio	Systems undergoing CE after the first SN tend to produce more symmetric compact binaries, while CE before the first SN tend to produce more asymmetric compact binaries
Zevin et al. (2020)	Mass Ratio	Both Classic CE channel and OSMO channels might contribute to the formation of unequal mass ratio BBHs
Bavera et al. (2021)	Spin	CE channels can produce spinning second-born BHs through tidal spin-up
Xing et al. (2024b)	Spin	CE BHNS have small spin tilt angles and small effective spins (peak at 0 and extend > 0) (at $Z_{\odot}$)
Olejak et al. (2024)	Mass ratio and spin	CE produces BBHs with relatively equal component masses ($q \sim 1$) and low or a range of BH spins.
Andrews et al. (2015)	Period	All (short-period) BNS systems form via CE after SN1.
Neijssel et al. (2019)	Delay time	CE preferentially produces shorter delay times than SMT
van Son et al. (2022)	Delay time	CE contributes more strongly to rapidly merging BBHs than SMT
Andrews et al. (2015)	Delay Time	BNS channels involving case BB MT lead to BNS with shorter periods
Xing et al. (2024b)	Mass Ratio and Period	Most CE BHNS have periods ≥ 50 d and mass ratios ≥ 0.5 at solar metallicity
Andrews et al. (2015)	eccentricity	BNS channels involving electron-capture SN (low natal kicks), lead to BNS with lower eccentricities

TABLE 4
QUALITATIVE STATEMENTS IN THE LITERATURE ABOUT THE OBSERVATIONAL IMPRINT OF THE CHE CHANNEL.

Study	Observable	Main statement
Li et al. (2025a)	Mass distribution	CHE produces a mass peak around $\sim 30 M_{\odot}$
Li et al. (2025a)	Mass distribution	CHE contributes to the high-mass part of the BBH population relative to CE
Chattaraj et al. (2026)	Donor structure	The evolutionary state of the donor at CE onset determines whether systems merge within a Hubble time
Chattaraj et al. (2026)	Natal kicks	Low supernova kick velocities are required to reproduce the observed DNS population

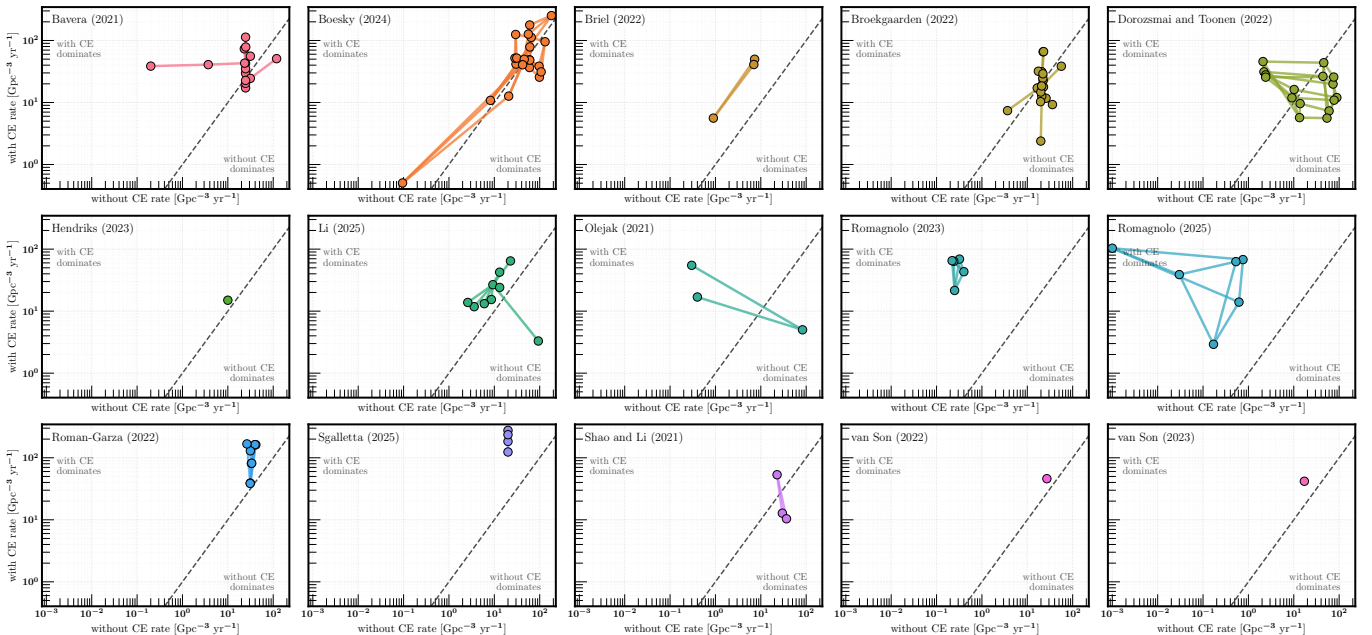


FIG. 16.— Without-CE versus with-CE intrinsic BBH merger rates for the compiled BBH population-synthesis simulations. Scatter symbols, colors, and connecting lines are as in the bottom panel of Figure 9. The dashed diagonal line indicates equal contributions from with-CE and without-CE formation channels.

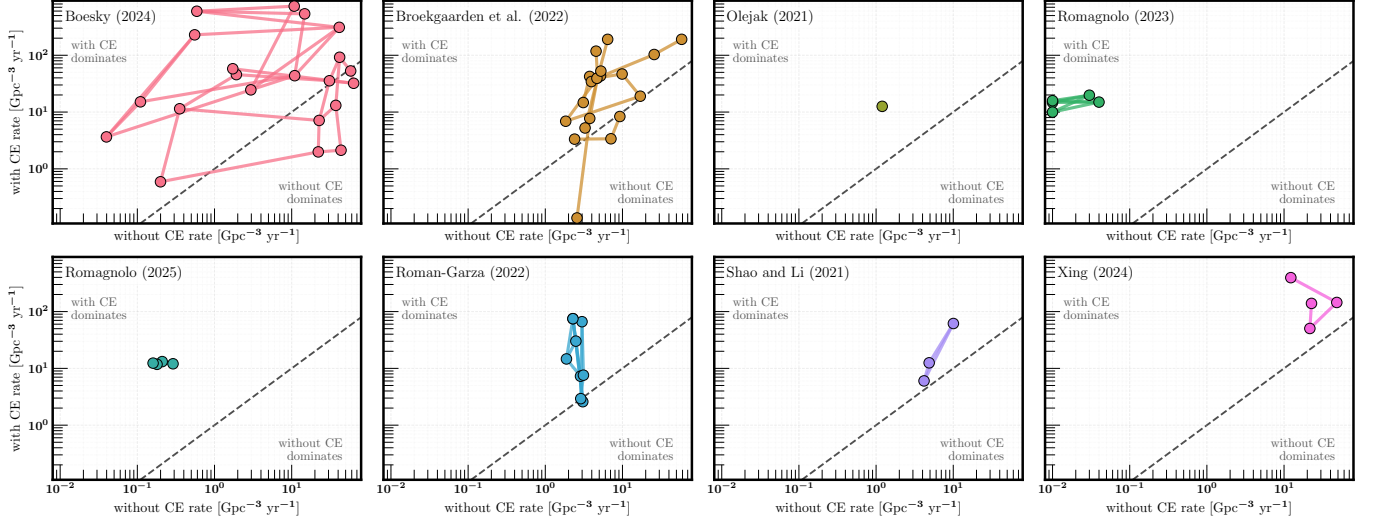


FIG. 17.— Same as 16 for BHNS mergers.